



Machine Vision Engineering Handbook

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1. Machine Vision Engineering Handbook

Welcome

The Machine Vision Engineering Handbook is the principal technical knowledge resource of the Machine Vision Academy.

Warning

*In order to make content available as soon as it is developed and reviewed, this handbook has been published with structural placeholders. Individual chapters will be released and marked complete as notated in the **revision history table below**. Chapters with content that has not been marked as released in the Revision History table should not be assumed to be complete nor reviewed.*

Who is this handbook for?

The handbook is intended for:

- engineers designing or specifying industrial vision systems;
- automation and controls engineers integrating vision equipment;
- computer-vision and machine-learning practitioners working in industry;
- students and lecturers studying machine vision;
- quality, validation and regulatory professionals;
- technical reviewers evaluating machine-vision projects.

The material is organised so that readers can use it either as a sequential engineering reference or as a source for individual design, implementation and validation topics.

Revision history

The current revision history is maintained online and is not reproduced in the printed editions of this handbook. The latest revision history is available at:

<https://machinevisionacademy.org/handbook/#revision-history>

Acknowledgements

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

Notation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

Part I.

Foundations

i Part status: Structural placeholder.
This part introduction will later summarise its scope, prerequisites and learning objectives.

2. Vision engineering: scope, definitions and history

2.1. Machine vision, computer vision and image processing

The terms machine vision, computer vision and image processing overlap, and their boundaries change with professional context. This handbook therefore does not attempt to impose a universal taxonomy. Instead, it establishes a working vocabulary for industrial automation, while recognising the broader usage found in computer science, signal processing and academic research.

2.1.1. Working definitions and industrial success criteria

For this handbook, machine vision is defined as follows:

Important

“Machine vision is the engineering discipline concerned with specifying, designing, integrating and validating an image-based function that observes a physical process and produces information used for inspection, measurement, identification, guidance or control.”

Its scope can therefore include the definition of requirements and reference truth; part or scene presentation; illumination; optics; sensing and acquisition; image processing and computer-vision algorithms; communications and control interfaces; the resulting process action; and evidence that the complete system performs adequately in production.

This systems view is consistent with industrial treatments of machine vision. The European Machine Vision Association (EMVA) presents a chain extending from handling, lighting and optical image formation through acquisition and processing to classification and process feedback [1]. The VDI/VDE/VDMA 2632 series places requirements specification and acceptance testing explicitly within the engineering lifecycle [2].

Caution

“A production vision system does not begin when an image reaches memory, and it does not end when an algorithm returns a score.”

Requirements deserve particular attention because a vision system cannot reliably automate a criterion that has never been made explicit. Where an existing human inspection provides the starting point, agreement studies can help establish whether the judgement is sufficiently repeatable to become an automated requirement. The detailed methods for requirements engineering, measurement-system analysis and validation are developed later in this handbook.

Industrial success is therefore judged at system level rather than by an algorithm metric alone. A strong offline classification result cannot compensate for unstable image formation, inadequate cycle time, poor reject timing, unrepresented production variation or an ambiguous acceptance criterion. In practice, performance combines the quality of the decision or measurement with robustness, timing, integration and maintainability.

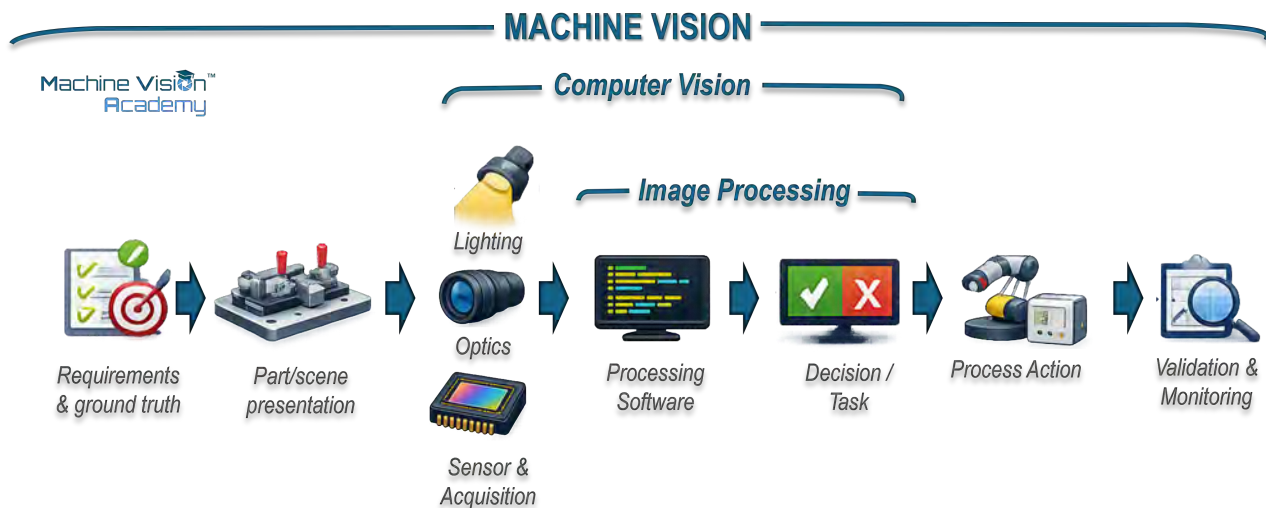


Figure 2.1.: Machine vision as an end-to-end engineering lifecycle

2.1.2. The four fields: image processing, computer vision, machine vision and AI

The following definitions are deliberately operational. They identify the centre of gravity of each field rather than drawing impermeable boundaries. A single industrial application can occupy all four fields at once.

Table 2.1.: Working meanings and typical emphasis of image processing, computer vision, machine vision and artificial intelligence.

Field	Working meaning in this handbook	Typical emphasis
Image processing	Mathematical or computational operations on image data, including correction, enhancement, filtering, transformation, segmentation and representation.	The image and its transformation.
Computer vision	Computational methods that infer information about objects, scenes, geometry, events or states from visual data.	Perception and inference from images.
Machine vision	The engineering of an image-based system that must perform a useful function within a physical or automated process.	The complete requirements-to-action system.
Artificial intelligence	A broad field concerned with systems that perform capabilities associated with intelligence. In later vision chapters, “AI-based vision” will normally denote data-trained models, particularly machine-learning and deep-learning approaches.	Learned inference and decision methods, while recognising that formal AI is broader than deep learning.

These meanings are not universal. Academic computer vision may include acquisition and image processing within its own scope; image processing is a substantial discipline in signal processing; and machine vision has historically also been used for practical artificial vision more generally. Davies discusses the historical convergence of computer vision and machine vision [3]. The British Machine Vision Association (BMVA) provides a visible example of the overlap: its remit explicitly includes machine vision, image processing and pattern recognition [4].

The same caution applies to AI. ISO/IEC terminology treats artificial intelligence as a broad field and does not equate it with neural networks or deep learning [5]. For clarity in the engineering chapters, the more useful distinction will usually be between classical or explicitly engineered vision methods and learned vision methods. Deep neural networks are a major family of learned methods, but they are not the formal boundary of AI.

Note

“The labels are useful only when they clarify engineering responsibility. They become misleading when a field name is treated as evidence of a capability that has not actually been demonstrated.”

2.1.3. Relations, overlaps and typical confusions

Terminological overlap matters most when expertise is assessed. A computer-vision specialist may have considerable depth in segmentation, detection, pose estimation, deep learning or open-source algorithm development while having limited experience of illumination, lenses, triggering, industrial communications or reject mechanisms. An experienced machine-vision integrator may show the reverse profile: strong acquisition and production-system engineering, but less depth in advanced computer-vision research. Neither profile is inherently more complete; the useful question is which part of the problem has actually been engineered.

Tip

“When assessing machine-vision expertise, ask which parts of the requirements-to-action chain the person has designed, commissioned and validated—not only which algorithms or products they have used.”

A second confusion arises from the word *train*. Classical pattern-matching systems can require a template image and may expose a command called *Train*, but this does not by itself make them machine-learning systems. Cognex PatMax, for example, represents a trained pattern using geometric features and their spatial relationships [6], [7]. HALCON shape-based matching similarly searches for instances of a shape model that has been created previously [8]. These are classical model- or template-based techniques rather than neural models fitted to a labelled dataset.

By contrast, learned vision methods estimate model parameters from training data; deep learning is a prominent example. The distinction matters because AI is now also a commercial descriptor. The UK Advertising Standards Authority has documented the widespread use of the term in advertising and the associated regulatory implications [9]. For this handbook, an AI claim is therefore more informative when it states what is learned from data, what model is trained and how performance on unseen data is demonstrated.

The practical relationship is simple enough to carry forward: image processing and computer vision provide computational techniques; machine learning adds families of data-trained inference techniques; and machine vision integrates whichever techniques are appropriate into a physical system that must satisfy an industrial requirement. Later chapters develop acquisition engineering, algorithms, learned vision, integration and validation in detail.

2.2. Historical development: several timelines, one discipline

Machine vision does not have a single birthday. One historical clock started when photographs became arrays of numbers; another when researchers began asking computers to recover shape and meaning from those numbers. A third started on the factory floor, where the decisive question was not whether a machine could understand an arbitrary scene, but whether it could inspect, measure or guide reliably at production speed. Consumer products then made many of the same technologies familiar, while standards emerged to make an increasingly diverse industrial ecosystem interoperable.

Those clocks do not tick at the same rate. A technique can exist in a research paper for years before sensors, processors, software tools and interfaces make it practical in production. Equally, a supposedly new consumer feature may have a surprisingly old industrial ancestor. The history is therefore presented here as a set of parallel timelines rather than a single sequence of inventions. The purpose is orientation: later chapters return to the individual technologies, architectures and validation methods in engineering detail.

2.2.1. The innovation timeline: when images became computable

A convenient starting point is 1957. At the US National Bureau of Standards, now NIST, Russell Kirsch and colleagues used a rotating-drum scanner with the SEAC computer to digitise a photograph of Kirsch's infant son, Walden. The image was only 176 pixels on each side, but it established the essential abstraction behind modern vision engineering: a picture could be represented as numbers and therefore manipulated by a computer [10].



Figure 2.2.: Image of Walden Kirsch, first digital picture 1957

The next step was to do useful work to those numbers. During the Ranger and Surveyor lunar programmes, engineers at NASA's Jet Propulsion Laboratory digitised and processed spacecraft imagery to reduce noise, compensate for distortions and improve visibility. Ranger 7 alone returned 4,308 images in the final minutes before lunar impact on 31 July 1964 [11], [12]. The lineage from spacecraft image correction to industrial flat-field correction, geometric calibration and computational photography is not direct in every implementation, but the engineering idea is familiar: image formation is imperfect, and computation can make the measurement more useful. Later in this handbook we, cover techniques to enhance images for industrial applications.

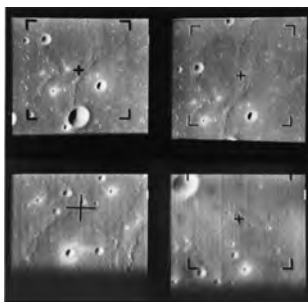


Figure 2.3.: Enhanced images of the moon taken by Ranger 7 - 1964

At roughly the same time, computer vision was becoming a research problem in its own right. Lawrence Roberts' 1963 MIT thesis analysed simple three-dimensional solids from two-dimensional images [13]. In 1966 Seymour Papert wrote the MIT Artificial Intelligence Memo commonly known as the Summer Vision Project, proposing that summer students should connect a camera to a computer and tackle a sequence of visual tasks [14]. The story is often retold as a single optimistic instruction from Marvin Minsky; the surviving proposal is Papert's. Either version captures the mood: researchers had not yet discovered how much of intelligence is hidden inside apparently simple acts of seeing.

The sensing hardware was changing just as quickly. Willard Boyle and George Smith described the charge-coupled device (CCD) in 1970 after its invention at Bell Laboratories in 1969 [15]. Five years later Steven Sasson at Kodak assembled a self-contained solid-state digital still camera using a CCD. The prototype recorded roughly 0.01 megapixels to a cassette and took about 23 seconds to store an image [16]. It was not a product, but it made a future without photographic film physically demonstrable.



Figure 2.4.: First digital camera prototype developed by Steve Sasson at Kodak (Credit: Getty Images) - 1975

i A 0.01-megapixel glimpse of the future

Sasson's 1975 camera is useful precisely because it looks so unlike a modern camera: about the size of a toaster, with a cassette recorder and a separate playback arrangement. The important innovation was not convenience; it was that the complete image chain - sensor, digitisation, storage and display - had become electronic. An engineering prototype can be historically important long before it is commercially sensible; this one looked like a kitchen appliance and stored its pictures on the same cassette tapes people used for mix-tapes [16]. The prototype was rejected by Kodak in an attempt to protect their dominance in the analog photography business.

Digitisation was only the beginning. From the 1960s to the 1980s, a classical toolbox accumulated that is still recognisable in industrial vision software. Hu's 1962 moment invariants provided compact shape descriptors designed to remain useful under changes of position, scale and orientation [17]. Rosenfeld and Pfaltz's 1966 sequential picture operations included connected-component labelling - the basis of what machine-vision engineers still call blob analysis [18]. Otsu's 1979 threshold-selection method gave

2. Vision engineering: scope, definitions and history

a principled way to separate foreground from background [19], while Ballard’s 1981 generalised Hough transform extended voting-based detection from simple analytic curves to arbitrary shapes [20]. Fischler and Bolles’ RANSAC, also published in 1981, made geometric model fitting robust to outliers [21]. Canny’s 1986 edge detector and Harris and Stephens’ 1988 corner detector then supplied local features that became enduring building blocks [22], [23]. The terminology sounds historical; the underlying operations - segment, label, describe, fit, locate and measure - remain everyday machine-vision tasks.

Some of the mathematics behind recognition was older than digital imaging itself. Pearson’s 1901 closest-fit formulation is an origin point for principal-component analysis (PCA) [24]. Much later, Turk and Pentland’s 1991 “eigenfaces” showed how face images could be represented in a much lower-dimensional appearance space [25]. The method is not a direct ancestor of every modern learned embedding, but the engineering instinct is familiar: retain the variation useful to a decision while discarding dimensions that contribute little.

Alongside this practical toolbox, the field was acquiring a language for thinking about vision. David Marr’s posthumous 1982 book framed visual processing at computational, algorithmic/representational and implementation levels [26]. In 1986 Rumelhart, Hinton and Williams published the influential back-propagation paper showing how multilayer networks could learn internal representations by adjusting connection weights [27]. Neither event immediately transformed factory inspection, but both helped define computational and learning frameworks on which later work would build.

A parallel strand asked not only what a machine could see, but what it could do with the result. SRI’s Shakey project (1966-1972) combined a television camera with perception, planning and mobile action [28]; in 1979 the Stanford Cart used stereoscopic vision to cross a chair-strewn room autonomously, although its 20-metre journey took about five hours [29]. In industry, Westinghouse’s 1983 APAS research used machine vision for positioning, orientation and inspection in flexible assembly [30]. Tsai and Lenz’s 1989 hand-eye calibration method provided a practical mathematical bridge between camera and robot coordinate frames [31]; Besl and McKay’s 1992 ICP method became foundational for aligning measured 3D shapes with models [32]; and by 1996 visual-servo research had formalised how image measurements could close the motion-control loop [33]. The recurring systems problem was already clear: a visual result becomes useful to a robot only when it can be converted into geometry and feedback the machine can act upon.

i Twenty metres in five hours

The Stanford Cart is a useful reminder of how quickly expectations move. Its 1979 autonomous run through a room was a landmark in vision-guided navigation, yet the cart stopped roughly every metre while its computers spent ten to fifteen minutes interpreting stereo images and choosing the next move [29]. What reads today like an impractically slow robot was, at the time, evidence that a machine could use vision to plan its own path. It advanced at roughly the pace of a contemplative tortoise; by the standards of 1979, that was astonishing.

Sensor integration then changed the economics. Eric Fossum’s work on CMOS active-pixel sensors at JPL in the early 1990s led to the “camera-on-a-chip” architecture reviewed in his 1997 IEEE paper [34]. Meanwhile, convolutional neural networks were becoming practical for constrained recognition tasks: LeCun and colleagues reported gradient-based document-recognition systems in 1998 [35]. These systems still addressed tightly framed problems, but they demonstrated that useful visual features could increasingly be learned rather than entirely hand-designed.

Hand-engineered recognition was also maturing. Viola and Jones’ 2001 boosted cascade brought fast face detection into real-time products [36], while Lowe’s SIFT formulation in 2004 made local image features much more robust to changes in scale, orientation and viewpoint [37]. This was the mature feature-engineering era: carefully designed representations, geometric consistency and robust matching rather than end-to-end learned features.



Figure 2.5.: Still from the 1989 demonstration of handwritten digit recognition using an early convolutional neural network architecture (a precursor to LeNet) at AT&T Bell Laboratories. *Image courtesy of Lawrence D. Jackel / AT&T Bell Labs (Now Nokia Bell Labs). Restored and documented by Yann LeCun*

2. Vision engineering: scope, definitions and history

The next discontinuity came from scale rather than a single new mathematical idea. ImageNet, introduced in 2009, supplied millions of labelled images organised around a large semantic hierarchy [38]. In 2012 Alex Krizhevsky, Ilya Sutskever and Geoffrey Hinton used a deep convolutional network and GPUs to win the ImageNet challenge by a striking margin [39]. The result accelerated a shift already under way: instead of engineers specifying most visual features by hand, increasingly large parts of the representation could be learned from data. Industrial products would absorb that change more cautiously, because a production line values repeatability, latency, traceability and failure handling as much as benchmark accuracy.

AlexNet was a turning point, not an endpoint. R-CNN showed in 2014 how deep features could drive object detection [40]; U-Net in 2015 established an influential encoder-decoder architecture for dense segmentation [41]; ResNet made substantially deeper models trainable through residual connections [42]; and YOLO reframed object detection as a single real-time network [43]. By 2020-2021, Vision Transformers showed that self-attention could compete with convolution as a core visual representation [44]. Later chapters return to these architectures in detail; here they simply mark how quickly learned vision expanded from classification into localisation, detection, segmentation and more general visual representations.

A parallel software timeline made those methods much easier to reproduce and reuse. Intel's OpenCV project reached its first public release in June 2000, putting a broad collection of computer-vision routines into an open-source library that could run on ordinary computers [45]. Google open-sourced TensorFlow on 9 November 2015, turning infrastructure developed for large-scale machine learning into a generally available framework [46], [47]. PyTorch development began at Facebook/Meta in 2016 and the framework was released publicly in January 2017 [48], [49]. These libraries did not originate as industrial machine-vision products, but they greatly lowered the barrier between research code and deployable vision software. OpenCV in particular became a common foundation for bespoke computer-vision applications, while TensorFlow and PyTorch accelerated the spread of trained neural models into both research and commercial vision.

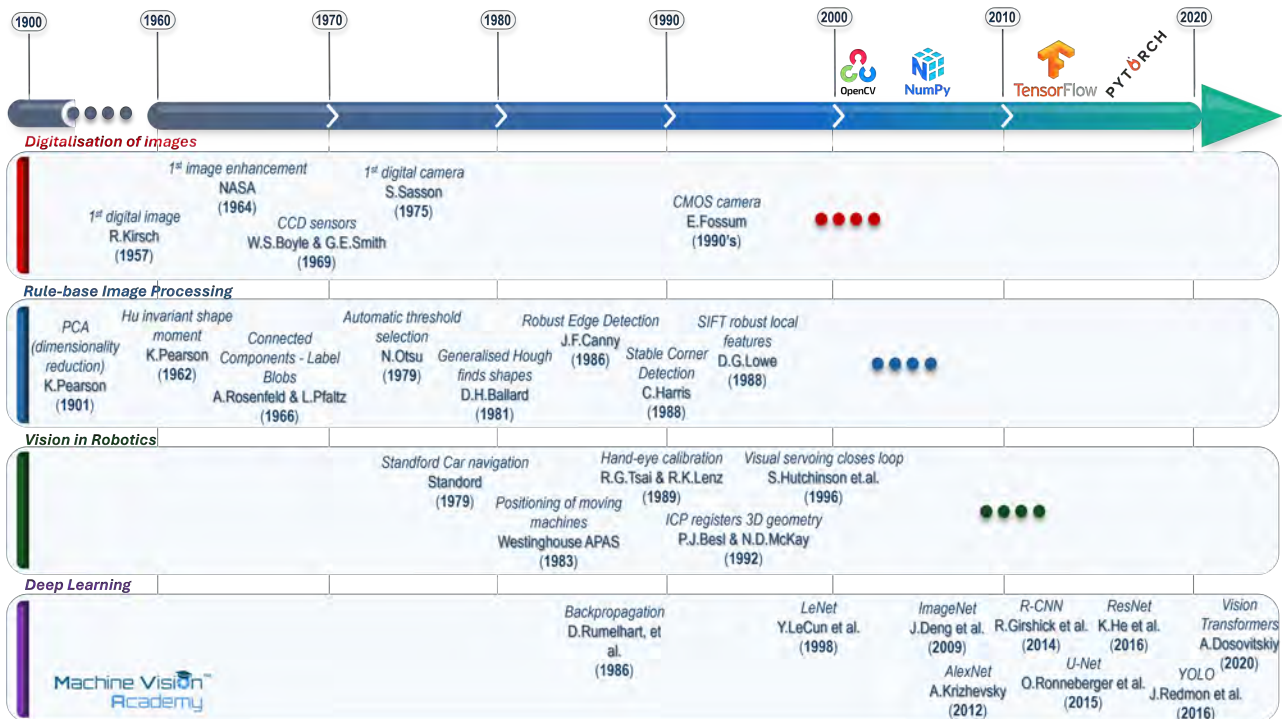


Figure 2.6.: Four parallel tracks describing the innovation timeline: Digitalisation, Rule base, Robotics & Deep Learning

2.2.2. The industrial timeline: from bespoke lab prototypes to deployable vision

Industrial machine vision followed a different rhythm from academic computer vision. Early systems of the 1970s and 1980s were often sizeable, expensive and specialised. A3's historical accounts describe an era in which VME backplanes and dedicated image-processing hardware were commonplace, while suitable cameras were scarce and costly [50]. The important industrial breakthrough was therefore not one algorithm but the gradual availability of an entire stack: reliable solid-state cameras, faster processors, frame grabbers, usable software, industrial I/O and engineers who knew how to make the image itself repeatable.

The creation of the Automated Imaging Association (AIA) in 1984 is a useful marker of that transition from experimental technology to recognisable industry [51]. By the late 1980s and early 1990s, camera suppliers were designing products specifically for machine-vision requirements, personal computers were displacing proprietary processing architectures, and systems were becoming sufficiently reliable for broader deployment. This is the period often remembered within the industry as machine vision's commercial coming of age.

The 1990s also changed where the computation lived. PC-based systems benefited from commodity processors and flexible expansion hardware; smart cameras moved processing closer to the point of acquisition. A3 notes that the first smart cameras appeared in the 1990s, although the boundary between a "vision sensor" and a "smart camera" has never been perfectly sharp [50]. This architectural choice - central computer, embedded controller or intelligence inside the camera - remains active today, only with much more computing power available in every category.

Standardisation was another quiet revolution. Camera Link, developed in 2000, standardised a high-performance digital connection between cameras and frame grabbers that could be purchased and install in an engineering PC. GigE Vision followed in 2006 and USB3 Vision in 2013, while Camera Link HS and CoaXPress addressed higher-bandwidth applications [52]. The first module of the EMVA 1288 sensor-characterisation standard was released in 2005 [53], while GenICam, whose working group began in 2003 and issued its first release in 2006, supplied a common software model for camera configuration across different transport technologies [54]. These standards mattered because they reduced ambiguity and the cost of changing components: machine builders could increasingly compare sensors more consistently and select cameras, interfaces and software from different suppliers without redesigning the whole acquisition architecture.

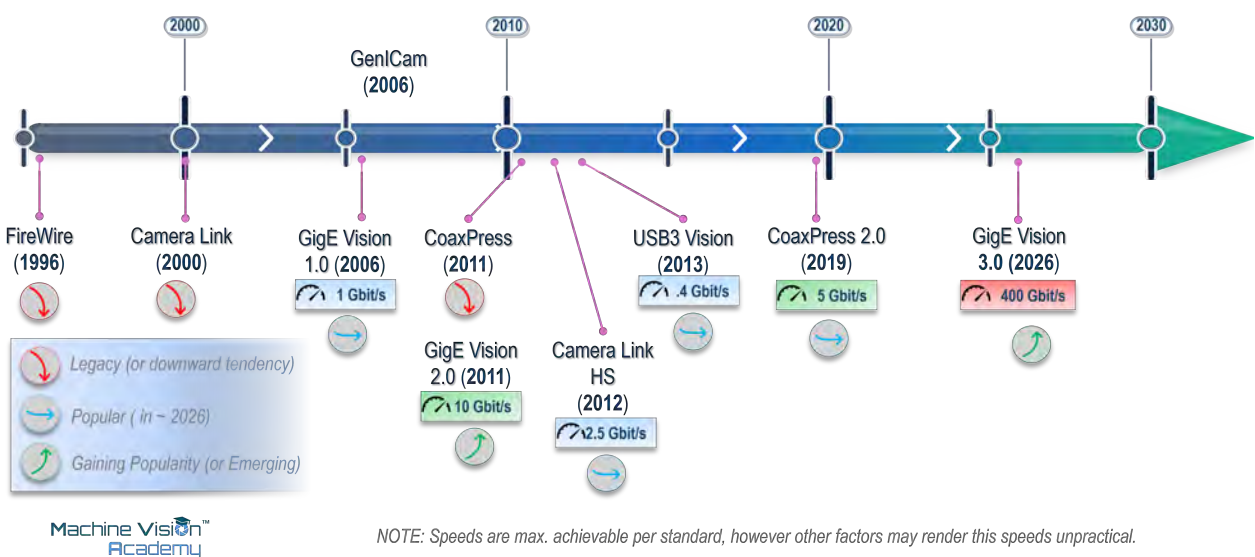


Figure 2.7.: Release timeline of some relevant communication standards for industrial cameras

Three-dimensional sensing then moved from specialist metrology towards mainstream in-line inspection. Laser triangulation profilers, time-of-flight cameras, structured-light systems and later compact 3D

smart sensors made height and shape available as ordinary production data rather than laboratory measurements. By the 2010s, products such as LMI’s all-in-one Gocator 3110 combined 3D point-cloud acquisition, embedded measurement tools and control decisions in a single sensor [55]. The historical direction is consistent: functions that once required a rack of hardware migrate into smaller, more standardised and more autonomous devices.

The same pattern is now visible with learned vision. Deep learning first changed academic benchmarks, then appeared as server- or GPU-based industrial software, and eventually moved into deployable smart-camera products. Cognex, for example, launched the In-Sight D900 in 2020 and described it as the first industrial smart camera with embedded deep-learning software [56]. OMRON introduced AI-based defect-detection functions into its FH platform in the same period [57]. The industrial adoption curve is slower than the research curve, but often more consequential: once a method survives the requirements of cycle time, maintenance and controlled change, it can be replicated across thousands of production assets.

2.2.3. The product timeline: how the industrial toolbox formed

Industrial machine vision is unusually shaped by product companies. Research establishes what is possible, but suppliers decide how that possibility reaches an engineer: as a camera, frame grabber, library, smart sensor, development environment, 3D profiler or integrated inspection platform. Table 2.2 therefore records representative milestones from companies that have had a visible influence on the industrial toolbox. It is not intended as a ranking or an exhaustive corporate history.

The trouble with claiming a first

Claims of “first” require care. Where the source is a company archive or press release, the wording below treats the claim as the supplier’s claim rather than as an uncontested historical fact. This matters because product categories evolve. A 1991 “smart image sensor”, a 1995 “industrial smart camera” and a later fully programmable smart camera can all plausibly be first under different definitions.

Table 2.2.: Representative milestones in the industrial machine-vision supplier ecosystem

Date	Company	Milestone	Why it mattered / source status
1968	Micro-Epsilon	Company founded; later specialised in precision displacement and profile measurement.	Its 1992 history entry records the launch of its first digital laser-triangulation sensors, followed by laser scanners and 3D inspection systems [58].
1974	KEYENCE	Company founded in Japan with a focus on automation and sensing.	Its corporate timeline marks the 2007 LJ Series 2D laser profiler as a high-precision non-contact measurement milestone [59].

2.2. Historical development: several timelines, one discipline

Date	Company	Milestone	Why it mattered / source status
1976	National Instruments	Company founded around computer-based instrumentation.	LabVIEW arrived in 1986; NI later extended the same “software-defined instrument” idea into PC-based machine vision [60].
1976	Matrox	Matrox founded; a dedicated imaging lineage developed alongside graphics and video.	Matrox records the 1986 MVP-AT as the world’s first PC-based image-processing accelerator and the 1993 Matrox Imaging Library (MIL) as an early hardware-independent vision library [61].
1980	DALSA / Teledyne DALSA	DALSA established in Waterloo, Canada.	In the late 1980s DALSA says it released the first commercially available high-performance 4K line-scan image-sensor chip, an important step for high-resolution industrial scanning [62].
1981-1982	Cognex	Cognex founded in 1981; DataMan followed in 1982.	Cognex describes DataMan as the world’s first industrial OCR system for direct-marked characters on parts - an early example of vision sold as a production tool rather than a research system [63].

2. Vision engineering: scope, definitions and history

Date	Company	Milestone	Why it mattered / source status
c. 1984	Adept Technologies	AdeptVision developed as an integrated commercial robot-vision system, building on earlier Unimation/Machine Intelligence Corporation vision work.	AdeptVision is an early example of vision becoming part of a commercial robot platform rather than a completely separate inspection system. A3 retrospective material describes vision as integrated into the Adept robot controller and records deployment in thousands of systems [64]. The exact product boundary and first-release date are treated as approximate.
1988	Basler	Basler founded.	The company later concentrated on digital industrial cameras; in 2016 it introduced its first 3D camera, a time-of-flight model [65].
Late 1990s	National Instruments	IMAQ Vision integrated machine-vision/image-processing functions with LabVIEW.	Vision became another software-defined engineering task inside a graphical automation environment rather than an isolated vision workstation [60], [66].
1991	DVT	SmartImage Sensor introduced.	DVT representatives later described this as the first smart camera. The claim is useful but definition-dependent [67].
1992	Micro-Epsilon	First digital laser-triangulation sensors in the company timeline.	Shows the parallel growth of optical metrology and machine vision rather than a purely 2D-camera history [58].
1993	Matrox Imaging	Matrox Imaging Library (MIL) introduced.	A long-lived example of the software-library model: one API and toolset abstracting changing acquisition hardware [61].

Date	Company	Milestone	Why it mattered / source status
1995	Vision Components	VC11 presented.	Vision Components describes the DSP-based VC11 as the first industrial smart camera suitable for series use [68].
1996	MVTec	MVTec founded as a spin-off of the Technical University of Munich and FORWISS.	HALCON became a major hardware-independent industrial machine-vision software environment, reinforcing software as a portable layer above cameras and frame grabbers [69].
1998	OMRON	F30 compact machine-vision system introduced.	Contemporary trade coverage describes the F30 as integrating camera, lighting and processor in a compact housing for simple inspection. OMRON also marketed the F10 pattern-matching sensor in this period, but the F10 used a sensor head plus amplifier/processing unit; calling both products single-body smart cameras would therefore be misleading [70], [71].
2000	Cognex	In-Sight family introduced; In-Sight 2000 announced in March 2000.	Cognex entered the low-cost factory-floor vision-sensor market with a spreadsheet-configured system that did not require a PC for operation. Cognex annual reports confirm the In-Sight family was introduced in 2000 [72], [73]. This is a major Cognex smart/embedded-vision milestone, but not an uncontested claim to the first smart camera.

Date	Company	Milestone	Why it mattered / source status
2002	Cognex	VisionPro software development environment introduced.	Cognex's 2002 Form 10-K states that VisionPro was introduced in early 2002 as an ActiveX-based environment exposing Cognex core vision tools in a PC/Microsoft environment [74]. The often-repeated 1998 date for a CVL-to-VisionPro transition was not supported by the primary material reviewed, so it is not used here.
c. 2004	KUKA / Cognex	KUKA.Vision for the KRC2 generation provided an integrated route for using Cognex In-Sight smart cameras with KUKA robots.	The exact first-release year has not been independently established in the sources reviewed. Cognex documentation confirms KUKA KR C2 and KR C2 Edition 2005 support and a KUKA-specific In-Sight communication mode [75]. Historically, the integration illustrates a partnership strategy in which a robot manufacturer adopted a specialist vision supplier rather than developing the complete vision stack internally.
2005	Cognex / DVT	Cognex completed its acquisition of DVT Corporation on 9 May 2005.	DVT was a significant supplier of low-cost vision sensors and had an established worldwide distributor network.

Date	Company	Milestone	Why it mattered / source status
2005	Matrox Imaging	Iris P-Series programmable smart cameras expanded.	Contemporary coverage records the Iris P300H in July 2005 and remote-head P-Series models in September. The cameras combined image sensing, embedded PC-class processing, Windows CE .NET and the Matrox Imaging Library API [76], [77].
2006	FANUC Robotics	iRVision introduced as FANUC's built-in robot-vision package for the R-J3iC controller family.	A contemporary FANUC announcement described iRVision as the company's first built-in vision package for the R-J3iC family, requiring only a camera and cable rather than separate processing hardware [78]. It represents the vertically integrated route: robot manufacturer, controller and vision environment supplied as one platform.
2007	KEYENCE	LJ Series 2D laser profiler launched.	Representative of laser profiling moving into accessible, in-line non-contact measurement [59].
2007-2008	KEYENCE	CV-5000 generation of controller-based vision systems entered service.	The CV-5000/CV-3000 families are controller-based vision systems rather than self-contained smart cameras. VS Series will later offer smart-camera formats [79].
2011	Teledyne / DALSA	Teledyne completed its acquisition of DALSA.	A marker of consolidation around portfolios spanning sensors, cameras, acquisition and imaging components [80].

Date	Company	Milestone	Why it mattered / source status
c. 2012-2013	KUKA / Cognex	KUKA.VisionTech brought Cognex VisionPro-based 2D vision functionality into the KUKA robot environment for localisation, inspection and code/OCR tasks.	Cognex identifies KUKA.VisionTech as integrating VisionPro software with KUKA robots [81]. A KUKA VisionTech 2.1 manual is dated October 2013 [82]; the initial release date is therefore shown approximately. It illustrates deep integration of a specialist third-party machine-vision stack with a robot manufacturer's platform.
2013	LMI Technologies	Gocator 3110 launched.	LMI described it as the industry's first all-in-one 3D smart snapshot sensor, combining acquisition, measurement and control decisions [55].
2016	Basler	First Basler 3D camera, using time-of-flight technology.	Shows industrial-camera suppliers expanding beyond conventional 2D acquisition into depth sensing [65].
2017	Cognex / ViDi Systems	Cognex acquired ViDi Systems SA on 4 April 2017.	The acquisition brought deep-learning software developed specifically for industrial machine vision into Cognex's Vision Products business and was an important step in adding learned inspection to its commercial portfolio [83].
2018	OMRON	FHV7 smart camera announced.	OMRON highlighted an integrated multi-colour light and high-resolution sensor as a way to make one camera adapt to high-mix inspection [84].

Date	Company	Milestone	Why it mattered / source status
2020	Cognex	In-Sight D900 deep-learning smart camera launched.	Cognex claimed the first industrial smart camera with deep learning embedded in the device [56].
2020	OMRON	FH Series gained AI defect-detection capability.	OMRON described the release as an industry-first defect-detection AI for its vision platform; the “first” is a vendor claim [57].
2022	Zebra / Matrox Imaging	Zebra completed the acquisition of Matrox Imaging.	The transaction brought Matrox’s frame grabbers, machine-vision software and systems into Zebra’s industrial automation portfolio [85].

Who invented the smart camera? It depends what counts

DVT literature dates its SmartImage Sensor to 1991 and describes it as the first smart camera [67]. Vision Components dates the VC11 to 1995 and describes it as the first industrial, series-ready smart camera [68]. A3 itself notes that the border between smart cameras and vision sensors has long been grey [50]. The apparent contradiction is historically useful: “first” depends on whether the defining feature is onboard processing, stand-alone operation, programmability, industrial I/O, or readiness for series deployment. In this chapter, contested firsts are therefore attributed rather than presented as universal facts.

Seen together, these product histories show several different routes by which machine vision matured. NI and MVTec helped make vision programmable and reusable through systems, libraries and development environments. KEYENCE and OMRON pushed highly integrated sensors and smart cameras towards users who might not be vision specialists. Cognex and Matrox (now part of Zebra) have pursued both product strategies, PC based libraries and integrated-edge devices, with increased focus on smart cameras in recent years. DALSA and Basler illustrate the camera and sensor lineage, while LMI and Micro-Epsilon show how optical metrology and 3D profiling became part of the same industrial ecosystem. Robot manufacturers also brought vision closer to the automation platform: FANUC introduced its controller-integrated iRVision in 2006 [78], while KUKA’s collaboration with Cognex is visible in KRC2-era In-Sight integration and later in KUKA.VisionTech around 2012-2013, which integrated Cognex ecosystem into the robot environment [75], [81]. These different approaches - proprietary integrated vision, specialist third-party software and increasingly autonomous smart cameras - illustrate how the boundaries between vision, robotics and automation have steadily blurred as suppliers have added software, AI, 3D and embedded processing.

i Robot vision predates these product milestones

Robot-integrated vision predates the commercial products listed here. The Adept, KUKA and FANUC entries should therefore be read as representative approaches to packaging vision within robot automation platforms, not as claims for the first vision-guided industrial robot.

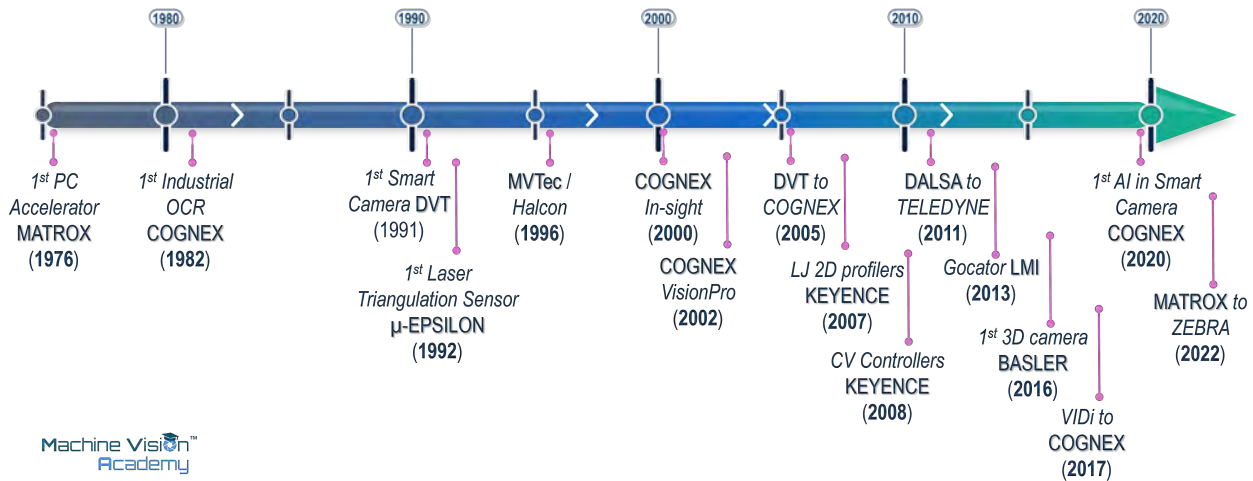


Figure 2.8.: From component to appliance: representative industrial machine-vision product milestones

2.2.4. The everyday consumer timeline: when vision escaped the laboratory and factory

Some of the best historical anchors to get a true sense of the timescale are ordinary everyday consumer devices or services. On 26 June 1974, the first installation of UPC supermarket scanners entered service at a Marsh supermarket in Troy, Ohio. A packet of Wrigley’s chewing gum became the first purchase made with the new scanner system [86]. The system did not “understand” groceries; it read a deliberately designed visual code with exceptional reliability. That is an industrial lesson in miniature: when the environment and target can be engineered, a narrow visual task can become commercially transformative.

The same inversion appears in the history of the QR Code. DENSO WAVE introduced it in 1994 for manufacturing and logistics, where conventional barcodes were becoming too limited for the amount and speed of information required. Position-detection patterns at three corners made the symbol easy to locate and orient. DENSO WAVE then promoted open use of the standardised code, helping a factory identification technology migrate into tickets, payments and - eventually - restaurant menus [87].

i From a Toyota-group factory problem to the cafe table

The QR Code is a useful reminder that consumer adoption can come long after industrial invention. DENSO WAVE developed the code in 1994 for production and logistics; camera-equipped phones helped it spread in the 2000s, and later mass use made the square pattern one of the most recognisable pieces of machine-readable imagery in everyday life. A code engineered to survive the grime of a car-parts plant turned out to be equally at home surviving a laminated menu [87].

A less consequential but more charming example came from Cambridge. In late 1991, researchers in the University of Cambridge Computer Laboratory pointed a camera at the Trojan Room coffee pot so colleagues elsewhere in the building could avoid a wasted trip when it was empty. The local network application was connected to the World Wide Web in 1993 and became the first webcam .. and the first reality TV shows [88]. The historical significance is disproportionate to the problem: remote visual sensing had become a service rather than a laboratory experiment.

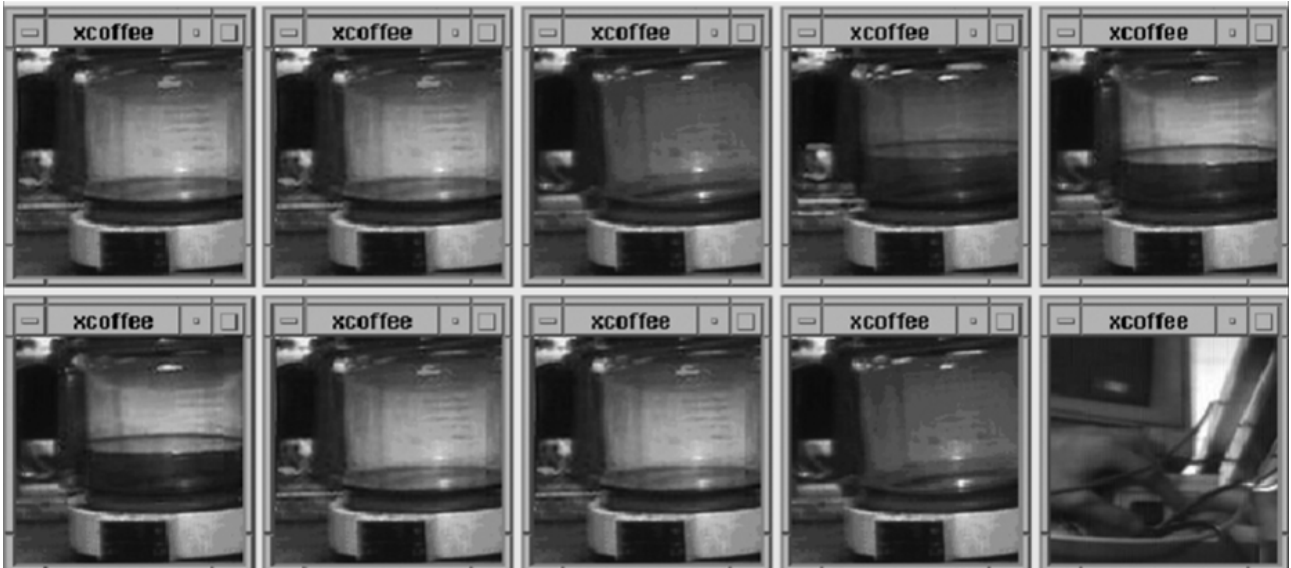


Figure 2.9.: The Trojan Room coffee pot: subject of the first webcam (1991), connected to the World Wide Web in 1993

i The first webcam monitored caffeine, not people

The Trojan Room system is a good antidote to technological grand narratives. The first webcam was not designed for telepresence, social media or surveillance; it solved what may be the most relatable problem in the history of computer vision: sparing researchers the walk across a building only to find the pot empty. It was switched off in August 2001, after a decade of faithful service [88].

By the 2000s, visual computation was increasingly disappearing inside everyday products. Fast face detectors allowed compact cameras to draw focus boxes around faces, while embedded image processing began correcting photographs before the user even saw them.

In 2007 real-time flash-eye detection was introduced in smart phones' firmware by Photonation [89]. For many readers, particularly those born in the new millennium, red-eye may be an unfamiliar photographic artefact precisely because its detection and correction have become so commonplace. For earlier generations, however, the unintended vampire look was a familiar feature of the family photo album. Developments such as these marked a subtle but important transition: the camera was becoming more than a sensor; it was beginning to interpret and modify its own images.

Kinect brought a different kind of machine vision into the living room. Microsoft launched Kinect for Xbox 360 on 4 November 2010, allowing users to control games through body movement and voice [90]. Guinness World Records records eight million units sold in its first 60 days, making it the fastest-selling gaming peripheral at the time [91]. For millions of people, depth sensing and body tracking arrived not as a robotics demonstration or a metrology system, but as the instruction to stand in front of a television and move.

By the middle of the next decade, semantic vision had become an almost invisible service. Google Photos launched in 2015 with automatic organisation by people, places and things, without requiring users to tag every image manually [92]. Two years later Apple's iPhone X made 3D face recognition an everyday authentication gesture: Face ID combined a dot projector, infrared camera and flood illuminator, projecting more than 30,000 invisible infrared dots and using neural networks to build a mathematical representation of the face [93]. These features are easy to take for granted precisely because the vision system is no longer presented as a separate application; it is simply part of using the product.

2. Vision engineering: scope, definitions and history

In medicine, the same broad progression towards automated visual decision-making reached a much more consequential threshold in 2018, when IDx-DR was authorised in the United States as an autonomous AI-based diagnostic system for diabetic retinopathy. Its pivotal study compared autonomous analysis of retinal photographs with a specialist reference standard [94]. The mechanisms behind a supermarket scanner, automatic red-eye correction, body tracking, face authentication and retinal diagnosis are very different, but the systems pattern is recognisable: capture an image, extract useful information, make a decision and determine what happens next.

2.2.5. What the history reveals

Viewed as parallel timelines, the history is less a sequence of replacements than a sequence of layers. Digital image processing did not disappear when computer vision arrived; classical geometry did not disappear when statistical learning became fashionable; PC-based architectures did not disappear when smart cameras became practical. Industrial systems continue to combine techniques from different decades because the correct choice is governed by the task, not by historical novelty.

A second pattern is migration. The same enabling technology repeatedly crosses domains: space programmes helped advance digital image enhancement and CMOS sensing; factory identification produced the QR Code that later became a consumer interface; research in learned representations eventually became a tool for production defect inspection. The path rarely runs in one direction from university to factory to consumer. Vision engineering is an ecosystem in which ideas, components and constraints circulate.

The third pattern is that industrial progress is often less dramatic than research progress and more dependent on integration. A better detector matters, but so do a stable light, a faster interface, a camera that can be replaced without rewriting software, a 3D sensor that performs its own measurement, and a validation method that explains what “good enough” means. This is why the history of machine vision cannot be told only through algorithms. It is equally the history of sensors, optics, computing architectures, software tools, standards, companies and the production problems that forced them to work together.

! The recurring pattern behind the milestones

Vision technology becomes transformative when three things coincide: a useful visual method, hardware inexpensive and reliable enough to deploy, and an engineering environment that makes the method repeatable. Research changes what can be imagined; product engineering changes what can be installed; standards and validation change what can be scaled and trusted.

2.3. Industrial application families

The previous chapter established what machine vision is and how the discipline developed. In this chapter we address what machine vision is for. On a factory floor, a technology is valuable because it solves an industrial task reliably, at production speed and, in some sectors, within a regulated quality system. This section is therefore organised around tasks rather than technologies.

A large proportion of industrial vision applications can be grouped into four families: ***measurement***, ***defect inspection***, ***identification*** and ***guidance***. *Measurement* returns a dimension or other quantitative value against a stated tolerance. *Inspection* determines whether a part is complete, correctly assembled and free of unacceptable defects. *Identification* reads a code or character string and associates the item with a record. *Guidance* locates a feature or object and provides coordinates that an automated system can act upon. Many deployed systems combine two or more of these functions in sequence.

The four application families considered here deliberately reflect the industrial focus of this handbook; they do not represent the full scope of machine and computer vision. Important neighbouring domains include *biometrics*, such as face, fingerprint and iris recognition; *medical imaging and computer-aided diagnosis*; *advanced driver-assistance and autonomous systems*, where visual perception supports functions such as lane keeping, object and pedestrian detection and navigation; and *remote sensing and autonomous robotics*, where visual information is used to interpret and act within less controlled environments. These fields are outside the principal scope of this handbook, but their development remains relevant to industrial vision. As the historical development of the discipline has already shown, technological progress is rarely linear or confined to one application domain: advances in sensors, learned representations, three-dimensional perception, real-time processing and validation frequently migrate between research, consumer, medical, automotive and industrial applications. Technologies being developed in these neighbouring fields can therefore be expected to influence the future capabilities and architectures of industrial automation.

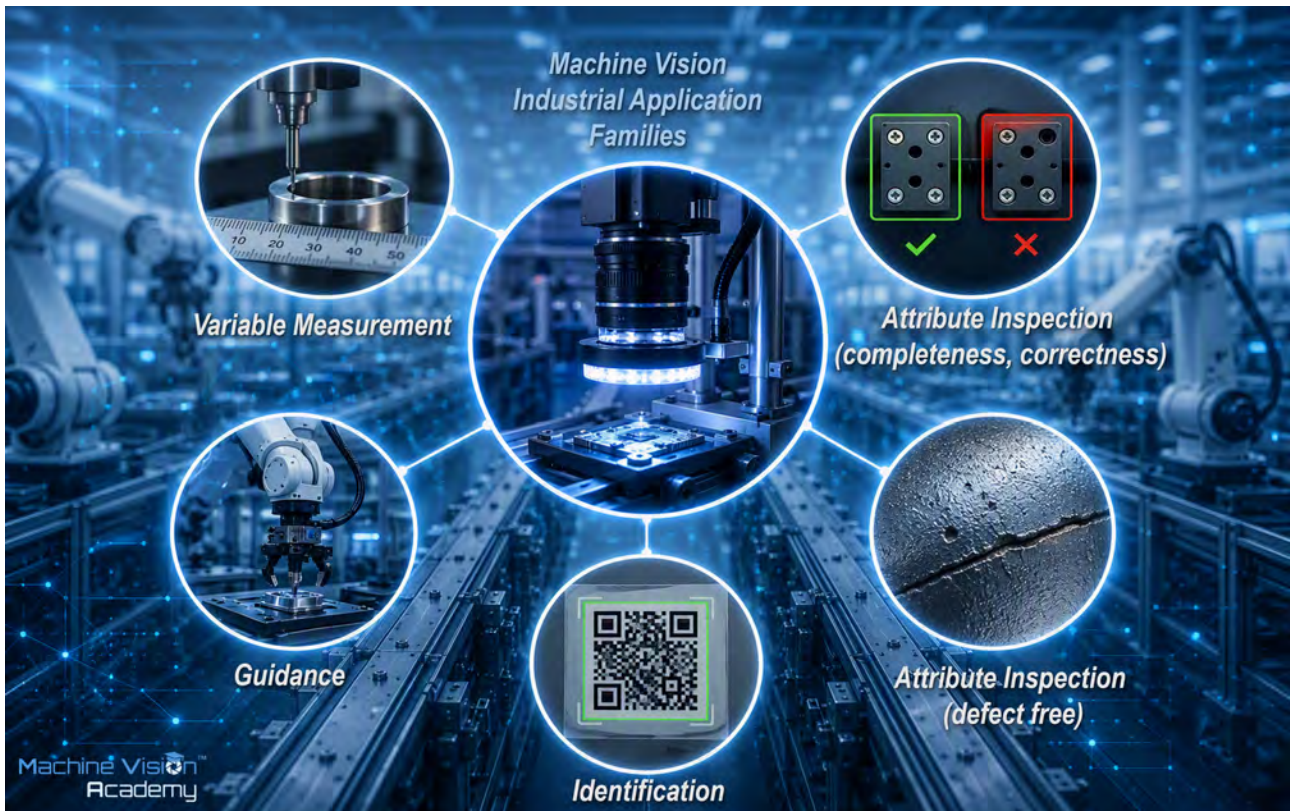


Figure 2.10.: Machine vision industrial application families

The motivation for automating visual inspection in the factory floor is longstanding. Manual visual inspection is repetitive, can be affected by fatigue and variation between inspectors, is difficult to document consistently, and may not keep pace with high-throughput production. Automated visual inspection can apply a defined decision process consistently under controlled conditions and can retain a record of its measurements and decisions [95].

An alternative but complementary classification, borrowed from quality engineering, is useful because it often points towards the appropriate measurement and decision strategy: **inspection by variables** and **inspection by attributes**. ISO 3951 describes sampling procedures for inspection by variables, whereas ISO 2859 addresses inspection by attributes [96], [97].

A *variable inspection* produces a quantitative result on a scale—for example, a diameter, gap or angle—that can be compared with a specification limit and verified against a calibrated reference. The measurement tasks in Figure 2.10 commonly have this form. They are a natural fit for deterministic, rule-based vision: an edge is located, a distance is calculated, a calibration is applied and the result is compared with a tolerance.

2. Vision engineering: scope, definitions and history

An *attribute inspection* produces a categorical result such as present/absent, conforming/non-conforming or scratched/clean. The underlying decision can still be derived from quantitative measurements; what distinguishes the attribute case is the categorical result used for acceptance. Some attribute checks are readily implemented with hand-crafted features and explicit thresholds. Others involve appearance criteria that are difficult to express analytically and may benefit from labelled examples and learned models. The choice between rule-based and learned methods therefore depends on the nature of the feature, the available data and the required evidence, rather than on the attribute/variable label alone.

Is it an attribute? Is it a variable? Is it both

Where practical, a variable output is preferable because it preserves information beyond pass or fail. A measured value allows acceptance thresholds to be revised and, more importantly, supports statistical analysis and process monitoring: a population can drift towards a limit long before the reject rate changes.

The difficulty is establishing a meaningful scale. A variable measurement should ideally be calibratable against a recognised reference, such as millimetres or degrees; derived image features such as contrast or texture scores may be much harder to calibrate meaningfully. A useful variable acceptance criterion also requires process characterisation to establish how values on that scale relate to product performance. This is good engineering practice, but it requires experimental evidence and is not always practical.

In reality, the machine-vision engineer often inherits an inspection already defined as an attribute judgement, even when a variable formulation might have been possible.

2.3.1. Measurement and non-contact gauging

Measurement is the family that produces a quantitative value—a length, diameter, angle, gap or height—and a principal advantage of vision-based gauging is that it can be non-contact. A traditional coordinate-measuring machine obtains dimensions by probing a part physically; an imaging system obtains information optically. This can be advantageous for delicate, sterile or soft parts and can permit in-line measurement of every item rather than off-line sampling. The resulting measurements can also feed statistical process control, turning inspection into a continuous source of process information.

Accurate gauging depends strongly on image formation. Telecentric optics can reduce magnification changes caused by object-distance variation; back-lighting can create a high-contrast silhouette; and sub-pixel edge localisation can estimate edge position more finely than the pixel sampling interval. These techniques are among the considerations that distinguish a traceable measurement system from one that merely displays a precise-looking number [98].

A measured number is not the same as a true one

A vision gauge initially derives a value from image coordinates; calibration against a traceable reference establishes the relationship to physical units, while measurement-uncertainty analysis determines how confidently the result can be used. A non-contact optical measurement and a contact CMM measurement of the same nominal feature can differ because the two systems interact with and sample the part differently. ISO 10360-7 specifies acceptance and reverification tests for Cartesian CMMs equipped with imaging probing systems operating in discrete-point probing mode [99]. Repeatability, reproducibility, bias and uncertainty are therefore as important as the nominal image resolution and would be covered in later chapters.

The measurement toolkit is predominantly deterministic: caliper and edge-pair gauges, line and circle fitting, blob geometry and sub-pixel contour analysis in two dimensions; and laser triangulation, profilometry, structured light and other optical ranging methods for height and shape. Learned

methods can assist some localisation or interpretation tasks, but precise dimensional gauging still depends on calibration, traceability and a defined measurement model; industry guidance likewise distinguishes this from appearance-based AI inspection [100]. This is the clearest application of the inspection-by-variables principle [96].

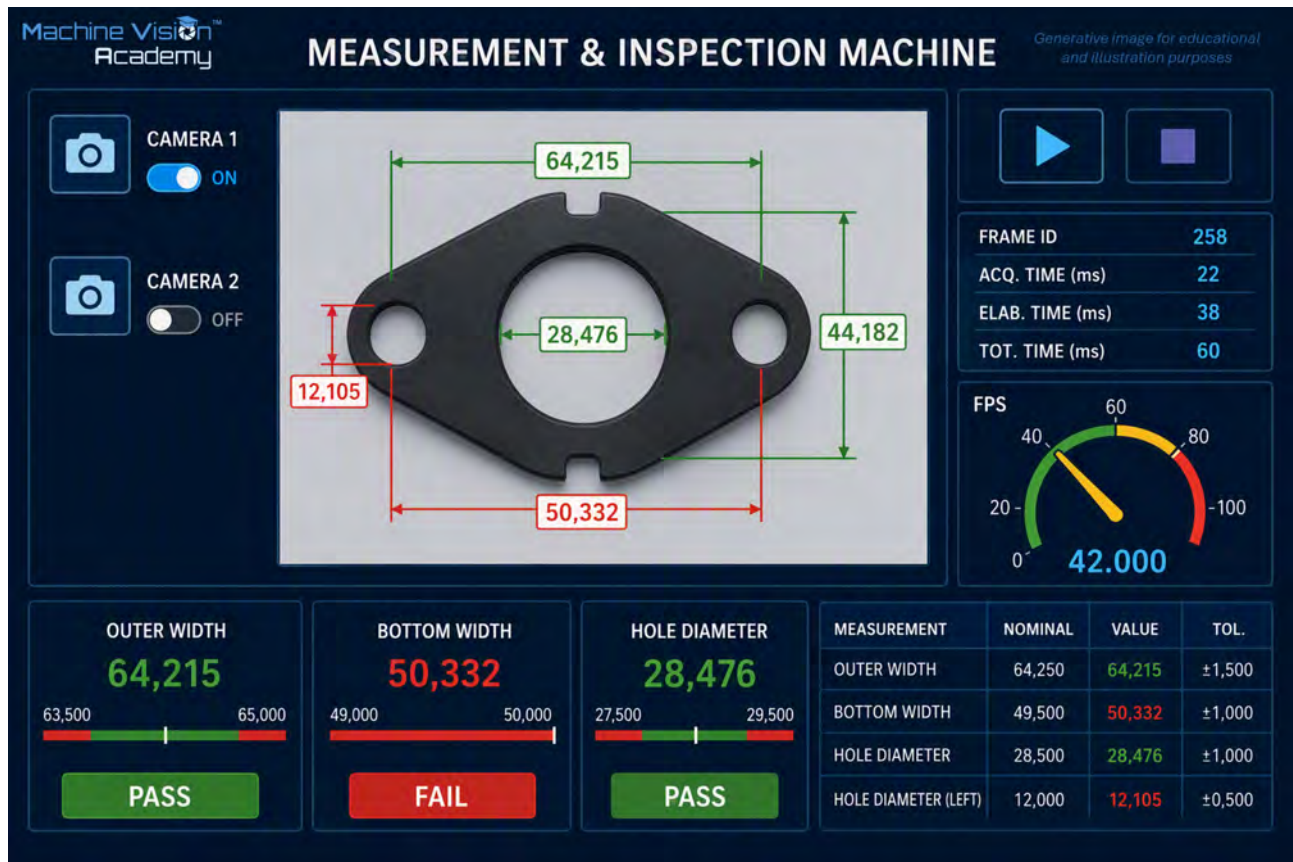


Figure 2.11.: Application example: Non-contact gauging for dimensional inspection

2.3.2. Defect inspection

Defect inspection is the family most commonly associated with machine vision on a factory floor. Typical tasks include *completeness* (e.g. presence of components), *correctness* (e.g. position and orientation, material or variant verification) and *cosmetic defect inspection* (e.g. surface condition and texture, colour, presence of foreign material). What ties these tasks together is a decision on which the process must act within the available cycle time, for example by rejecting the part or intervening to correct the fault.

Most tasks in this family are attribute checks, although an attribute inspection can contain variable sub-steps. A colour check, for example, may calculate a histogram or colour-distance value on a continuous scale and then convert that value into a categorical decision. The family describes the decision delivered by the inspection, not every mathematical operation used internally.

The traditional approach to implementing these inspections is to translate the acceptance criterion into explicitly engineered image features and decision logic. Typical rule-based tools include thresholding and connected-component analysis, template and geometric pattern matching, edge and contour analysis, and classical texture or colour measures. These methods are particularly effective when the characteristics that distinguish acceptable from defective parts can be expressed consistently through a manageable set of measurable image features. They are deterministic, generally require relatively few examples during development, and make the relationship between the observed feature and the

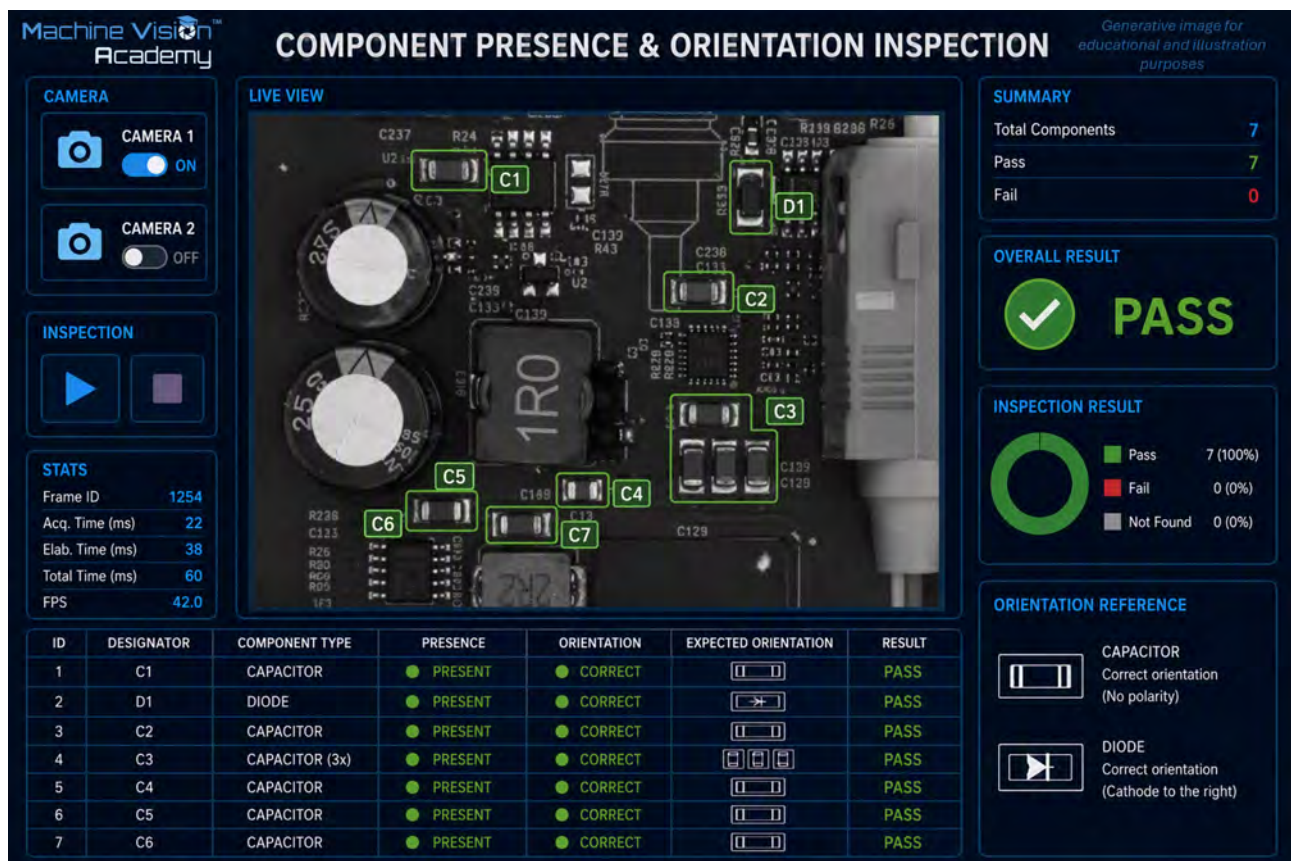


Figure 2.12.: Application example: Component presence and orientation inspection

resulting decision explicit. Their limitations become apparent when acceptable product appearance and defect variation are too complex to describe reliably through hand-crafted features and thresholds.

Deep Learning approaches provide an alternative when that visual variation is difficult to describe explicitly. Depending on the inspection task, these include image classification, object detection, semantic or instance segmentation and anomaly detection. Their flexibility, however, shifts a significant part of the engineering problem from feature design to data: representative examples must be collected, labelled and shown to cover the relevant production variation. This can be particularly difficult in manufacturing because genuine defects are often rare, resulting in small and highly imbalanced datasets. Published work on sterile-barrier seal inspection illustrates this problem and the resulting need to evaluate model behaviour carefully under such conditions [101], [102]. Scarcity of defective samples and class imbalance increase the risk of overfitting and make validation particularly demanding, especially in regulated industries. Rule-based and learned approaches are developed in detail later in the handbook; here they should be regarded as complementary strategies whose suitability depends on the inspection requirement and the available evidence.

2.3.3. Identification of unique parts: codes, characters and traceability

Identification answers the question: *which item is this, and can its history be traced?* It reads machine-readable and human-readable marks—one- and two-dimensional codes, printed text and direct-part marks—and associates each item with a record. In regulated industries this capability supports traceability, authentication and recall processes. Medical-device Unique Device Identification and pharmaceutical serialisation are examples of regulatory contexts in which machine-readable identifiers and verification are important [103], [104]. Industry accounts also emphasise the value of integrating code reading and logging directly into automated production [105].



Figure 2.13.: Application example: Surface inspection

Data Matrix, standardised by ISO/IEC 16022, is widely used where a compact two-dimensional identifier is required. Direct-part marking by laser, dot-peen, inkjet or other methods can provide a persistent identifier without a separate label. Reading low-contrast marks on curved, reflective, damaged or contaminated surfaces can require full machine-vision techniques rather than a simple idealised barcode-reading scenario [104].

A related pair of tasks is optical character recognition (OCR) and optical character verification (OCV). OCR determines text whose content is not known in advance, whereas OCV checks that an expected string has been produced correctly and legibly [106]. A traceability station can combine these functions: verify printed text, decode a machine-readable identifier, compare the two and reject a mismatch.

The practical difficulty of identification is therefore robustness and speed. Industrial marks may suffer from low contrast, glare, curvature, print variation or direct-marking distortion. Code quality can itself be graded so that the process controls not merely whether a symbol is readable at the current station, but whether its quality is sufficient for downstream use.

2.3.4. Guidance and robot vision

A conventionally taught robot repeats positions defined relative to its coordinate system. If the workpiece is not presented at the expected pose, the operation can fail. Vision guidance reduces that dependence on precise fixturing by locating a feature or object in two or three dimensions and providing pose information that the robot or motion system can use. The industrial value is flexibility: components can tolerate greater variation in presentation, which can reduce dedicated fixturing and support higher product mix.

Two-dimensional guidance estimates position and in-plane rotation when the relevant geometry can be represented on a plane, for example locating a component on a conveyor or determining an offset for



Figure 2.14.: Application example: Product identification and OCR inspection

placement. Three-dimensional guidance adds depth and full spatial pose. A representative application is bin picking, in which the system identifies candidate parts in a container, estimates their poses and supports selection of a feasible pick [107].

The difficult engineering problems extend beyond recognition. Reflective, dark or transparent surfaces can challenge depth acquisition; parts may overlap and occlude one another; the complete perception-and-motion sequence must meet cycle time; and the camera coordinate system must be related accurately to the robot coordinate system. Hand-eye calibration is therefore fundamental. Relevant methods include two-dimensional pattern location and pose estimation, three-dimensional acquisition using structured light, time-of-flight or stereo, point-cloud registration, and learned grasp-point selection where object variation makes explicit modelling difficult.

2.4. System architectures

Having looked at what vision systems are asked to do, the next question is where the image-to-decision process should live. This is partly a development question - which tools, skills and computing resources are needed to engineer the solution - but it is primarily a deployment question. Production architecture must satisfy cycle time, physical space, integration, reliability, cybersecurity and long-term maintainability while supporting the availability, performance and quality targets that ultimately contribute to OEE.

Whatever the hardware arrangement, the logical chain remains familiar: **part presentation -> lighting -> optics -> sensor/image acquisition -> image processing -> decision -> process action**, as illustrated in Figure 2.1 and discussed in later chapters. Architectures differ less in what the vision system is trying to achieve than in where processing occurs, how the processing is packaged, what other systems it depends upon and how widely failures can propagate.

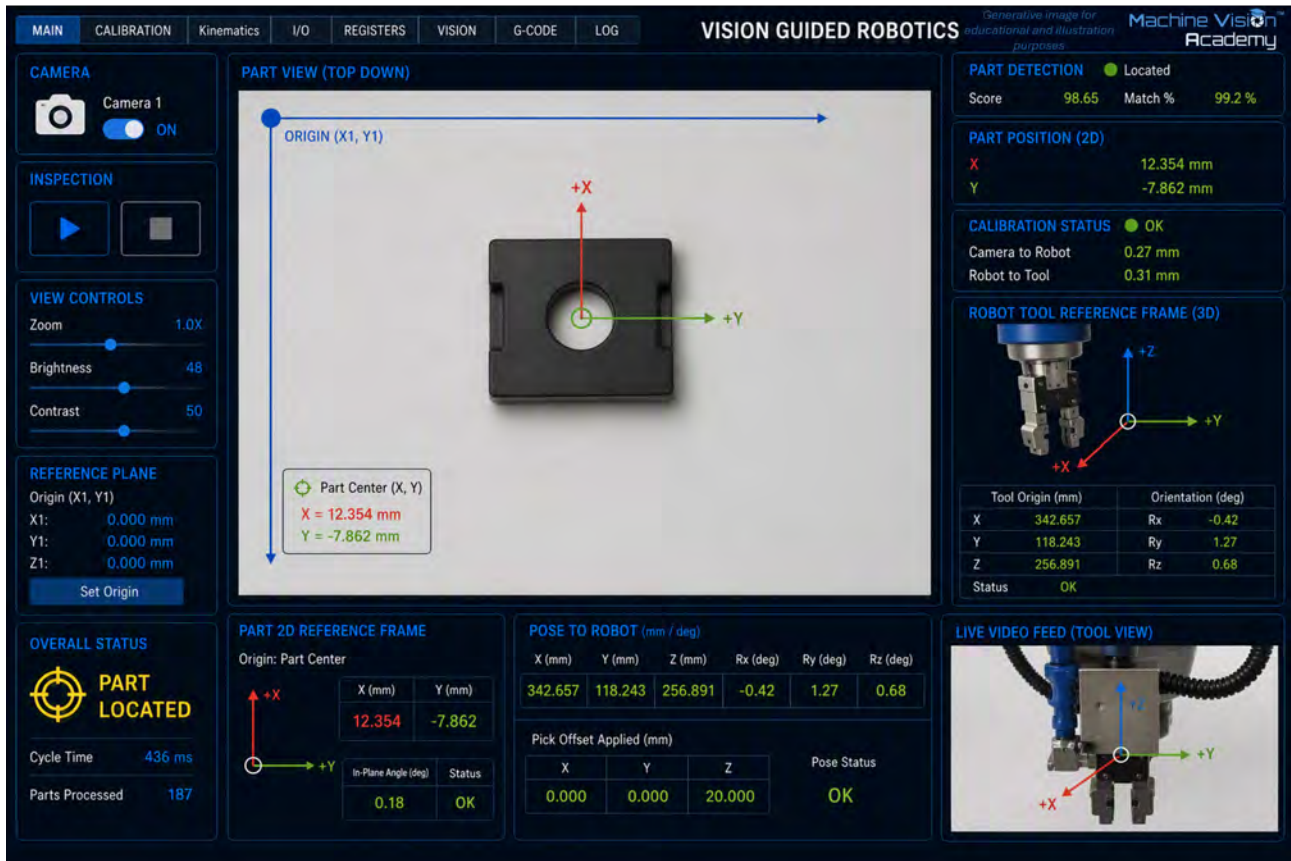


Figure 2.15.: Application example: Robotic guided part picking

A useful way to visualise deployment is as a spectrum extending outwards from the camera. **Device Edge Computing** keeps acquisition and processing inside the imaging device. **Machine Edge Computing** separates the camera from a controller, industrial PC or specialised AI computer located at or beside the machine. **On-Premises Vision Server** centralises processing within the organisation, usually so that compute, storage or software services can be shared. **Cloud Inference** moves the production inference itself to remotely hosted infrastructure. GPUs, NPUs and other accelerators are not separate positions on this spectrum; they are computational choices that can appear at several of these levels.

Hybrid architectures cut across this spectrum in two ways. Production functions can be divided between locations - for example, local inference with central storage and monitoring - and development can be separated from deployment, such as training a model in the cloud and deploying it later to a smart camera or industrial PC. The sections that follow therefore work from the camera outwards, before comparing the principal trade-offs and offering practical selection guidance.

2.4.1. Device Edge Computing

Device Edge Computing places image acquisition, processing and decision-making within the imaging device itself. In machine vision this is represented mainly by the **smart camera, smart vision sensor or smart 3D sensor**: a compact industrial device containing the sensor, processor, memory, vision runtime and communications interfaces, often with digital I/O capable of interacting directly with the machine. The inspection loop can therefore be remarkably short: trigger, acquire, process, decide and communicate - or even switch an output - without an external vision computer in the production path [108], [109].

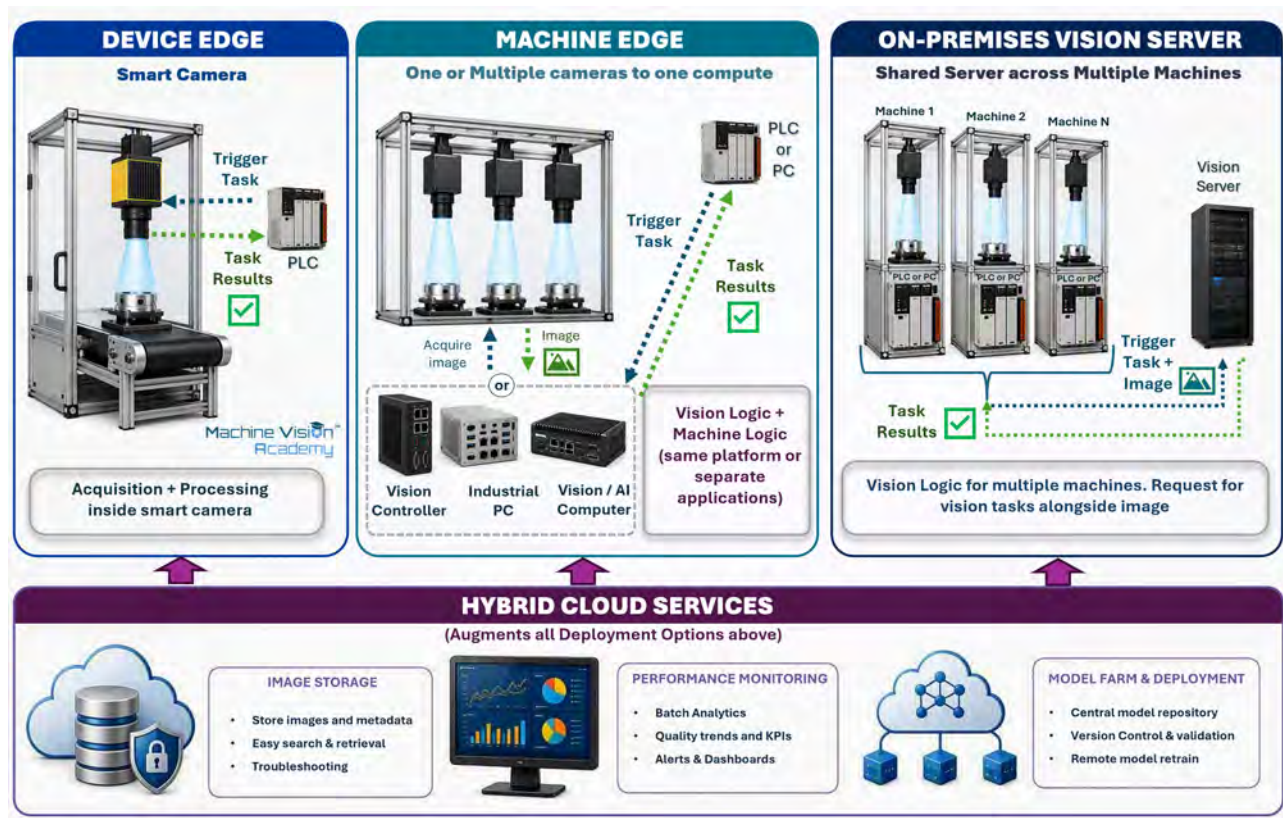


Figure 2.16.: The spectrum of vision architectures: where the processing lives

i Representative Device Edge vision products and families

A non-exhaustive selection of **2D smart-camera families** includes **Cognex** In-Sight, **KEYENCE** VS Series, **Zebra** VS and Iris GTX, **OMRON** FHV7 and MicroHAWK F-Series, **SICK** InspectorP6xx, **Teledyne DALSA** BOA/BOA Pro, **Balluff** BVS SC, **Banner Engineering** VE Series and **wenglor** B60 uniVision [79], [110], [111], [112], [113], [114]. Device Edge also includes specialised **3D smart sensors**, such as **LMI Technologies** Gocator laser profilers (laser triangulation) and snapshot sensors (structured-light 3D) [115], and **Micro-Epsilon** scanCONTROL SMART laser-line profilers [116], where acquisition and at least part of the measurement or inspection processing execute within the sensor

Edge computing is not new to industry. Machine vision applications have used processing close to the process for decades. What has changed is how much capability can now be packaged into the device. An example that illustrates this evolution is the DataMan smart sensors, the first product released by Cognex in 1982, soon after the company was founded. This was the predecessor of the In-sight smart camera and it had one only function, OCR. Contemporary smart cameras combine traditional operations - thresholding, blob analysis, pattern matching, metrology and identification - with increasingly capable deep learning tools.

The advantages follow directly from this integration. With no external computer in the normal inference path, communications latency and network dependencies are reduced, timing can be coupled closely to the machine, and production can continue independently of remote IT or Internet services. Installation is simple and compact: for a suitable application the vision system may require little more than power, triggering, machine communications and the optical arrangement. A3 similarly characterises smart cameras as self-contained systems particularly well suited to distributed, independent inspection points [109].

Smart cameras are also frequently designed around **configuration or no-code/low-code environ-**

ments rather than conventional software programming. This can shorten development for established inspection tasks and makes useful vision functions accessible to a range of engineers who are not software specialists. No-code does not necessarily mean that programming is unavailable: many platforms expose expressions or limited scripting, while products such as Zebra Iris GTX, SICK InspectorP6xx and wenglor B60 provide more extensive customisation routes [111], [113].

The smart-camera paradox

Making a vision application easy to configure does not make machine vision itself easy. No-code environments can produce convincing results very quickly, but they can also hide the assumptions behind optics, image formation, algorithms and validation. From a practitioner perspective, this accessibility is valuable when it reduces implementation effort for a well-understood task; it becomes risky when ease of configuration is mistaken for a substitute for vision knowledge. A robust architecture should be selected because it fits the application, not simply because it is the only environment the developer already knows.

The limitations are largely the consequence of the same integration. Processing power, memory, storage, thermal dissipation, sensor choice and expansion capability are bounded by the camera platform. Scaling often means adding or replacing devices rather than expanding a shared computational resource, and tightly coordinated multi-camera applications can become expensive or awkward. Hardware and software are also commonly tied to one vendor ecosystem. **There are notable exceptions:** freely programmable platforms such as the Baumer AX series and Vision Components VC Z series retain the Device Edge location while exposing Linux-based development and third-party libraries. Baumer supplies an Ubuntu/Jetson software stack and SDKs, while Vision Components supports embedded Linux and HALCON deployment [117], [118]. These products trade some appliance simplicity for much greater algorithmic freedom.

Device Edge is therefore most attractive when the complete inspection can be contained comfortably within one or a few largely independent devices. The familiar description of a smart camera as a complete vision system in a compact package remains useful; the engineering judgement is to make sure that the application - including plausible future growth - remains inside the performance and software envelope of that package.

Strengths and best fit

- **Short and predictable image-to-decision path** - acquisition, processing and decision remain within the device; particularly valuable where the interval between image capture and machine action is very short.
- **High production autonomy** - little or no dependence on external PCs, servers or IT infrastructure during normal operation.
- **Simple and compact architecture** - fewer components, interfaces and software dependencies to install, diagnose and maintain.
- **Rapid development for established tasks** - integrated no-code/low-code environments can substantially reduce implementation effort.
- **Good failure isolation** - a device failure normally affects one inspection point rather than a whole group of machines
- **Straightforward lifecycle support** - hardware, runtime and configuration can often be managed and replaced as one vendor-supported device

Best fit: Self-contained applications using one or a small number of **independent cameras**, where the processing fits comfortably within the device and priorities include low latency, local autonomy, compact installation and straightforward maintenance. Typical examples include presence/completeness checks, code reading, straightforward gauging, localised defect inspection and simple robot guidance.

Watchouts

- **Finite computational envelope** - processing performance, memory, storage and thermal capacity are bounded by the selected device.
- **Limited scalability** - increasing processing capacity commonly means adding or replacing devices; this matters where an initially simple application is expected to grow substantially.
- **Multi-camera coordination can become cumbersome** - especially when several views must be synchronised or processed jointly
- **Vendor dependence is usually higher** - conventional smart cameras couple sensor, processor, runtime and development environment, although open programmable platforms provide exceptions
- **No-code can constrain complex applications** - bespoke algorithms, complex sequencing and data handling can become difficult to implement and maintain.
- **Ease of configuration can conceal engineering difficulty** - no-code reduces implementation effort, not the need to understand lighting, optics, variation and validation.

Rule of thumb:

Choose Device Edge because the application benefits from being **self-contained**, not simply because a smart camera is the easiest architecture to programme. If scope or complexity is likely to grow, consider the future software and performance requirements before committing to a tightly integrated platform.

2.4.2. Machine Edge Computing

Machine Edge Computing separates image acquisition from the principal processing hardware while keeping that processing within, beside or directly associated with the production machine. Cameras acquire the images, but the inspection executes on a separate local platform. Compared with Device

Edge, this introduces additional components and interfaces; in return it provides much greater freedom in computational resources, camera selection, software architecture and machine integration while retaining most of the latency, availability and data-locality benefits of local processing [108].

Three useful variations can be distinguished within this architecture.

The first is the **dedicated vision controller**. Cameras connect to a purpose-built controller supplied as part of an integrated vision ecosystem. Current examples include the **KEYENCE CV-X**, **OMRON FH**, **Cognex In-Sight 6900** and **Zebra 4Sight** controller families [119], [120], [121], [122]. These products retain much of the appliance-like character of a smart camera - coordinated cameras, I/O, industrial communications and graphical vision software - while supporting more cameras, greater processing capacity or broader acquisition options. Dedicated controllers are often a useful middle ground between an all-in-one camera and a fully bespoke IPC.

The second variation is the **PC-based vision system**, in which one or more industrial cameras connect to a general-purpose industrial PC or workstation. This is usually the most flexible Machine Edge form because the camera technology, acquisition interface, processor, GPU and software environment can be selected relatively independently.

Representative PC-based Machine Edge vision software

A non-exhaustive selection of commercial PC-based machine-vision software includes **Cognex** VisionPro; **Zebra** Aurora Design Assistant and Aurora Imaging Library; **MVTec** HALCON and MERLIC; **NI** LabVIEW with the Vision Development Module; and **Teledyne DALSA** Sherlock. These range from graphical/no-code environments to complete programmable libraries, but all support PC-based Machine Edge deployment [123], [124], [125], [126]. Some environments can also target **Device Edge or embedded hardware**, which can delay part of the deployment decision while the application is being developed. **Zebra** Aurora Design Assistant can deploy to Iris GTX smart cameras; **MVTec** supports HALCON/MERLIC on selected embedded platforms; and **Teledyne DALSA** has offered Sherlock on BOA smart-camera systems [111], [114], [123], [124]. This portability is product-specific: **Cognex** VisionPro applications, for example, are not simply deployed as In-Sight jobs. A less common variation is **NI** LabVIEW with the Vision Development Module, where supported applications can target PCs and real-time controllers and a subset of image-processing functions can execute on FPGA hardware [125]. Open-source alternatives include **OpenCV** for classical image processing and computer vision, and **TensorFlow** and **PyTorch** for learned vision.

The use of deep learning can itself influence architecture selection when model size, memory, acceleration or lifecycle requirements exceed the practical envelope of a smart camera. Once a sufficiently capable Machine Edge platform has been selected, however, classical and learned vision do not normally need separate computing engines. The same IPC can combine calibrated metrology, conventional pattern matching, 3D processing and GPU-accelerated neural-network inference, allowing each part of the inspection to use the method best suited to it.

The third variation uses a **specialised vision or AI computer**. Compact platforms based on NVIDIA Jetson and similar accelerators combine CPU and GPU/NPU computing in hardware designed for edge deployment. Architecturally this remains Machine Edge: a Jetson running TensorRT and an x86 IPC containing a discrete GPU occupy the same deployment location. They differ in processor architecture, power and cooling, expansion, serviceability and software ecosystem rather than in where the production decision is made.

Machine Edge consequently offers a particularly broad range of engineering choices. Processing can range from a modest fanless controller to a high-performance multicore computer with discrete GPUs. Cameras, optics and lighting can usually be selected independently, using standard interfaces such as GigE Vision and USB3 Vision or dedicated acquisition hardware for CoaXPress and Camera Link HS. This flexibility is well suited to multi-camera systems, high-resolution imaging, computationally

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demanding 3D or deep-learning applications and projects requiring close integration with PLCs, robots, databases and other software.

Development flexibility is similarly broad. At one end are graphical and no-code environments; at the other are libraries and application frameworks supporting C++, C#, .NET, Python and other languages. This becomes especially valuable when vision is only one part of a larger machine application. A semi-automated process might ask an operator to select a region on an HMI, acquire an image, review an intermediate result, move a mechanism and then continue the vision sequence. Smart-camera logic can implement some of these interactions, but a PC-based architecture generally provides much greater freedom when vision, operator interaction, machine sequencing and data management must behave as one coordinated application.

The price of that freedom is greater system responsibility. The designer may need to manage operating systems, drivers, camera interfaces, GPU runtimes, application software, cybersecurity and updates, and must keep the complete combination supportable throughout the machine lifetime. More components create more interfaces and potential failure points than an appliance-style smart camera, although they also allow cameras, frame grabbers, computers and accelerators to be replaced or upgraded independently. Development effort can therefore range from a short controller configuration exercise to a substantial software-engineering project.

For demanding manufacturing applications this trade-off is often worthwhile. Machine Edge preserves local production autonomy while providing considerably greater computational, software and integration flexibility than Device Edge. In practice it becomes the natural architecture when vision is no longer an isolated inspection but part of a more complex machine-level application.

Strengths and best fit

- **High computational flexibility** - processing can range from a dedicated controller or fanless IPC to multicore CPUs, GPUs and specialised AI accelerators
- **Deep machine-software integration** - vision can form part of a wider application incorporating machine sequencing, operator interaction, robotics, databases and other functions.
- **Broad hardware choice** - cameras, sensors, acquisition interfaces and processing hardware can usually be selected independently.
- **Excellent multi-camera capability** - several cameras can share processing resources and their acquisitions can be synchronised or analysed jointly.
- **Local production autonomy** - time-critical inference remains at the machine without depending on remote servers or cloud connectivity.
- **Expandable architecture** - memory, storage, processing hardware and accelerators can often be upgraded without replacing the complete imaging system.

Best fit: Applications requiring **multiple coordinated cameras, substantial integration with the wider machine application, or processing beyond the practical envelope of Device Edge**. Typical examples include multi-view assembly inspection, high-resolution inspection, 3D bin picking, advanced metrology, demanding deep-learning inference and semi-automated workflows combining vision with operator interaction.

⚠ Watchouts

- **Greater integration responsibility** - operating systems, drivers, acquisition hardware, GPU runtimes, application software and industrial communications may all become part of the design.
- **Larger software lifecycle** - cybersecurity, updates and dependency compatibility become part of maintaining the production machine.
- **Slower initial deployment** - additional hardware and software elements must be selected, integrated, commissioned and supported.
- **Complexity can grow quickly** - an open software environment enables sophisticated applications but can also produce software that is difficult to validate and maintain if architecture is not controlled.
- **A larger failure domain** - where one computer serves several cameras or inspection functions, loss of that computer may disable the complete machine-vision application.

Rule of thumb:

Choose Machine Edge when the application is already moderately or highly complex, **future requirements are uncertain**, or significant growth is expected. It is particularly appropriate for multi-camera coordination, demanding 3D or deep-learning processing, or close integration of vision with the machine software and operator workflow.

2.4.3. On-Premises Vision Server

An **On-Premises Vision Server** moves the principal processing away from the individual machine while retaining it within the factory or organisational computing infrastructure. Cameras or local acquisition systems transmit images to a server, workstation or compute cluster that provides processing services to one or more machines, cells or lines.

This architecture should not be defined simply by the physical size of the computer. The useful distinction is **centralisation and service scope**. A large GPU workstation dedicated to one inspection machine is still Machine Edge even if it sits outside the machine cabinet. An on-premises server, by contrast, is normally managed as shared infrastructure and provides compute, storage, model serving or other software services to multiple production processes.

A common reason to centralise is simply that the required compute no longer belongs comfortably on the factory floor. Large deep-learning or 3D workloads may need high-power GPUs, substantial memory and storage, and cooling arrangements that are awkward to package beside a machine. In dusty, hot, wet or space-constrained production areas, accommodating such hardware can require a large protected or cooled enclosure. If the process can tolerate a networked inference path, moving this hardware into a clean, controlled server room can be more practical than forcing data-centre-class compute into the machine.

The second attraction is **resource sharing**. Several machines may call different models on the same GPU infrastructure or use multiple instances of a common model. NVIDIA Triton Inference Server is designed for this model-serving pattern, supporting model repositories and concurrent model instances across CPU and GPU resources [127]. Pooling can improve utilisation and lets compute capacity be expanded independently of each production asset rather than sizing every station for its own peak requirement.

Centralisation also makes **lifecycle management and monitoring** easier. Model versions, software libraries, images, inspection results, logs and performance metrics can be managed from common infrastructure. This is useful even where no regulatory requirement applies, and it becomes more

important where AI governance is formalised. Under the EU AI Act, providers of **high-risk AI systems** must establish post-market monitoring that systematically collects and analyses performance information throughout the system lifetime [128]. Centralised inference is not required to meet that obligation - edge systems can report telemetry to separate monitoring services - but server architectures naturally support enterprise monitoring, audit trails, version control and controlled deployment of model updates.

The trade-off is that the production network becomes part of the inspection path. Images must reach the server and results return within the available decision time. High-resolution or high-frame-rate systems can generate considerable network traffic, and latency is now influenced by transport, switching, queuing and shared server load as well as image acquisition and inference. Where very short or tightly bounded response is required, keeping the time-critical decision at Device or Machine Edge may remain preferable even if storage, monitoring or model management is centralised.

Centralisation also enlarges the failure domain. Failure of a Machine Edge computer normally affects one machine; failure of a shared server or network component may affect every station that depends on it. As more production relies on shared infrastructure, redundant networking, compute, storage and failover become progressively more valuable. Troubleshooting likewise crosses the OT/IT boundary and can require expertise in vision, networking, servers, cybersecurity and production automation.

Security and data sovereignty remain easier to control than with remote cloud inference because images, models and production data can stay inside the organisation. The architecture is therefore most compelling when compute is impractical to house at the machine, resources can be shared effectively, or centralised model management and monitoring provide enough value to justify the network dependency.

Strengths and best fit

- **High-performance compute away from the machine** - large GPUs, memory and storage can be installed where power, cooling and physical access are easier to engineer
- **Better suited to harsh production environments** - sensitive high-power computing can remain in a clean, controlled room rather than a large protected machine cabinet.
- **Shared computational resources** - expensive CPU/GPU capacity can serve several machines rather than being duplicated at every station.
- **High computational scalability** - server hardware and accelerator capacity can be expanded independently of individual machines.
- **Centralised software, model and monitoring services** - repositories, versions, logs, metrics and image archives can be consolidated.
- **Strong data locality compared with cloud deployment** - production images, intellectual property and models remain within organisational infrastructure.

Best fit: Applications requiring **compute that is impractical to install at the machine**, or where several machines can benefit from shared GPU resources, central model services, large image repositories or enterprise-level monitoring. It is particularly attractive for demanding deep-learning workloads where local-network latency is acceptable.

⚠ Watchouts

- **The production network enters the inspection loop** - bandwidth, latency, congestion and network availability can directly affect inspection performance.
- **Timing becomes less predictable** - response includes image transmission, server queuing, workload contention and result communication as well as inference.
- **The failure domain becomes larger** - a server, switch or network failure can affect multiple machines or production lines.
- **Redundancy may become necessary** - production-critical shared infrastructure may require redundant networking, compute, storage and failover.
- **Raw imaging can consume enormous bandwidth** - central processing may be inefficient if many stations continuously transmit large image streams.
- **OT and IT responsibilities converge** - commissioning and support can require machine-vision, network, server and cybersecurity expertise.
- **Resource sharing requires capacity engineering** - efficient average utilisation does not guarantee acceptable response during simultaneous peaks.

Rule of thumb:

Choose an On-Premises Vision Server when **compute requirements, physical/environmental constraints, resource sharing or centralised monitoring justify moving processing away from the machine**, and the application can tolerate the resulting local-network dependency. As centralisation grows, redundancy and capacity planning should grow with it.

2.4.4. Cloud Inference

Cloud Inference places the production vision processing outside the local manufacturing infrastructure. Images or other vision data are transmitted over a wide-area network to a remotely hosted service, where the model or application executes and returns a prediction or decision. This must be distinguished from **cloud-based development, model management or monitoring**, all of which can use cloud infrastructure while production inference remains at Device Edge, Machine Edge or on an On-Premises Vision Server.

There is no shortage of technically capable hosted inference services. **Amazon SageMaker AI** provides managed real-time endpoints with monitoring and automatic scaling [129]; **Microsoft Azure Machine Learning** offers managed online endpoints and customer-managed Kubernetes endpoints [130]; **Google Vertex AI** provides online prediction endpoints for deployed models [131]; and **LandingAI's LandingLens** offers hosted inference as well as Docker and industrial-edge deployment routes [132]. LandingAI was founded by Andrew Ng in 2017, which helped make its data-centric vision workflow particularly visible within industrial AI [133].

The important point is that these services solve a somewhat different scaling problem from a typical production machine. A web or enterprise service may receive one request now and thousands a few seconds later, so **elasticity and request concurrency** are central design requirements. AWS explicitly positions serverless inference for intermittent or unpredictable traffic [134], while LandingLens describes its hosted option as appropriate for variable inference loads and labels the cloud route as higher latency than its local Docker and LandingEdge options [132]. A manufacturing station is usually the opposite: the number of cameras, approximate image rate and peak workload are known, but a result may be required repeatedly within a defined process window. For that workload, predictability can matter more than elasticity.

This difference explains why Cloud Inference remains unusual for tightly coupled factory-floor inspection. Wide-area network latency and variability, service queues, external connectivity and service availability become part of the inspection loop. A cloud provider can legitimately describe an endpoint as real-time because it answers synchronous low-latency API requests; that does not imply the deterministic timing

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normally expected of a production control path. Continuous transfer of high-resolution images can also create significant bandwidth costs and extends the cybersecurity and data-governance boundary to production imagery that may reveal product or process intellectual property.

None of this makes cloud inference impossible. It can be sensible where timing is relaxed, request volume is intermittent or highly variable, the image payload is modest, or a remote service provides a capability that would be uneconomic to maintain locally. The practical warning is simply not to confuse **technical possibility with architectural fit**. A cloud endpoint designed to serve a changing population of users is solving a different optimisation problem from a camera inspecting the same product every 300 ms for ten years.

In industrial vision, cloud platforms are often more compelling in **hybrid architectures**. AWS documents manufacturing patterns in which SageMaker is used to build and train models in the cloud while AWS IoT Greengrass performs inference locally for latency-sensitive inspection [135], [136]. LandingLens similarly separates hosted inference from self-hosted Docker and LandingEdge, the latter connecting directly to industrial cameras and PLCs [132]. Cognex OneVision provides another contemporary example: AI models are developed, managed and governed using cloud services while runtime inspection executes on supported In-Sight systems at the edge [137]. In these arrangements, loss of cloud connectivity may affect monitoring, synchronisation or model distribution without necessarily stopping the immediate production decision.

It is also useful to distinguish **development platforms from serving platforms**. Tools such as Voxel51 FiftyOne are valuable for dataset curation, applying models to datasets and model evaluation, but are better viewed here as development/evaluation infrastructure rather than as a managed factory-floor inference endpoint [138]. Likewise, a product carrying a cloud label may use cloud licensing, collaboration or training while its runtime remains local. The architecture should therefore always be classified by the location of the production decision, not by the marketing label attached to the development environment.

Cloud development can be especially attractive for learned vision because training and experimentation often need substantial GPU resources only intermittently, whereas production inference may need smaller, predictable resources continuously. The development and deployment architectures therefore do not need to match: datasets can be annotated and models trained centrally, then the validated model can be deployed to an IPC, embedded AI computer, supported smart camera or on-premises server. This separation preserves local production autonomy while still exploiting shared development infrastructure.

i Cloud architectures in industrial machine vision

- **Cloud Inference - technically capable, but still unusual for tightly coupled factory-floor control.** Hosted endpoints are attractive for elastic or variable request loads, but WAN dependency, timing variability, image bandwidth and governance make them a poor default for deterministic production inspection.
- **Hybrid Deployment - local inference with cloud services.** The time-critical decision remains at Device/Machine Edge or on-premises, while cloud services provide storage, monitoring, dataset/model management, collaboration and fleet-level visibility.
- **Cloud Development - develop centrally, deploy locally.** Annotation, training and experimentation can use scalable cloud resources while the resulting model is deployed to the production architecture that best meets the machine requirements.

A cloud-based machine-vision platform does not imply that production inference should execute in the cloud. First decide where the image-to-decision path needs to run; then decide which development and supporting functions benefit from cloud infrastructure.

2.4.5. Comparison and selection criteria

When selecting an architecture, a useful starting question is: **where should the production image-to-decision process actually run?** Once that is understood, development tools can be selected to support the deployment rather than allowing familiarity with a particular software package to decide the architecture by accident. It is a common project trap: a convenient prototype becomes the production platform before anyone has asked whether it is the best place for the system to live.

There is rarely one objectively correct answer. This section focuses on technical criteria that can be compared reasonably generally - timing, processing requirements, camera topology, integration, reliability, maintainability and growth. Real projects are also shaped by criteria that are organisation- and site-specific: approved vendors, internal skills, procurement, spare-parts strategy, legacy systems, cybersecurity rules, support agreements and corporate standardisation. Those constraints can be every bit as decisive as benchmark performance.

 Use this guidance as a decision aid

The comparisons below are intended to support an engineering decision, **not prescribe one**. Similar applications can reasonably use different architectures because the surrounding organisation, site and lifecycle constraints differ. Technology also moves quickly; the useful objective is not to select the architecture that looks most advanced, but the one whose limitations are easiest to live with for the intended production life.

The following table summarises the main technical characteristics of the three architectures normally considered for **on-site production inference**. Cloud Inference is omitted from the table because it remains an exception rather than the default for tightly coupled factory-floor decisions; where it is genuinely appropriate, its WAN, service and governance constraints should be assessed explicitly rather than treated as another local compute option.

 Traffic-light guide for Table 1.2

favourable · neutral · unfavorable

Table 2.3.: Inference architecture selection summary. Qualitative ratings are comparative guidance rather than absolute performance guarantees.

Selection criterion	Device Edge	Machine Edge	On-Premises Vision Server
Image-to-decision latency	Very low	Low	Network-dependent
Timing predictability	Very high	High	Depends on network and shared workload
Processing capacity	Device-bounded	High to very high	Very high
Compute scalability	Limited; add/replace devices	Good; upgrade local hardware	Very good; expand shared resources
Complex application logic	Limited to platform	Good to very high	Good for services; less suited to tight HMI sequencing
Camera flexibility	Integrated/supported sensors	High	High, subject to acquisition architecture
Multi-camera coordination	Limited	Very good	Good where networked acquisition is practical

Selection criterion	Device Edge	Machine Edge	On-Premises Vision Server
Hardware / accelerator flexibility	Low	High	Very high
Software / custom-code flexibility	Low to medium	High to very high	High
Machine-controller integration	Very direct	Very good	Indirect / networked
Physical footprint at machine	Very small	Moderate	Small locally; infrastructure elsewhere
Production autonomy	Very high	Very high	Depends on local network/server infrastructure
Network dependency for inference	None or minimal	None or minimal	Local production network
Failure isolation	Usually one inspection point	Usually one machine/cell	Shared infrastructure can affect several machines
Data locality	Very high	Very high	Within organisational infrastructure
Implementation complexity	Low	Moderate to high	High
Software / infrastructure lifecycle burden	Low; appliance-like	Moderate to high	High; combined IT/OT infrastructure
Expansion / future growth	Limited	Good	Very good
Centralised resource sharing	None	Limited	Very good
Enterprise monitoring / model governance	Limited unless hybrid	Good with added services	Very good

The table is useful for comparison, but architecture selection often becomes clearer when approached from the opposite direction: **what characteristics of the application should make a particular architecture attractive?**

💡 When to consider Device Edge

- **The task is bounded and relatively simple.** The complete inspection sits comfortably inside the device software and compute envelope.
- **A very short and predictable image-to-action path matters.** Acquisition, processing and potentially digital I/O can remain within one device.
- **Only one or a few largely independent cameras are required.** Each inspection point can operate autonomously.
- **Compactness and simplicity are priorities.** Reducing computers, operating systems and interfaces has real lifecycle value.

Think twice when: the application could grow substantially, requires bespoke software or tight multi-camera coordination, or coding sequences and tools start becoming complex and hard to troubleshoot.

💡 When to consider Machine Edge

- **The vision application is complex.** The project benefits from independently selecting camera, acquisition, compute and software.
- **Requirements are uncertain or likely to grow.** An IPC/controller preserves significantly more processing, memory, storage and software headroom.
- **Several cameras must act as one system.** Multi-view inspection and synchronised acquisition are easier to coordinate centrally at the machine.
- **Processing is demanding.** High-resolution imaging, 3D, deep learning or mixed classical/learned pipelines justify CPUs, GPUs or other accelerators.
- **Vision must integrate deeply with the wider machine software.** Robotics, databases, HMI interaction and machine sequencing can be coordinated within one application.

Practical warning sign: poorly defined or evolving inspection requirements. Where substantial uncertainty remains, PC-based development - or a software environment that can later target an embedded device - can preserve options while the real requirements become clearer.

💡 When to consider an On-Premises Vision Server

- **The required compute is awkward to install at the machine.** Large GPUs, power and cooling are easier to manage in controlled infrastructure.
- **Several machines can share expensive resources.** Central CPUs, GPUs, memory or storage can serve more than one inspection station.
- **Central model or software services provide value.** Common models, libraries and versions are easier to manage centrally.
- **Monitoring, storage and governance are important.** Images, results, logs and model performance can be consolidated across machines.
- **The inspection can tolerate a networked processing path.** Network and server response can be kept within the process timing envelope.

Main question: do the benefits of centralisation justify enlarging the failure domain? As more machines depend on the same infrastructure, capacity planning, redundancy and failover become correspondingly more important.

The architecture selected for production does not have to dictate the rest of the vision lifecycle. Local inference can be combined with cloud development, model management, monitoring, image storage and cross-site services. These hybrid arrangements are best considered **after the production inference path has been placed**, rather than treated as another competing location for the time-critical

decision.

A practical closing principle is to favour **the simplest architecture that satisfies the production requirements with adequate margin and an acceptable lifecycle burden**. Simplicity does not mean choosing the device that is easiest to configure in a demonstration. It means avoiding compute, software, infrastructure and dependencies that production does not need, while retaining enough headroom for the system to remain maintainable as requirements and technology evolve.

2.5. The vision system as an engineered chain

Previous sections considered what machine-vision systems are asked to do and where their processing can be deployed. This section turns to how the pieces fit together. Whatever the application family or architecture, a working vision system is an engineered chain: each stage creates the conditions required by the next, and weaknesses introduced early in the chain are difficult to recover later. The discussion begins with the stages themselves, then considers the order in which they should be engineered, shows how a requirement propagates through the complete system, and closes with the professional roles, standards and credentials that support the discipline.

2.5.1. The seven stages: presentation, illumination, optics, sensor, processing, decision and action

Machine-vision texts commonly describe a path from image acquisition through image processing to a decision, while industry guidance similarly treats lighting, optics, sensors, processing and communications as interdependent elements of a complete system [95], [139]. For the purposes of this handbook, it is useful to make that chain slightly wider. The five stages that form the core imaging chain - illumination, optics, sensor, processing and decision - are bracketed by two application stages: presentation before the image is acquired, and action after the decision has been made. This gives seven stages in total. The distinction is deliberate: a technically excellent image-processing algorithm is of little value if the part is not presented in a usable way, or if the final result never produces the required machine response.

The five core stages can be stated briefly because each is developed later in the handbook. Illumination controls how the relevant features appear. Optics form and scale the image on the sensor. The sensor converts the optical signal into sampled image data. Processing transforms those pixels into measurements, features, locations or learned outputs. The decision stage then interprets those outputs against the application requirement: for example pass/fail, a dimensional result, a class, a pose or a confidence value. These are the stages most often associated directly with machine vision, but they do not by themselves make a complete automated application.

The first application bracket is **presentation**. For inspection and measurement, repeatable presentation is usually an advantage: the feature should remain within the usable field of view, focus range and lighting geometry, and acquisition should occur at a controlled point in the process. Guides, nests, rails, fixtures, controlled motion and robot poses are therefore not merely mechanical details; they directly determine how much variation the vision system must tolerate [140]. Triggering and illumination synchronisation form part of the same problem. At production speed, even small timing errors translate into positional variation, which is why industrial cameras commonly provide dedicated trigger and illumination-control interfaces for deterministic acquisition [141].

Presentation also determines how many views of the part must be acquired and how those views are created. If the complete inspection can be performed from one viewpoint, the decision is simple. Where features lie on different faces, at substantially different heights, or around the circumference of a component, the engineer must instead decide whether to **move the part, move the camera, or use multiple cameras**. This choice should be made early because it shifts complexity between

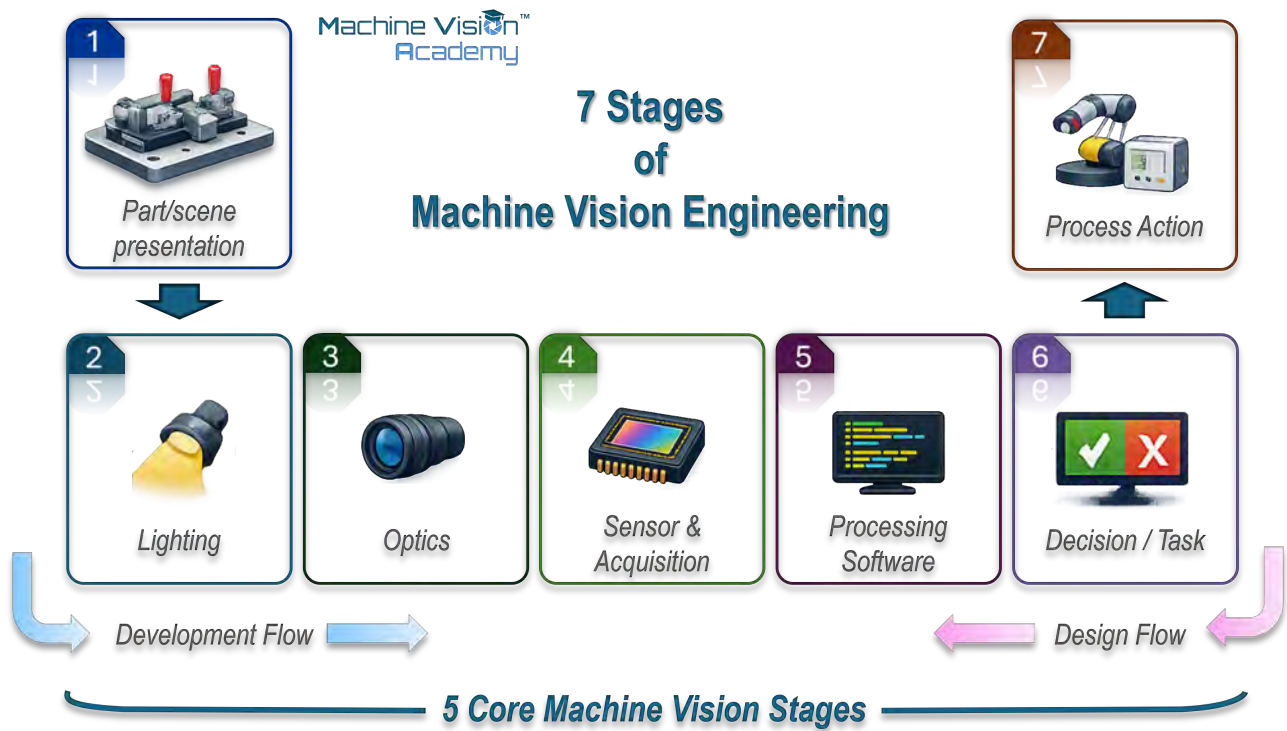


Figure 2.17.: The vision system as a seven-stage engineered chain

the mechanical system and the vision system and can influence camera count, calibration, processing architecture and cycle time.

A useful order of preference is to start with the resources already present in the design of the equipment or station. If a robot or positioning mechanism already handles the component, presenting several faces to a fixed camera may add relatively little hardware and can be an economical solution. Where only a small displacement is required—perhaps between two nearby inspection planes—a simple linear or pneumatic movement of either the part or camera may also be practical. As the required motion becomes larger or more complex, however, several fixed cameras often become more attractive, particularly when cycle time does not allow sequential movement and acquisition. Multiple cameras increase acquisition, calibration and software complexity, but avoid adding moving mechanisms to the inspection cycle.

The surrounding machine design matters just as much. Moving a part or camera introduces actuators, guarding and safety considerations; moving a camera additionally requires suitable cable routing and management. These additions may be quite reasonable inside an already guarded automated cell, but difficult to justify on an otherwise simple open station. Applications involving calibrated measurement also deserve particular care: a fixed camera provides a stable imaging geometry, whereas moving either the camera or part requires the relevant position and transformation to remain sufficiently repeatable. Robot-mounted cameras are entirely feasible, but they introduce an explicit hand-eye calibration between the camera and robot coordinate systems. Current machine-vision and robotics tools support both fixed-camera and moving-camera configurations, but their calibration becomes part of the engineered system rather than an incidental setup step.

💡 Choosing how to obtain multiple views

If more than one view is required, a useful starting guide is:

- **Move the part** when it is already manipulated by a robot or positioning mechanism and additional poses can be introduced without compromising cycle time.
- **Move the camera** when the part is difficult to manipulate and the required camera movement is simple, repeatable and compatible with calibration, guarding and cable management.
- **Use multiple fixed cameras** when several views are required simultaneously, cycle time is short, movement would be mechanically complex, or maintaining a fixed calibrated geometry is particularly important.
- **Consider specialised multi-view optics** before adding motion or cameras where the part geometry permits it. Pericentric, catadioptric, hypercentric and multi-mirror optics can capture several surfaces or viewpoints in a single exposure, sometimes replacing a multi-camera or mechanically rotated arrangement.

There is no universal hierarchy: the objective is to place the necessary complexity where it is easiest to control and maintain—mechanics, optics or vision software.

Specialised optics deserve particular mention because they provide a fourth option that is easily overlooked. Opto Engineering, for example, offers **360° and multi-view lenses** designed to capture several surfaces of a component using one camera [142]. Pericentric and catadioptric optics can image the top and lateral surfaces of suitable objects, while multi-mirror designs can place several side views within the same image. For cylindrical parts, caps, bottles, threads and similar geometries, this can remove both mechanical rotation and the need to synchronise several cameras. The trade-off is that these are specialised optical arrangements with constraints on object geometry, working distance, field of view and image utilisation, so they should be considered as another engineering option rather than a universal substitute for multiple cameras.

A second principle of presentation is to **remove as much variation as possible that is not inherent to the product or the inspection requirement**. Uncontrolled changes in position, orientation, background, illumination, surface condition or surrounding equipment increase the range of appearances that the vision system must interpret without adding useful information. This matters for rule-based vision, but it is equally important for deep learning. A model can learn substantial variation when that variation is represented during training—and predictable changes can sometimes be reproduced through augmentation—but unexpected production conditions introduce *domain shift* and can cause significant performance degradation even in otherwise capable models. In practice, these variations are particularly troublesome because they may appear late in development, during validation or only after deployment, leading to additional data collection, retraining and revalidation. The aim is therefore not to make every image identical, but to ensure that the variation reaching the vision system is intentional, understood and representative of the conditions against which the system has been designed and validated.

The second application bracket is **action**. A verdict that remains inside the camera or PC has not completed the automation task. The result must be delivered to the machine and associated with the correct product at the correct time: a reject mechanism may need to fire, a robot may need a coordinate, a process may need to be adjusted, an image may need to be stored for traceability, or an operator may need to be alerted. Depending on the architecture, this hand-off may use digital I/O, an industrial Ethernet protocol, a robot interface, OPC UA, a database or higher-level manufacturing software. The specific technology matters less than the engineering requirement: the result must reach the right destination, remain associated with the right product, and produce the intended response.

A recurring example from practice illustrates why this wider system boundary matters. Consider an automated inspection station that identifies a defective, non-serialised product but the system is designed to rely on an operator to remove it manually once the vision identifies a defect and notifies it by means of a visual alarm or status indicator. The vision algorithm may be performing perfectly, yet the original business objective - preventing defective product from continuing through production - still depends on a fallible human action and therefore can fail as an end-to-end solution if the operator simply gets distracted and forgets to dispose of the defective part.

Presentation and **action** reach into mechanics, robotics, controls, data systems and process engineering, so this handbook does not attempt to cover them with the same depth as illumination, optics, sensors and image processing. They are included in the chain because the vision engineer must nevertheless understand and influence them. Machine vision is rarely installed in isolation; it succeeds as part of a machine that performs an end-to-end task.

The seven stages can also be used as a compact checklist when first framing an application:

Table 2.4.: The seven stages of the engineered vision chain

Stage	What it does	Question to answer	Developed in
1 Presentation	Places the part or scene in a usable state at a known acquisition instant	Is the required feature observable with controlled or intentionally bounded variation?	This section; Ch. 17
2 Illumination	Controls how the relevant feature interacts with light	Can the feature be made visible with useful and repeatable contrast?	Chapter 4
3 Optics	Forms and scales the image on the sensor	Is the required detail resolved across the necessary field and depth?	Chapter 3
4 Sensor	Converts the optical image into sampled image data	Are resolution, sensitivity, speed, dynamic range and noise suitable?	Chapter 5
5 Processing	Extracts measurements, features, locations or learned outputs	Can the required information be recovered reliably from the image?	Chapters 6–10
6 Decision	Maps the processed result to an application outcome	Does the result satisfy the acceptance, classification or localisation requirement?	Ch. 9, 16, 18
7 Action	Transfers the result and causes the required process response	Does the correct result drive the correct response for the correct product, in time?	This section; Ch. 17

2.5.2. Constraint-first engineering: solve the image before the algorithm

If there is one design principle worth carrying through the rest of this handbook, it is to improve the image in the physical world before compensating for it in software. Image acquisition is not a neutral

front end: illumination, surface interaction, optics, focus, exposure, motion and sensor characteristics determine what information is actually recorded [95]. Once a feature has been hidden by glare, clipped by saturation, blurred by motion or never resolved by the optics, downstream processing has far less to work with. Some processing can enhance weak information; it cannot reliably recreate information that the acquisition system never captured.

Lighting normally provides the greatest leverage because the camera observes light from the scene rather than the object in the abstract. Changing the direction, spectrum, diffusion or polarisation of that light can suppress irrelevant structure and make the required feature dominant. Industrial guidance repeatedly makes the same practical point: stable, application-specific illumination simplifies camera configuration and algorithm design, whereas uncontrolled illumination creates variability that must be handled later [143]. Optics and presentation are part of the same argument. A better lens, a more suitable working distance, a polariser, a controlled background or a small change in part angle can turn a difficult image-processing problem into a straightforward one.

💡 The order of attack: physics before software

When a feature is difficult to detect, first ask what can be changed before the image becomes pixels:

- reduce unnecessary variation in part presentation or background;
- change light direction, wavelength, diffusion or polarisation;
- use bright-field, dark-field, backlight, coaxial or another suitable illumination geometry;
- adjust working distance, magnification, aperture, focus or optical filtering;
- shorten exposure and synchronise illumination if motion blur is the limitation;
- only then decide how much algorithmic complexity is genuinely required.

This is not an argument against sophisticated processing. It is an argument for giving that processing the best possible signal, so that the end-to-end solution has a better chance to work consistently and repeatedly after deployment.

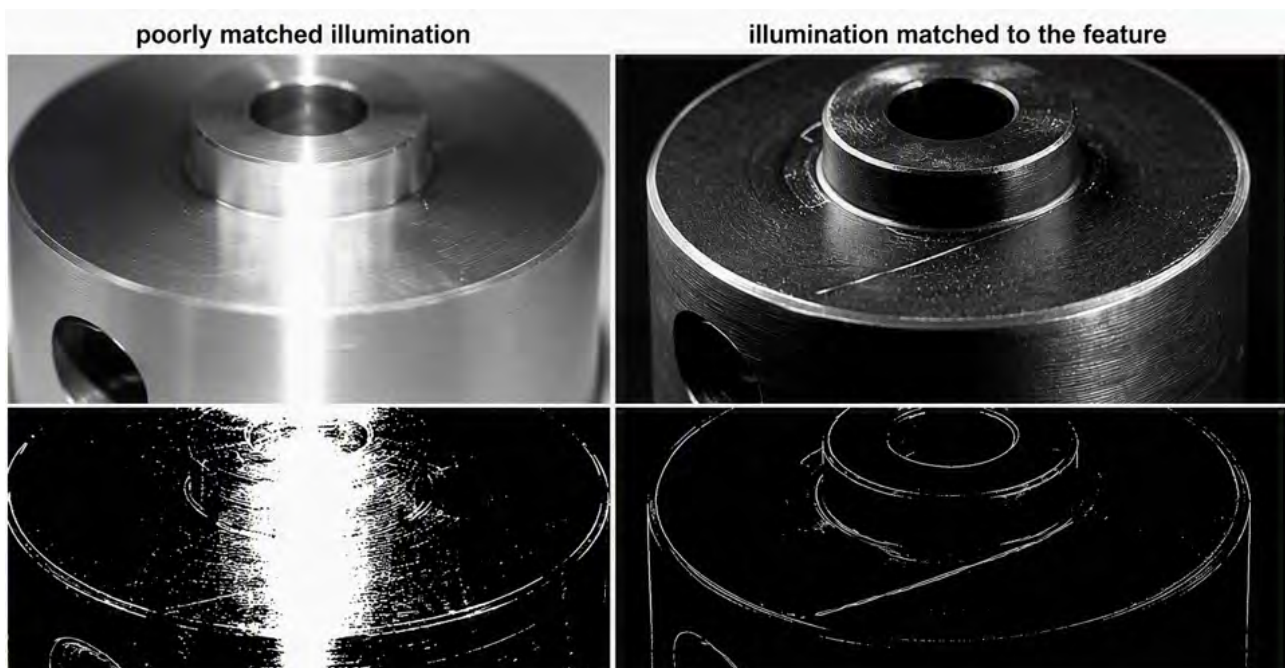


Figure 2.18.: The same feature under poor and well-chosen lighting after applying Sobel edge detection.

Deep learning makes this principle more important, not less. Learned models can tolerate forms of variation that would be awkward to model with hand-designed rules, but they remain dependent on the

information and statistical consistency of their input images. Research on computer-vision robustness shows that blur, noise, illumination change and other common image corruptions can materially reduce model performance [144]. Experimental work with controlled active illumination has also shown that making image appearance more consistent can reduce the amount of training data required to reach a given level of performance [145]. These studies are not industrial quality-inspection recipes, but they reinforce the same idea: robustness in the model does not remove the value of robustness in image formation. This should not be interpreted to mean that better and more consistent images make very small training datasets sufficient. Rather, reducing unnecessary and unpredictable variation makes dataset creation more efficient, allowing the available samples to concentrate on the product and process variation that the model genuinely needs to learn. A small dataset containing little variation can produce apparently strong training or validation results while encouraging overfitting and providing poor generalisation to conditions not adequately represented in the data. Dataset diversity, overfitting and generalisation are important topics in their own right and are examined in greater depth in later chapters.

The temptation in a deep-learning project is easy to understand. As soon as images can be acquired, it is possible to start collecting data and training models. That can create two expensive traps:

- **Poor images become a poor long-term asset.** The image dataset can become one of the most valuable and reusable outputs of a vision application, supporting model retraining, root-cause analysis, future inspections and wider process studies. Its value is particularly high when it has been developed following sound engineering principles: the sources of variation are understood and documented, sufficient quantities of real production data have been collected, and the dataset is representative of the conditions the system will encounter in production. Building such a dataset can require considerable time and cost, which makes preserving its information content especially important. A well-formed image may also contain useful information beyond the original inspection requirement; an image that is blurred, saturated or dominated by uncontrolled reflections has already discarded some of that opportunity. Dataset design, representativeness and management are covered in greater depth later in the handbook.
- **Weak image formation can be mistaken for insufficient training.** In a rule-based project, a small set of images often reveals quite quickly that a feature is not visible enough. In deep learning, the team may instead respond by collecting more samples, increasing training time, trying larger models or tuning augmentation before questioning the acquisition setup. Because model performance usually changes gradually rather than failing for one obvious physical reason, the point at which the image itself is inadequate can be harder to recognise.
- **The model may learn nuisance variation that the machine could have removed.** If background, illumination, reflections or pose vary unnecessarily in the training set, a model may devote capacity to those factors instead of the feature that matters. Data augmentation and broader training data are valuable tools, but they should represent variability the deployed system genuinely needs to handle, not compensate by default for an avoidably unstable imaging station.

Deep learning does not repeal image formation

In our experience, one of the easiest mistakes in a deep-learning vision project is to start model development as soon as the first usable images appear. Resist that pressure. Spend enough time proving that the image formation is as strong, stable and informative as it reasonably can be before committing to large-scale data collection.

Waiting for a different light, lens, filter or fixture can feel like lost development time. Collecting and labelling thousands of images only to discover later that the feature was poorly illuminated is usually much more expensive.

2.5.3. How the chain interacts: propagate the requirement end to end

The chain is drawn from **presentation** to **action** because that is the direction in which information flows during operation. Design, however, starts from the required outcome. A vision engineer should not begin with the question “which camera should I use?” but with a measurable description of what the system must decide or measure, under what production conditions, and with what timing and confidence.

A solution looking for a problem

One of the easiest engineering traps is to start with a solution we want to use and then find a problem that fits it. Good engineering works in the opposite direction: understand the problem first, then let the requirements lead you to the solution.

From there, each stage constrains the stages around it. This systems view is consistent with the broader computer-vision engineering principle of modelling the image-formation process and testing the complete method under the conditions in which it is expected to operate [146].

Consider a simplified requirement to detect a surface defect approximately 50 micrometres across. That feature must first be represented with enough spatial information and contrast to distinguish it from normal surface variation. The required field of view and sampling then constrain sensor resolution and optical magnification. Magnification, sensor size, working distance and aperture influence lens choice and depth of field; depth-of-field requirements feed back to how tightly the part height or pose must be controlled. The material and defect geometry constrain illumination because a scratch, dent, contamination mark and colour change interact with light differently. Throughput then constrains exposure time, triggering and processing time. A requirement that initially sounds like a single number therefore becomes a coupled set of optical, mechanical, electrical and computational constraints.

The same coupling appears in less obvious forms. Increasing magnification may improve spatial sampling but reduce field of view and depth of field. Shortening exposure may reduce motion blur but require more illumination or sensor sensitivity. A wider aperture may increase signal but reduce depth of field. Relaxing mechanical fixturing may simplify the machine but increase the search range and algorithmic burden. Moving processing to a shared server may increase compute capacity but add network timing and a larger failure domain. None of these choices is inherently good or bad; the engineering task is to make the trade-offs explicit and keep the whole chain inside the requirement envelope.

This is why feasibility work should try to expose contradictions before hardware and software become fixed. If the required field of view and smallest feature imply an optical setup that cannot maintain focus across the actual part-height variation, the useful result of the feasibility study is not a clever algorithm - it is discovering that the requirement, presentation or imaging geometry must change. The later design framework formalises this process, but the habit starts here: trace important requirements through every stage and ask where they become difficult, sensitive or mutually incompatible.

i Requirements flow backwards; information flows forwards

During operation, the system runs from presentation to action. During design, work backwards from the requirement:

1. What decision or measurement must the system deliver, and what action follows?
2. What image information is necessary to support that result?
3. What illumination, optics and sensor conditions are required to capture that information?
4. What presentation, timing and environmental controls are needed to keep those conditions valid in production?

Then run the reasoning forwards again to verify that the complete image-to-action chain closes within the required cycle time, accuracy and reliability.

2.5.4. Professional roles, standards and certification landscape

Machine vision sits across several engineering disciplines. A single deployment may involve illumination and optics, image sensors, mechanics, electrical design, software, industrial networking, robotics, controls, metrology, data handling and production validation. It is therefore common for different organisations to divide the work differently. In a small team, one engineer may cover most of the chain; in a larger programme, responsibility may be distributed across specialists.

Typical roles around a machine-vision project include:

- Vision or applications engineer - develops the imaging concept and normally owns much of the illumination, optics, acquisition and image-processing chain.
- Systems integrator or machine builder - combines vision with mechanics, controls, safety, motion and the wider machine architecture.
- Controls or automation engineer - integrates triggering, sequencing, communications, interlocks and the resulting machine actions.
- Metrology or quality engineer - defines measurement traceability, acceptance criteria, capability studies and validation where the vision result is used quantitatively.
- Data, AI or software engineer - increasingly involved where deep learning, dataset infrastructure, deployment pipelines or substantial application software form part of the solution.
- End user, process engineer and maintenance team - provide the production knowledge that defines real variation, failure modes, maintainability and long-term ownership.

The boundaries between these roles are deliberately porous. A good vision engineer does not need to be the design team controls expert or the mechanical designer, but must understand enough of those disciplines to recognise when they constrain the image or the final action. Equally, software expertise alone is not a substitute for understanding image formation. That breadth is one reason machine vision is often learned through a mixture of formal study, vendor and association training, laboratory work and production experience. In this sense, professional accreditation is neither essential nor a substitute for practical experience. It can, however, be valuable when applying for machine-vision roles by providing evidence of a recognised level of knowledge while also demonstrating commitment to, and genuine interest in, the discipline.

At the time of editing, one of the most established broad professional credentials in the machine-vision industry is the Association for Advancing Automation (A3) Certified Vision Professional (CVP) programme [147]. CVP-Basic covers the foundations of machine vision through courses in fundamentals, optics, lighting, cameras and image sensors, image processing and vision-system design [148]. CVP-Advanced extends the scope into topics including advanced optics and lighting, 3D vision, machine

2. Vision engineering: scope, definitions and history

learning, line-scan imaging, high-speed systems, vision-guided robotics, non-visible imaging, colour, metrology, advanced image processing and system integration [149]. A3 currently offers proctored Basic and Advanced examinations, and states that individual CVP certification is valid for five years [147].

Table 2.5.: Examples of current machine-vision professional certification routes.

Programme	Purpose / scope	What it signifies
A3 CVP-Basic	Broad foundation: machine-vision fundamentals, optics, lighting, cameras/sensors, image processing and system design.	Individual has demonstrated a common baseline of machine-vision knowledge.
A3 CVP-Advanced	Advanced application topics including 3D, machine learning, line scan, high speed, robotics, metrology and integration.	Individual has demonstrated broader and deeper knowledge across advanced vision topics.
EMVA 1288 User / Expert	Specialist knowledge of EMVA 1288 camera/sensor characterisation and use of its measurement data.	Individual has demonstrated knowledge of this specific camera-characterisation standard, not a general machine-vision qualification.

In Europe, the European Machine Vision Association (EMVA) has a different but equally important role. It develops and supports industry standards and educational resources rather than offering a direct European counterpart to the broad CVP qualification. Its EMVA 1288 programme provides User- and Expert-level certification specifically around camera and image-sensor characterisation [150]. The EMVA also hosts GenICam, the generic camera-interface standard used across interfaces such as GigE Vision, USB3 Vision and CoaXPress [151]. At the time of editing, the long-running EMVA 1288 characterisation standard is also being progressed towards the international ISO 24942 standard; ISO lists ISO/DIS 24942 as under development in 2026 [152]. These details will inevitably evolve, so readers should treat the named programmes as examples of the current professional landscape rather than a permanent catalogue.

💡 Certification is a foundation, not the finish line

A recognised qualification is valuable because it establishes vocabulary, fundamentals and a common technical baseline. It does not replace application judgement.

Machine vision is learned by repeatedly connecting theory to real images, real parts and real production constraints: seeing how a surface changes under a different light, discovering why a nominally correct lens does not provide enough depth of field, or tracing an intermittent false reject back to timing, vibration or presentation. Certification can structure that learning; experience turns it into engineering judgement.

There is also a broader educational challenge. Machine vision is used widely in manufacturing, yet access to structured, vendor-neutral learning remains uneven across universities, companies and regions. One of the longer-term aims of the Machine Vision Academy is therefore to make reusable educational material openly available so that individuals, educators and academic institutions can build courses around a common industrial baseline. At the time of writing this remains an ambition rather than an established accreditation route, but the direction is intentional: make the fundamentals easier to learn, then allow institutions and professional bodies to build recognised learning pathways on top of them. Current project material is intended to be released through MachineVisionAcademy.org as that programme develops.

The broader message of this section is that machine vision is not a sequence of isolated component choices. It is an engineered chain, owned by people who must reason across the boundaries between

image formation, computation and automation. The next section turns that principle into a more explicit method for approaching a new vision problem from the requirement onwards.

3. Light, colour and radiometry

Everything a vision system knows about the physical world reaches the sensor through light. A camera does not observe an object directly; it records optical radiation that has been emitted, reflected, transmitted, scattered or otherwise modified by the scene and then admitted by the optical system. The appearance of a scratch, edge, printed mark, transparent layer or coloured component is therefore governed first by the physics of light and matter, and only later by image processing.

Describing that light is less straightforward than it may initially appear. Depending on the problem, physics represents light using rays, electromagnetic waves or photons, while modern quantum optics provides a deeper description from which some of these behaviours emerge. These models are complementary rather than competing: each exposes different aspects of the same physical phenomenon and becomes useful for different engineering questions. This chapter introduces them at the level required for machine vision, while deliberately avoiding theoretical depth that does not contribute to understanding or designing an imaging system.

The previous chapter argued for constraint-first engineering: improve the image in the physical world before asking software to compensate for a weak one. This chapter develops the physical language behind that principle. It is deliberately more theoretical because practical decisions about illumination geometry and wavelength, optical filtering, exposure, colour, polarisation and sensor selection ultimately depend on the behaviour of light introduced here.

From light physics to machine-vision contrast

The theory in this chapter becomes useful when it helps us manipulate the difference between a feature and its background. Most practical illumination problems can be approached through a small set of physical levers:

- **geometry and angular structure** — decide which reflected or transmitted rays can enter the lens;
- **wavelength and spectrum** — exploit differences in reflectance, transmission, absorption or fluorescence;
- **filtering and polarisation** — reject unwanted wavelengths, ambient light or specular components;
- **optical exposure** — deliver enough useful photons, uniformly and repeatably, within the available motion and cycle-time constraints.

This chapter explains the physics behind these levers; the later optics and illumination chapters turn them into lens, filter and lighting choices.

For readers who want a parallel treatment that keeps the complete automated-inspection system in view, Beyerer, Puente León and Frese develop light, image formation and radiometry directly in the context of machine vision rather than as isolated optical theory [95].

3.1. Physical foundations of light

Before quantifying light, we first need to decide how to describe it. No single model is equally convenient for every problem, and machine vision routinely moves between several.

The simplest is the **ray model**, which represents light as travelling along idealised paths. It deliberately ignores wave effects such as phase and diffraction, but is remarkably powerful when the dimensions of the imaging system are much larger than the wavelength of light. Ray geometry allows us to reason about illumination direction, reflection, refraction and image formation without describing the underlying electromagnetic field. Perhaps its simplest demonstration is the **pinhole camera**: straight-line geometry alone can explain how points in a three-dimensional scene form an inverted two-dimensional image. The pinhole model is developed in the optics chapter, where it becomes the starting point for perspective, magnification, field of view and lens-based image formation.

The **electromagnetic-wave model** provides the classical physical description needed when wavelength, phase or polarisation matters. It explains phenomena such as interference and diffraction that cannot be represented adequately using rays alone. At the generation and detection ends of the imaging chain, the **photon description** becomes particularly useful because optical energy is exchanged with matter in discrete quantities. This ultimately underpins LED and laser emission, semiconductor photodetection, quantum efficiency and photon shot noise. Modern **quantum optics** goes deeper still by describing the electromagnetic field itself as quantised, although this level of theory is rarely required for conventional machine-vision engineering.

The useful principle is therefore not to search for one model to use everywhere, but to choose the simplest description that captures the phenomenon being considered. The models used throughout the handbook can be summarised as follows:

Table 3.1.: Useful descriptions of light in machine vision

Description	What it represents	Machine-vision questions it answers
Geometrical/ray optics	Light represented by idealised propagation paths; phase is ignored.	Where does illumination travel? Which reflected rays enter the lens? What field of view and image geometry result?
Classical electromagnetic waves	Continuous electric and magnetic fields governed by Maxwell's equations.	How do wavelength, phase, interference, diffraction, refraction and polarisation affect the image?
Photon / quantum description	Optical energy exchanged in discrete quanta with energy $E = h\nu$.	How much energy does each photon carry? How does a semiconductor detect light? What sets quantum-efficiency and shot-noise limits?
Quantised electromagnetic field	Modern quantum-optical description in which field modes are quantised and photons are excitations of those modes.	Foundations of quantum optics, single-photon statistics and non-classical states; rarely required for conventional machine-vision design.

i Reference: wave-particle duality without the historical shortcut

The familiar statement that light is “both a wave and a particle” is useful shorthand, but it can suggest that light literally switches between two classical forms. Modern quantum optics instead treats the electromagnetic field as quantised; photon language describes discrete excitations and energy exchange, while wave-like interference and propagation emerge from the field description. Classical Maxwell theory remains an excellent macroscopic limit for most ordinary imaging conditions.

For readers who want the formal transition from the classical electromagnetic field to quantum optics, Loudon’s *The Quantum Theory of Light* develops field quantisation, photon states and quantum-optical phenomena in substantially greater depth [153].

3.1.1. Electromagnetic radiation, photons and the optical spectrum

For most of the propagation problems in this handbook, the classical electromagnetic-wave description is the natural starting point. In a homogeneous medium, light can be represented as coupled electric and magnetic fields propagating through space. A monochromatic plane wave has a sinusoidal field variation characterised by amplitude, wavelength, frequency, phase and polarisation. The electric and magnetic fields are transverse to the direction of propagation, and their orientation is the basis of the polarisation concepts developed later in this chapter.

In vacuum, wavelength and frequency are linked by the speed of light:

$$c = \lambda\nu, \quad c = 2.99792458 \times 10^8 \text{ m s}^{-1}.$$

where λ is the vacuum wavelength and ν is frequency. In a medium with refractive index n , the phase velocity becomes

$$v = \frac{c}{n}.$$

The frequency is fixed by the source and remains continuous across an interface, so the wavelength in the medium becomes

$$\lambda = \frac{\lambda_0}{n}.$$

This distinction becomes important when refraction, interference, optical thickness or wavelength-dependent material properties are considered.

Maxwell’s equations are the four fundamental equations of classical electromagnetism. At a high level, they describe how electric charges and currents produce electric and magnetic fields, how changing electric and magnetic fields generate one another, and how those fields propagate through space. One of their most important consequences is that a changing electric field can sustain a changing magnetic field, and vice versa, allowing an electromagnetic disturbance to propagate even through empty space. Light is such an electromagnetic wave.

The complete Maxwell equations are not required for most machine-vision calculations. Instead, it is useful to examine one important result that follows from them. In a homogeneous, source-free and non-dispersive region, the electric and magnetic fields satisfy the wave equations

$$\nabla^2 \mathbf{E} - \frac{1}{v^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0, \quad \nabla^2 \mathbf{B} - \frac{1}{v^2} \frac{\partial^2 \mathbf{B}}{\partial t^2} = 0.$$

3. Light, colour and radiometry

Here, $\mathbf{E}$ and $\mathbf{B}$ are the electric- and magnetic-field vectors, t is time and v is the propagation velocity of the wave through the medium. In vacuum, $v = c$. The symbol ∂ denotes a partial derivative, used when a quantity depends on several variables; ∂^2 , the second derivative with respect to time describes how the rate of change of the field itself changes with time.

i Reading the symbol

The inverted triangle ∇ is called **nabla** or **del**. It is a vector differential operator used to describe how quantities change through space.

In the wave equations it appears as ∇^2 , called the **Laplacian** or *del squared*. In Cartesian coordinates,

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}.$$

Applied to $\mathbf{E}$ or $\mathbf{B}$, it describes the spatial curvature or variation of the field around a point. It is not necessary to calculate the Laplacian to follow the rest of this chapter; recognising what the symbol represents is sufficient.

The structure of the wave equations is more important here than their derivation. They connect how the electromagnetic field varies **through space** with how it varies **through time**. For the electric field, this can be read conceptually as

$$\underbrace{\nabla^2 \mathbf{E}}_{\text{variation in space}} \longleftrightarrow \underbrace{\frac{1}{v^2} \frac{\partial^2 \mathbf{E}}{\partial t^2}}_{\text{variation in time}}.$$

The same relationship applies to the magnetic field. This mathematical structure permits disturbances in the fields to propagate through space as waves.

The wave equations admit many possible solutions. A particularly useful idealisation for understanding optical behaviour is the **monochromatic plane wave**:

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \phi).$$

This expression describes the electric field at any position $\mathbf{r}$ and time t . Its individual terms have useful physical meanings:

- $\mathbf{E}(\mathbf{r}, t)$ is the electric field at the position and time being considered;
- $\mathbf{E}_0$ defines the **amplitude and orientation** of the electric field;
- $\mathbf{r} = (x, y, z)$ is the **position vector**;
- $\mathbf{k}$ is the **wave vector**: its direction gives the direction of propagation and its magnitude describes how rapidly phase changes with distance;
- $\mathbf{k} \cdot \mathbf{r}$ is the dot product that relates position to phase along the propagation direction;
- ω is the **angular frequency**, measured in radians per second;
- ϕ is the **phase offset**, defining where within its oscillation cycle the wave begins.

The wave vector and angular frequency are connected to the more familiar wavelength and frequency through

$$|\mathbf{k}| = \frac{2\pi}{\lambda}, \quad \omega = 2\pi\nu.$$

A complete oscillation corresponds to 2π radians. The first relationship therefore says that the phase changes by 2π over one wavelength λ : shorter wavelengths have larger wave numbers $|\mathbf{k}|$ and change

phase more rapidly with distance. The second expresses the same idea in time: a field oscillating at ν cycles per second has an angular frequency of $2\pi\nu$ radians per second.

The plane-wave equation can consequently be read more intuitively as

$$\mathbf{E}(\mathbf{r}, t) = \underbrace{\mathbf{E}_0}_{\text{amplitude and orientation}} \cos \left(\underbrace{\mathbf{k} \cdot \mathbf{r}}_{\text{position}} - \underbrace{\omega t}_{\text{time}} + \underbrace{\phi}_{\text{phase offset}} \right).$$

If time is held fixed, the equation describes the sinusoidal field pattern **through space**. If position is held fixed, it describes the field oscillating **with time** at that location. Changing ϕ shifts the oscillation without changing its wavelength or frequency, while changing the orientation of $\mathbf{E}_0$ changes the polarisation of a linearly polarised wave.

For a propagating electromagnetic wave, the magnetic field follows the corresponding electric-field behaviour while remaining perpendicular to the electric field and to the direction of propagation. The interactive representation in Figure 3.1 provides a more intuitive way to explore these relationships before they are used later in the chapter.

💡 What matters for machine vision

You will rarely solve these wave equations when designing a conventional vision system. Their value is that they expose several quantities that later become practical engineering variables:

- **wavelength** controls spectral interactions and contributes to diffraction;
- **phase** determines interference behaviour and becomes important with coherent illumination;
- **electric-field orientation** gives rise to polarisation;
- **electromagnetic behaviour at material boundaries** produces reflection, transmission and refraction.

For a fuller derivation of electromagnetic-wave propagation from Maxwell's equations and its relationship to geometrical optics, see Hecht [154].

The photon description becomes useful when optical energy is generated or absorbed. A photon associated with frequency ν carries energy

$$E_\gamma = h\nu = \frac{hc}{\lambda},$$

with Planck constant

$$h = 6.62607015 \times 10^{-34} \text{ J s.}$$

A convenient engineering approximation is

$$E_\gamma[\text{eV}] \approx \frac{1240}{\lambda[\text{nm}]}.$$

Shorter-wavelength photons therefore carry more energy individually. In a semiconductor image sensor, an absorbed photon can transfer its energy to an electron, allowing that electron to move into a state where it can contribute to an electrical signal. This process leaves behind an empty state known as a **hole**; together, the mobile electron and the hole are called an **electron–hole pair**. The photon must carry sufficient energy for this transition, which is one reason different semiconductor materials respond to different wavelength ranges. At normal illumination levels it is usually more convenient to describe the incoming light as continuous optical power using radiometric quantities, but the camera signal

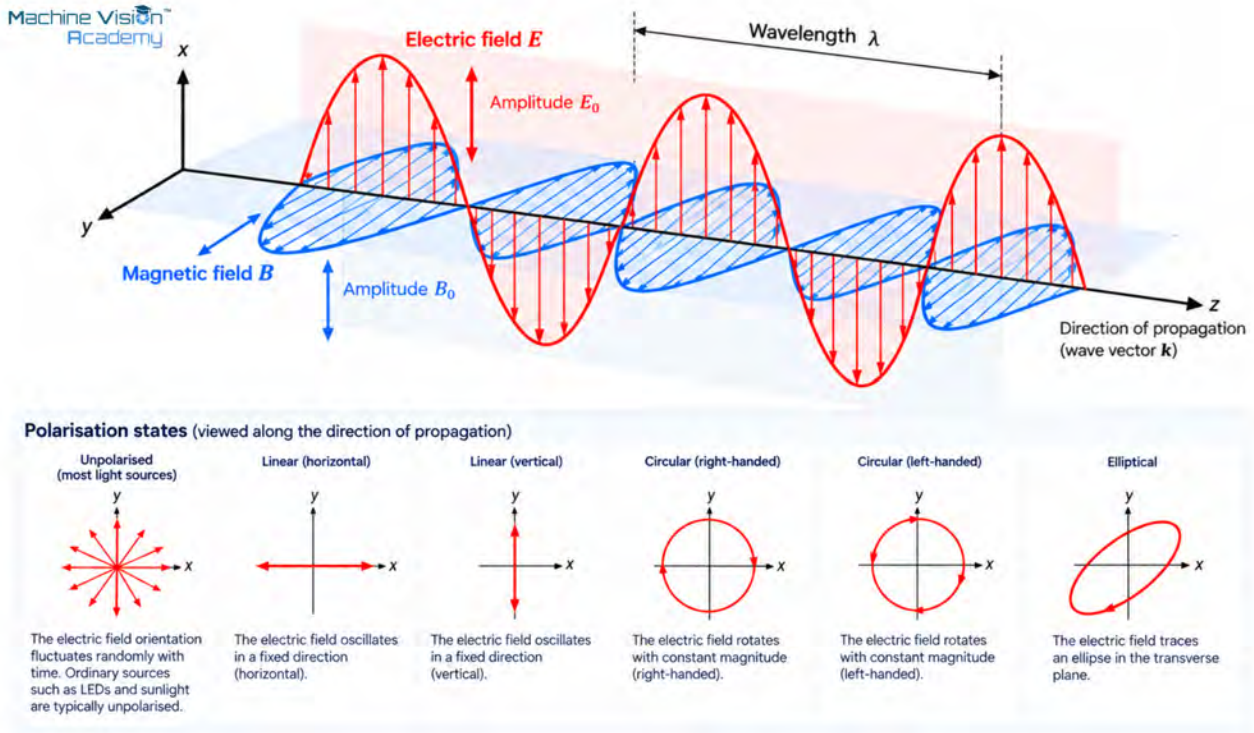


Figure 3.1.: Electromagnetic-wave representation. The online edition provides an interactive version; static formats show the corresponding print illustration.

ultimately results from a finite number of absorbed photons and the electrical charges they generate. This discrete nature of detection later becomes important when considering **shot noise**.

The same quantum viewpoint applies at the source. LEDs emit through electron-hole recombination, while lasers rely on stimulated emission. The machine-vision engineer normally does not need to model these processes from first principles; source spectra, radiometric output, stability and modulation behaviour are the useful engineering interfaces. The quantum description explains why those interfaces have the form they do, while classical wave and ray descriptions remain the efficient tools for designing how the emitted radiation reaches the part and camera.

For a deeper classical treatment of electromagnetic waves, interference, diffraction, polarisation and geometrical optics, Hecht is a useful companion text [154]. Readers who want to go beyond the engineering photon model into formal quantum optics can continue with Loudon [153].

The electromagnetic spectrum extends far beyond human vision. A particularly memorable demonstration came from William Herschel in 1800. While measuring heating across the colours produced by a prism, he also measured beyond the visible red end of the spectrum expecting to log a drop in temperature but found an even stronger heating effect where no light was visible to the eye, providing early experimental evidence of what is now called infrared radiation [155].

The boundaries between named spectral bands are conventions rather than physical discontinuities. Machine vision commonly uses ultraviolet (**UV**), visible (**VIS**), near infrared (**NIR**) and short-wave infrared (**SWIR**), with longer-wave infrared used for thermal imaging. Silicon sensors cover much of the visible spectrum and extend into the NIR; InGaAs detectors are widely used beyond the silicon cut-off in the SWIR. Exact useful ranges depend on sensor construction, optics, coatings, filters and the required signal-to-noise ratio rather than on the band name alone.

For machine vision, the useful spectrum is defined by the complete imaging chain. A light source may emit strongly at a wavelength that the object absorbs, the lens or protective window blocks, or the sensor detects poorly. Conversely, radiation invisible to the operator can produce a strong camera

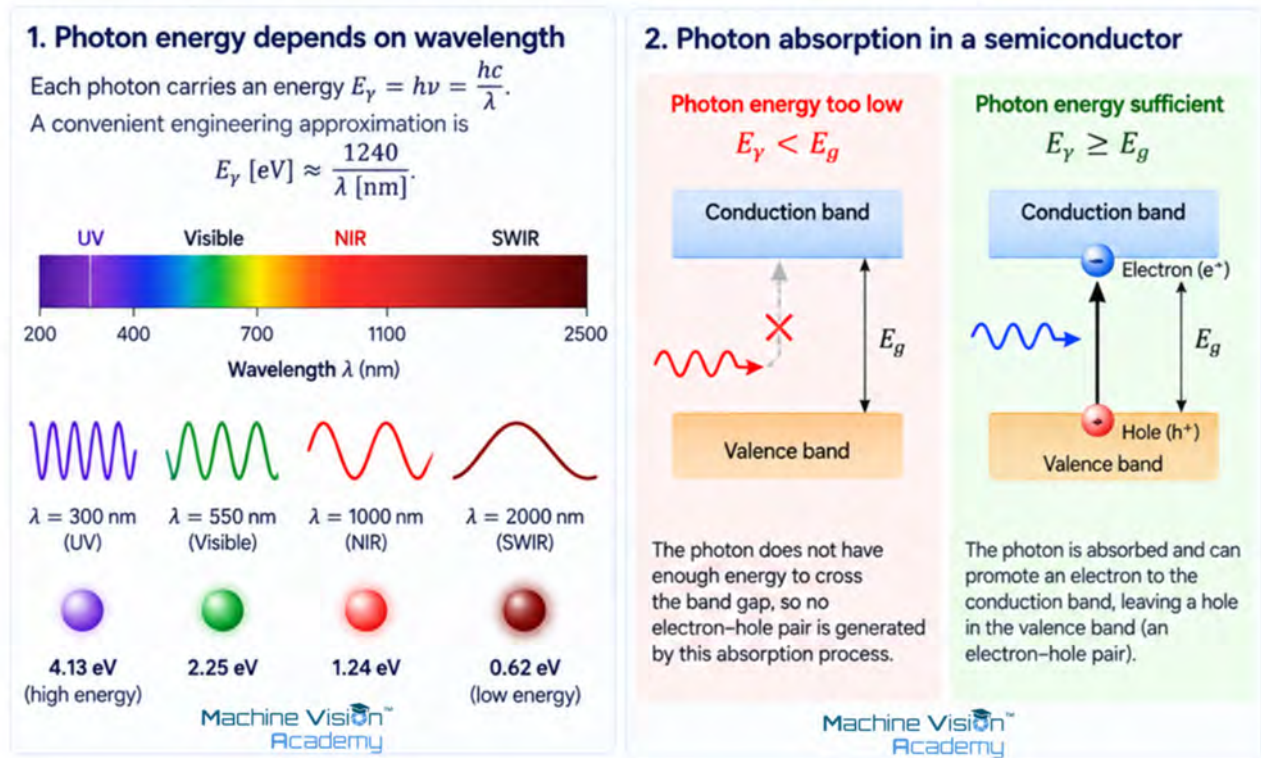


Figure 3.2.: Photon energy and semiconductor detection. The online edition provides an interactive version in which wavelength and detector material can be varied; static formats show the equivalent print illustration.

response. Wavelength should therefore be treated as a selectable engineering variable, not merely as the colour of a lamp.

! Machine vision does not see the same spectrum that you do

A visual check by eye is useful during setup, but it is not a measurement of what the camera receives. Always consider the source spectrum, object response, optical/filter transmission and sensor spectral sensitivity together.

- A bright-looking lamp can provide little signal in the wavelength band that matters.
- A near-infrared source can appear dark to a person while strongly exposing a silicon camera.
- Changing the camera, lens, protective window or filter can change contrast even when the light and part are unchanged.

Table 3.2.: Wavelength bands commonly encountered in industrial machine vision. Boundaries are approximate and application-dependent

Band	Approximate wavelength range	Typical machine-vision role	Common detector families
UV	~200–400 nm	Fluorescence excitation; coatings; surface and print contrast	UV-enhanced silicon; specialised UV detectors
Visible	~400–700 nm	General monochrome and colour imaging	Silicon CCD/CMOS

Band	Approximate wavelength range	Typical machine-vision role	Common detector families
NIR	~700–1100 nm	Contrast through some inks and coatings; material contrast outside normal human vision	Silicon CCD/CMOS
SWIR	~1000–1700 nm (common InGaAs range)	Moisture, fill level, semiconductor/solar inspection and material discrimination	InGaAs

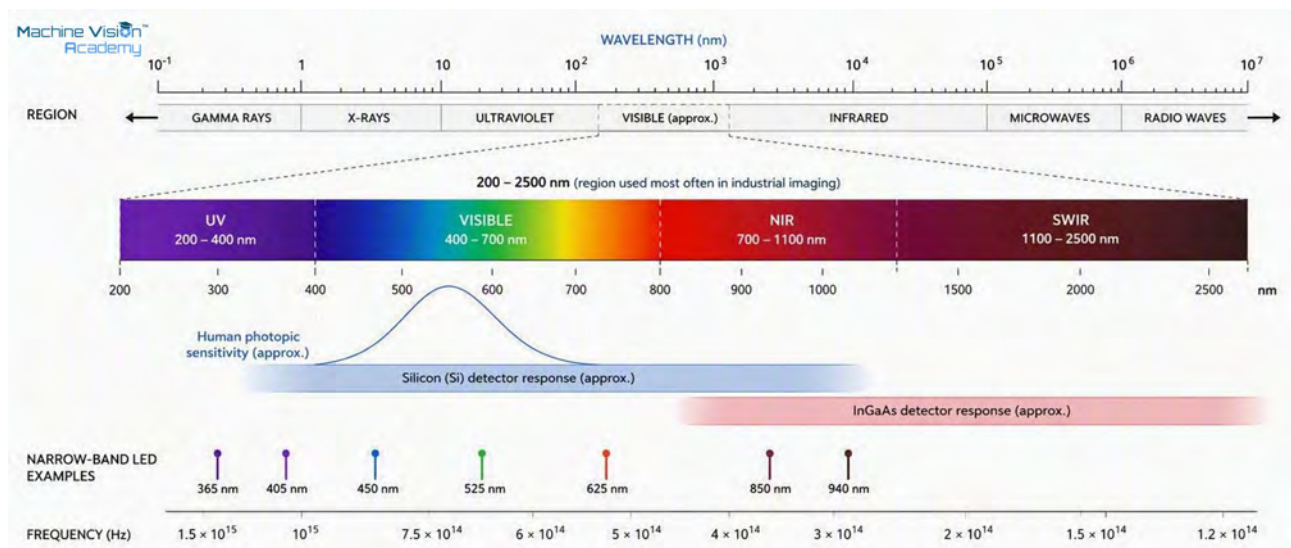


Figure 3.3.: Electromagnetic spectrum with machine-vision bands and representative detector materials.

Real light sources are rarely perfectly monochromatic. Instead of emitting all their optical power at one wavelength, they normally produce a **spectral distribution**: some wavelengths contain more optical power than others. Two sources can therefore have the same total radiant power but interact very differently with a material or camera because that power is distributed differently across the spectrum. For machine vision, the shape of the spectrum can be as important as the total amount of light.

Thermal emitters provide a useful example. Any object above absolute zero emits electromagnetic radiation as a consequence of its temperature. To describe this behaviour, physics uses the idealised **black body** as a reference. A perfect black body absorbs all electromagnetic radiation incident upon it, reflecting and transmitting none of it, and is also the maximum possible thermal emitter at a given temperature. Its emitted spectrum can therefore be described solely in terms of its temperature and wavelength, without the additional complication of radiation reflected from its surroundings. **Planck's law** describes how this thermal radiation is distributed across wavelength: as the temperature increases, the total emitted radiation increases and the distribution shifts towards shorter wavelengths. Real materials do not behave as perfect black bodies; their emission is modified by their wavelength-dependent **emissivity**, and they may also reflect radiation from their surroundings. These effects become particularly important in thermal imaging and are considered later in the handbook.

Planck's law therefore describes a thermal emitter as a distribution of radiation across wavelength, rather than as radiation emitted at a single wavelength. For a black body at any given temperature, this distribution has a wavelength at which its spectral radiance reaches a maximum. As the temperature changes, the position of this maximum also changes.

Wien's displacement law provides a simple way to calculate where that peak occurs:

$$\lambda_{\max} T = b,$$

or, more conveniently,

$$\lambda_{\max} = \frac{b}{T},$$

where:

- $\lambda_{\max}$ is the wavelength at which the black body's spectral radiance is greatest, measured in metres;
- T is the absolute temperature of the emitter, measured in kelvin (K);
- b is Wien's displacement constant,

$$b \approx 2.898 \times 10^{-3} \text{ m K}.$$

The inverse relationship is the important engineering point: **as an object becomes hotter, the peak of its thermal emission moves towards shorter wavelengths.**

For example, an object at approximately 300 K, close to room temperature, has a peak wavelength of

$$\lambda_{\max} \approx \frac{2.898 \times 10^{-3}}{300} \approx 9.7 \text{ } \mu\text{m}.$$

This is well into the infrared and helps explain something that is otherwise easy to overlook: **ordinary objects around us are continuously emitting electromagnetic radiation even though we cannot see it.** A long-wave infrared camera can detect this radiation because its detector operates in a spectral region inaccessible to human vision.

As temperature increases, the thermal spectrum not only becomes more intense but also shifts progressively towards shorter wavelengths. At sufficiently high temperatures, a significant part of the emission enters the visible spectrum and the object begins to glow. This is why a hot object may first appear dull red and then become progressively brighter as its temperature rises. The filament of an incandescent or halogen lamp operates at a sufficiently high temperature for its thermal spectrum to extend strongly into the visible region, although a substantial proportion of its radiation remains in the infrared.

For machine vision, Wien's law provides a useful connection between **object temperature, emitted spectrum and detector selection.** It helps explain why thermal cameras operate in infrared spectral bands, why incandescent and halogen sources produce substantial infrared radiation, and why visible appearance alone provides an incomplete description of a source's optical output. Planck's and Wien's laws are not normally calculations required when configuring a conventional visible-light inspection, but they provide the physical basis for understanding broadband thermal sources, colour temperature and thermal imaging developed later in the handbook.

For camera-oriented examples of how wavelength, illumination geometry and filtering affect machine-vision contrast, the Advanced Illumination practical guide and the CCS *Theory of Light and Color* series provide useful supplementary reading [156], [157]. They provide a practical complement to the more formal optical treatment in Hecht [154].

3.1.2. Radiometric quantities: radiant flux, irradiance and radiance

The previous subsection described optical energy at the level of individual photons. A photon of wavelength λ carries energy $E_\gamma = hc/\lambda$, and a camera signal ultimately results from a finite number of those photons being absorbed by the sensor. In most machine-vision applications, however, an enormous number of photons are emitted, reflected and detected during every exposure. Following each photon individually would be neither practical nor useful. Instead, we describe their **collective transport of energy** using **radiometry**.

The two descriptions are directly connected. If N is the **number of photons**, and the radiation is monochromatic so that every photon carries the same energy E_γ , their total radiant energy is

$$Q_e = NE_\gamma.$$

If those photons arrive over time, their **photon rate** is

$$\dot{N} = \frac{dN}{dt} \quad [\text{photons s}^{-1}].$$

The dot above N is important: N represents a number of photons, whereas $\dot{N}$ represents how many photons arrive per second. The corresponding radiant power, or **radiant flux**, is

$$\Phi_e = \dot{N}E_\gamma = \dot{N} \frac{hc}{\lambda}.$$

The units make the relationship particularly clear:

$$\underbrace{\text{photons s}^{-1}}_{\dot{N}} \times \underbrace{\text{J photon}^{-1}}_{E_\gamma} = \underbrace{\text{J s}^{-1}}_{\Phi_e} = \text{W}.$$

Equivalently,

$$\dot{N} = \frac{\Phi_e}{E_\gamma} = \frac{\Phi_e \lambda}{hc} \quad [\text{photons s}^{-1}].$$

💡 From photons to watts

Photon energy tells us how much energy each photon carries; radiometry tells us how much optical energy is being transported collectively.

Two monochromatic sources can therefore deliver the same radiant power while producing different numbers of photons per second. Longer-wavelength photons carry less energy individually, so more of them are required to transport the same optical power. This does not necessarily mean a larger camera signal: the sensor must also absorb those photons efficiently, and its quantum efficiency depends on wavelength.

Radiometry provides the practical language for following this optical energy through a machine-vision system. The terminology can initially appear excessive, but the quantities are easier to understand when treated as answers to different questions: **how much optical power is there, where is it travelling, over what area is it distributed, and how much energy eventually accumulates at the sensor?**

The starting quantity is **radiant energy**, Q_e , measured in joules. **Radiant flux**, Φ_e , often called **radiant power**, describes how quickly that energy is transferred:

$$Q_e \quad [\text{J}], \quad \Phi_e = \frac{dQ_e}{dt} \quad [\text{W}].$$

A radiant flux of 1 W therefore represents the transfer of one joule of optical energy per second. This is an **optical watt**, not the electrical power consumed by the illumination source. A LED consuming 10 W electrically, for example, will emit only part of that input power as optical radiation.

Power alone does not tell us where the radiation is going. Direction is described using a **solid angle**, the three-dimensional equivalent of an ordinary angle. For a small area dA viewed from distance r , where its normal forms an angle θ with the viewing direction,

$$d\Omega = \frac{dA \cos \theta}{r^2} \quad [\text{sr}].$$

A complete sphere contains 4π steradians. **Radiant intensity**, I_e , describes how radiant flux is distributed with direction:

$$I_e = \frac{d\Phi_e}{d\Omega} \quad [\text{W sr}^{-1}].$$

Once radiation reaches a surface, **irradiance**, E_e , describes the radiant power arriving per unit area:

$$E_e = \frac{d\Phi_e}{dA} \quad [\text{W m}^{-2}].$$

Irradiance is particularly useful in machine vision because it can describe how strongly an illuminator exposes a workpiece or how much optical power per unit area reaches the image sensor. The corresponding quantity for radiation **leaving** a surface is **radiant exitance**, M_e :

$$M_e = \frac{d\Phi_e}{dA} \quad [\text{W m}^{-2}].$$

The two therefore have the same units but describe opposite directions relative to the surface: **irradiance arrives; exitance leaves**.

A camera, however, does not collect all the radiation leaving a surface. It collects only the portion travelling within the directions accepted by its lens. For imaging, we therefore need a quantity that describes both where the radiation leaves the surface and in which direction it travels. This is **radiance**, L_e :

$$L_e = \frac{d^2\Phi_e}{dA \cos \theta d\Omega} \quad [\text{W m}^{-2} \text{ sr}^{-1}].$$

Radiance provides the natural connection between a scene patch and the optical system viewing it. An ideal passive optical system cannot arbitrarily increase radiance; more precisely, L_e/n^2 is conserved through lossless geometrical propagation when refractive-index changes are included. A lens can collect a larger portion of the available radiation and redistribute it onto the sensor, but it cannot create optical brightness from nothing.

Finally, the sensor accumulates optical energy throughout its exposure rather than responding to irradiance at only one instant. **Radiant exposure**, H_e , describes this accumulated energy per unit area:

$$H_e = \int_0^{t_{\text{exp}}} E_e(t) dt \quad [\text{J m}^{-2}].$$

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For constant irradiance,

$$H_e = E_e t_{\text{exp}}.$$

This closes the loop back to the photon description. At a fixed wavelength, greater irradiance means more photons arriving per unit area and unit time, while a longer exposure allows that photon flow to continue for longer. Provided the sensor remains within its linear operating range, either increases the number of photo-generated charge carriers and therefore the recorded signal. The sensor chapter develops this relationship further through quantum efficiency, shot noise, full-well capacity, gain and saturation.

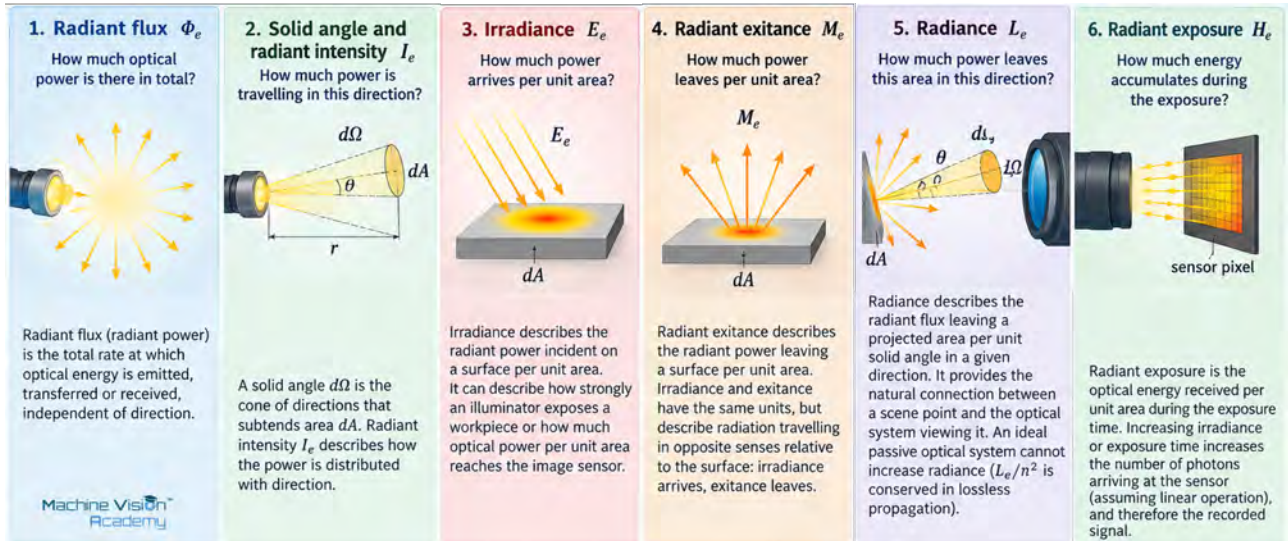


Figure 3.4.: Radiometric quantities through a machine-vision imaging chain. The online edition provides an interactive version in which each quantity can be selected and highlighted; static formats show the equivalent print illustration.

The quantities introduced above are summarised in Table 3.3. The important distinction is not simply their units, but what information has been added to the description of the optical energy: total power, direction, area, direction and area together, or accumulation over time.

Table 3.3.: Core radiometric quantities and their practical interpretation

Quantity	Symbol	SI unit	Practical interpretation
Radiant energy	Q_e	J	Light source - Total optical energy
Radiant flux	Φ_e	W	Light source - Optical power
Radiant intensity	I_e	W sr ⁻¹	Light source - Source power per direction
Irradiance	E_e	W m ⁻²	Object - Power arriving per unit area
Radiant exitance	M_e	W m ⁻²	Object - Power leaving an object per unit area
Radiance	L_e	W m ⁻² sr ⁻¹	Object - Power per projected area and direction

Quantity	Symbol	SI unit	Practical interpretation
Radiant exposure	H_e	J m^{-2}	Sensor - Energy accumulated per unit area during exposure

For broadband radiation, these quantities can also be expressed **spectrally**. For example, spectral irradiance $E_{e,\lambda}(\lambda)$ describes how irradiance is distributed with wavelength, and the total irradiance is recovered by integrating over the relevant spectral range:

$$E_e = \int E_{e,\lambda}(\lambda) d\lambda.$$

This spectral form becomes particularly important in machine vision because the source spectrum, material response, optical transmission and sensor quantum efficiency can all vary strongly with wavelength. At the sensor, the expected number of collected photoelectrons can be estimated from

$$N_e \approx A_p t_{\text{exp}} \int \frac{E_{e,\lambda}(\lambda) QE(\lambda) \lambda}{hc} d\lambda,$$

where A_p is the pixel area and $QE(\lambda)$ is the wavelength-dependent quantum efficiency. The factor $\lambda/(hc)$ converts radiant energy into photon number, reconnecting the radiometric description with the photon model introduced in Chapter 3. This relationship will be developed further when image sensors are considered.

A deeper way of understanding the conservation of radiance is through **etendue**, which describes how an optical bundle occupies both an area and a range of propagation directions. In simplified form,

$$G = n^2 A \Omega,$$

where A is the area occupied by the optical bundle, Ω is its solid angle and n is the refractive index of the medium. In an ideal lossless passive optical system, etendue cannot decrease. In practical terms, light collected from a large area or over a wide range of directions cannot be arbitrarily compressed into both a smaller area and a narrower angular range. This principle establishes fundamental limits on light collection and becomes relevant when considering lens aperture, numerical aperture, illumination coupling and optical throughput. For a deeper treatment of radiance and etendue, see Ohno [158].

The engineering rule

Passive optics can collect, redirect and redistribute available light, but they cannot create additional radiance.

A larger aperture may collect light over a wider range of angles, for example, but the lens has not made the original scene brighter.

For formal definitions of radiometric and photometric quantities, CIE 018 provides the internationally established framework [159]. The next subsection introduces **photometry**, which uses closely related quantities but deliberately weights optical radiation according to the spectral sensitivity of human vision.

3.1.3. Photometric quantities and the photopic response

The previous subsection introduced **radiometry** as an objective way of measuring optical energy and power. A watt of radiant power is a watt regardless of whether the radiation is ultraviolet, green, red or infrared. Human vision, however, does not respond equally to every wavelength. If equal radiant power is presented at different visible wavelengths, some will appear much brighter than others. Quantities intended to describe light as perceived by people therefore require a different system: **photometry**.

Photometry starts with the same physical radiation as radiometry but weights it according to the spectral sensitivity of human vision. Under normal light-adapted, or **photopic**, conditions this weighting is described by the standardised luminous-efficiency function $V(\lambda)$. As shown in Figure 3.5, human vision is most sensitive around the green-yellow region of the spectrum and becomes progressively less sensitive towards the blue/violet and red extremes.

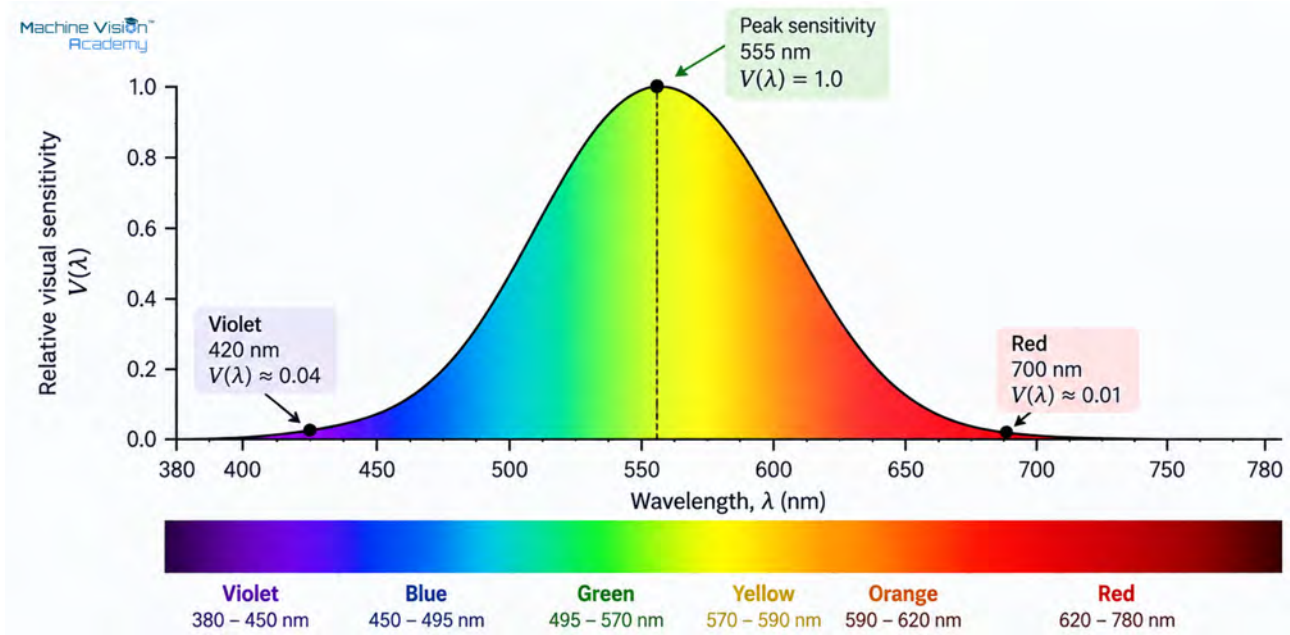


Figure 3.5.: Photopic response of the human eye. The CIE photopic sensitivity function $V(\lambda)$ represents the relative spectral sensitivity of human vision under well-lit conditions. For the same radiant power, wavelengths near 555 nm produce the greatest perceived brightness, while sensitivity decreases towards the blue/violet and red ends of the visible spectrum. The coloured wavelength scale provides an approximate correspondence between wavelength and perceived colour.

Photopic spectral sensitivity of the human eye. The response is normalised to its maximum near 555 nm and illustrates why equal radiant power at different wavelengths does not produce equal perceived brightness.

The relationship can be expressed mathematically by weighting the spectral radiometric quantity with $V(\lambda)$. In general,

$$X_v = K_m \int X_{e,\lambda}(\lambda) V(\lambda) d\lambda,$$

where $X_{e,\lambda}(\lambda)$ represents the radiometric quantity distributed across wavelength, $V(\lambda)$ represents the relative photopic sensitivity of the human eye, and K_m converts the weighted radiometric quantity into the corresponding photometric quantity. In the modern SI, the maximum photopic luminous efficacy is approximately

$$K_m = 683 \text{ lm W}^{-1},$$

associated with monochromatic radiation at a frequency of 540 THz, corresponding to approximately 555 nm [159].

The equation is useful mainly for understanding what photometric measurement means: **the spectrum is not treated equally**. Radiation near the maximum of $V(\lambda)$ contributes strongly to the photometric result, while wavelengths to which the eye is relatively insensitive contribute much less. Radiation outside the visible range contributes essentially nothing to conventional photometric quantities, even though it may carry substantial radiant power.

The radiometric quantities introduced previously consequently have human-vision-weighted photometric counterparts:

Table 3.4.: Radiometric quantities and their human-vision-weighted photometric counterparts.

Radiometric quantity	Photometric counterpart	Photometric unit
Radiant flux	Luminous flux	lumen (lm)
Radiant intensity	Luminous intensity	candela (cd)
Irradiance	Illuminance	lux ($\text{lx} = \text{lm m}^{-2}$)
Radiance	Luminance	cd m^{-2}
Radiant exposure	Luminous exposure	lx s

Of these terms, **illuminance in lux** is probably the one most frequently encountered by a machine-vision engineer because lux meters are readily available and lighting specifications are often expressed in lux. It is important, however, to understand what the instrument is actually measuring: a lux meter is designed to approximate the spectral sensitivity of human photopic vision, **not the spectral sensitivity of the camera**.

A machine-vision camera has its own wavelength-dependent response, determined by the detector material, sensor construction, colour filters, optical filters and other components in the imaging chain. Its response can therefore be very different from $V(\lambda)$. An 850 nm near-infrared illuminator, for example, can produce a strong response from a suitable silicon camera while contributing essentially nothing to a lux measurement. Conversely, two white illuminators can produce similar lux values but different camera signals if their spectral distributions differ.

 Lux is not a camera measurement

Photometric quantities describe light for human vision, not for the camera.

Lux can be useful for documenting ambient lighting or comparing visible illumination under controlled conditions, but it should not normally be treated as a direct measure of the optical signal available to a machine-vision sensor.

For camera performance, the relevant quantities are the **spectral output of the source**, the **spectral behaviour of the object and optics**, and the **spectral sensitivity or quantum efficiency of the sensor**.

It is also useful to keep three different uses of the word *power* separate. **Electrical power**, measured in watts, describes what an illuminator consumes. **Radiant power**, also measured in watts, describes the optical energy it emits per second. **Photometric quantities**, such as lumens and lux, describe that optical radiation after weighting it for human visual sensitivity. None can be converted reliably into another without additional information about source efficiency and spectral distribution.

This distinction is particularly important as machine vision moves outside the visible spectrum. UV, NIR and SWIR illumination may be extremely useful to a camera while being weakly perceived or

completely invisible to the operator. Photometry therefore remains important for understanding lighting specifications and human working environments, but **radiometric and spectral quantities are generally the more appropriate language when analysing the signal available to a machine-vision camera.**

For formal definitions of photopic, scotopic and other standard luminous-efficiency functions, see CIE 018 [159]. The CCS *Theory of Light and Color* series provides an accessible machine-vision-oriented comparison of human spectral sensitivity and camera spectral response [157].

3.1.4. The inverse-square law and cosine law

The radiometry subsection introduced **irradiance**, E_e , as radiant power arriving per unit area. For machine vision, knowing the total power of an illuminator is therefore not enough: the irradiance reaching a part also depends on **where the light is positioned and at what angle the surface receives it.** Two simple geometric relationships—the inverse-square law and the cosine law—provide useful first-order models of these effects.

Consider first an ideal **point source** emitting with radiant intensity I_e in the direction of the workpiece. As the distance from the source increases, the same emitted power is distributed over an increasingly large area. Because the area subtended by a fixed solid angle grows in proportion to r^2 , irradiance on a surface perpendicular to the illumination direction decreases as

$$E_e = \frac{I_e}{r^2}.$$

The relationship is inverse-square rather than simply inverse with distance. If all other conditions remain unchanged, doubling the source-to-surface distance gives

$$E_e(2r) = \frac{I_e}{(2r)^2} = \frac{1}{4}E_e(r).$$

Tripling the distance reduces the irradiance to one ninth. This provides useful engineering intuition: **small changes in working distance can produce much larger changes in illumination than the distance change alone might suggest.**

Distance is only part of the geometry. If the receiving surface is tilted by an incidence angle θ relative to the incoming radiation, the same optical power is distributed over a larger effective surface area. The resulting irradiance becomes

$$E_e = \frac{I_e \cos \theta}{r^2}.$$

At normal incidence, $\theta = 0^\circ$ and $\cos \theta = 1$, so the irradiance is maximum. As the surface tilts away from the illumination direction, the projected area decreases and so does the irradiance:

$$\begin{aligned}\theta = 0^\circ &\Rightarrow \cos \theta = 1.00, \\ \theta = 30^\circ &\Rightarrow \cos \theta \approx 0.87, \\ \theta = 60^\circ &\Rightarrow \cos \theta = 0.50.\end{aligned}$$

This is directly relevant to machine vision. A light illuminating a three-dimensional component does not necessarily expose every surface equally, even when the material is identical. Features farther from the source receive less irradiance, while surfaces tilted relative to the illumination receive less again. Changes in part height, orientation or presentation can therefore become changes in image intensity before material reflectance or camera processing are considered.

The distance and incidence effects can be combined into the useful first-order relation

$$E_e(r, \theta) = \frac{I_e \cos \theta}{r^2}$$

for a point source illuminating a planar receiving surface.

This equation is valuable for developing intuition, but its assumptions matter. Most machine-vision illuminators are **not point sources**. Ring lights, bar lights, area lights, backlights and domes have significant physical dimensions, and machine-vision applications frequently operate close enough that different parts of the source subtend substantially different angles and distances at the workpiece. In this **near-field** condition, irradiance does not necessarily follow a simple $1/r^2$ relationship. Collimated illuminators are different again because they are deliberately designed to restrict angular spread.

The inverse-square law should therefore be treated as a **first-order engineering model**, not as a universal rule for positioning industrial illumination. It is particularly useful for understanding trends and for sources that are small relative to their working distance. For close extended-source arrangements, irradiance and uniformity are better determined from manufacturer data, measurement or an appropriate optical model.

A second cosine relationship appears when considering the radiation **leaving** a diffuse surface. An ideal **Lambertian surface** has constant radiance with viewing direction, but its radiant intensity varies with angle according to

$$I_e(\theta) = I_e(0) \cos \theta.$$

Although the mathematics looks similar, this describes a different physical situation. The **incidence cosine law** describes how much radiation a tilted surface **receives**; Lambert's cosine law describes how an ideal diffuse surface distributes radiation **leaving** it.

Table 3.5.: Three geometric relationships that can appear similar mathematically but describe different effects

Relationship	What changes?	Physical interpretation
$E_e \propto 1/r^2$	Source-to-surface distance	Optical power spreads over increasing area
$E_e \propto \cos \theta$	Incidence angle	Incoming power is distributed over a larger receiving area
$I_e(\theta) \propto \cos \theta$	Observation/emission angle from a Lambertian surface	Emitted or reflected intensity decreases with angle while radiance remains constant

These relationships provide the theoretical basis for several practical illumination effects developed later in the handbook. They help explain why changing working distance affects illumination uniformity, why curved or tilted surfaces develop intensity gradients, and why the position and angular extent of a light can matter as much as its nominal optical power.

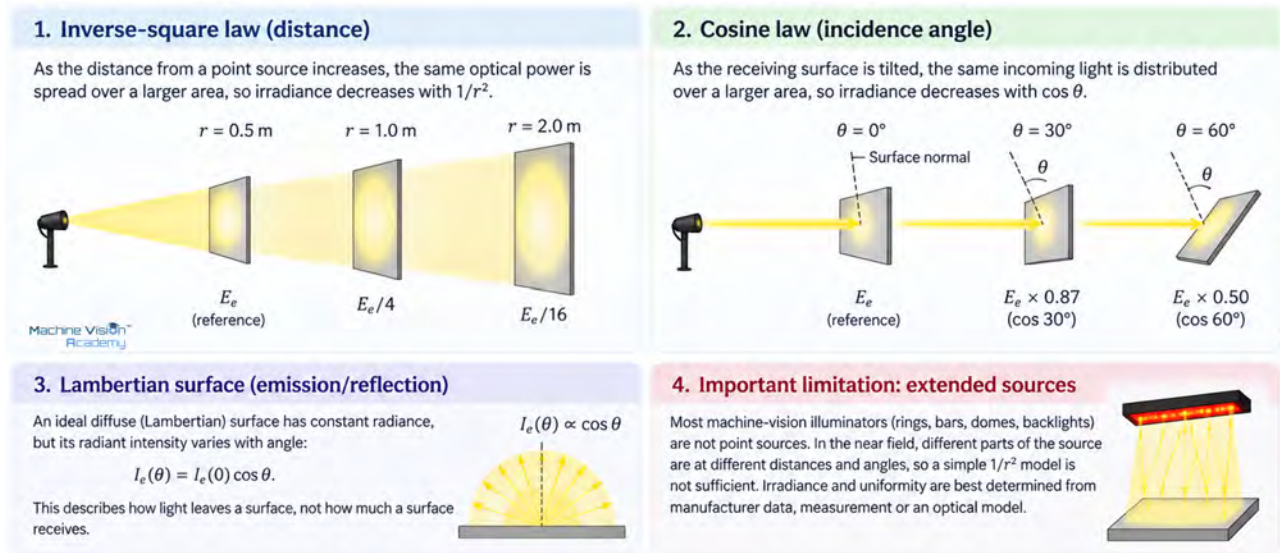


Figure 3.6.: Effect of distance and incidence angle on irradiance. The online edition provides an interactive version in which source-to-surface distance and incidence angle can be varied; static formats show the equivalent print illustration, together with the Lambertian and extended-source qualifications.

3.2. Interaction of light with matter

The previous section described light as electromagnetic radiation and introduced the quantities used to measure it. A machine-vision camera, however, never receives light in isolation: before reaching the sensor, that radiation has interacted with the part, its surface, the surrounding medium and often several optical interfaces. Those interactions are what create the contrast on which inspection depends.

At a material boundary or within a material, incident radiation can be reflected, transmitted, absorbed, scattered or re-emitted at another wavelength. Which mechanism dominates depends on wavelength, surface geometry, refractive index, roughness, thickness, internal structure and polarisation. For a vision engineer, the practical question is therefore not simply *how much light is available?* but *what does the feature do to that light differently from its background?*

$$R(\lambda) + T(\lambda) + A(\lambda) = 1,$$

where R , T and A denote spectral reflectance, transmittance and absorptance for the specified geometry. This compact energy balance is an excellent starting point, but the directional distribution of reflected and transmitted radiation is often just as important as the totals.

For machine vision, the objective is not simply to make the image bright. A useful illumination condition should increase the optical difference associated with the feature of interest, suppress contrast that is irrelevant to the inspection, and preserve that separation across expected production variation. This practical view complements the physics developed in this section: first identify how the feature changes the light, then choose geometry, wavelength, filtering or another optical lever that makes that change visible and repeatable.

i Practical companion: from physics to lighting trials

The NI *Practical Guide to Machine Vision Lighting* is a useful companion to this section because it approaches the same principles from an application perspective. It frames a robust lighting solution around three goals: **maximise contrast on the feature of interest, minimise irrelevant contrast, and maintain robustness to expected variation**. It also organises practical lighting decisions around geometry, projected structure, wavelength and filtering [160]. The detailed selection of lighting hardware is developed later in the illumination chapter; the guide is particularly useful here for seeing how the physical mechanisms in this section appear in real inspection images.

3.2.1. Specular and diffuse reflection; the BRDF

When light reflects from a surface, it can be useful to think of the reflected radiation as containing two broad components: **specular reflection**, in which light remains strongly concentrated around a particular direction, and **diffuse reflection**, in which the reflected energy is distributed over a much wider range of directions. The balance between them depends on properties such as surface roughness, microstructure, material composition and wavelength. Real industrial surfaces normally contain some combination of both: a painted panel that appears matte can still produce a specular highlight, while a polished metal surface can scatter measurable light because of scratches, contamination or microscopic roughness.

The direction of the specular component follows a simple geometrical rule. At the point where light reaches the surface, an imaginary line perpendicular to the surface is called the surface normal. The angle between the incoming ray and this normal is the angle of incidence, θ_i , while the angle between the normal and the specularly reflected ray is the angle of reflection, θ_r . The law of reflection states that these angles are equal:

$$\theta_r = \theta_i.$$

This law does not imply that a real surface reflects all of its energy along one perfectly defined ray. Rather, it identifies the direction around which the specular component is centred. A very smooth surface can concentrate much of its reflected energy close to this direction. As roughness and local slope variation increase, the reflected energy tends to spread over a broader range of directions. In some materials, light can also enter the surface and undergo subsurface scattering before emerging again, contributing further to the diffuse component.

This distinction is immediately relevant to machine vision. A camera does not receive all the light reflected by an object; its lens collects only a limited range of outgoing directions. A smooth surface can therefore appear very bright when its specular reflection is directed towards the camera and almost black when the same reflection is directed elsewhere. A scratch or defect can reverse that behaviour by changing the local surface angle or scattering light into directions that differ from those of the nominal surface.

The law of reflection tells us where the ideal specular direction lies, but it does not describe how much reflected energy travels in that direction or how the remaining energy is distributed. A more general description is provided by the **bidirectional reflectance distribution function (BRDF)**. The term *bidirectional* reflects the fact that surface appearance depends on two directions: where the illumination comes from and where the surface is observed.

A single angle is not enough to describe the direction of a ray in three-dimensional space. Imagine a light positioned above a flat workpiece. First, we need to know how far the light is tilted away from the surface normal; this is described by θ . But many different positions can have the same tilt: the light could be to the left, right, front or back of the part while remaining at the same angle from the

3. Light, colour and radiometry

normal. We therefore need a second angle, ϕ , called the azimuth angle, which describes the direction around the surface.

The incident illumination direction is therefore represented by (θ_i, ϕ_i) , while the direction in which the reflected light is observed is represented by (θ_r, ϕ_r) . Together, these four angles allow the BRDF to describe how much light arriving from one particular direction is reflected towards another particular direction, measured in inverse steradians, where the steradian (sr) is the SI unit of solid angle—the three-dimensional counterpart of the radian:

$$f_r(\theta_i, \phi_i; \theta_r, \phi_r) = \frac{dL_r(\theta_r, \phi_r)}{dE_i(\theta_i, \phi_i)} \quad [\text{sr}^{-1}].$$

The BRDF should therefore not be interpreted simply as *how reflective* a material is. It describes where the reflected light goes for a particular combination of illumination and observation directions.

It is useful to consider two idealised extremes to understand what the BRDF represents. At one extreme is a perfectly **diffuse, or Lambertian, surface**. It redistributes reflected light broadly rather than concentrating it in one particular viewing direction. Its BRDF is constant:

$$f_r = \frac{\rho}{\pi},$$

where ρ is the diffuse reflectance. A constant BRDF means that, for a given illumination condition, the surface radiance is independent of observation direction. This does not mean that equal optical power is emitted into every direction: the projected-area cosine term still governs the power distributed over the hemisphere.

At the opposite extreme is an ideal **mirror-like, or specular, surface**. Instead of distributing reflected light broadly, it concentrates the reflection into the specific direction predicted by the law of reflection, $\theta_r = \theta_i$, with the corresponding azimuthal geometry. Its BRDF is therefore not constant: it is effectively concentrated into a very narrow peak around the specular direction. A perfectly ideal mirror requires a more specialised mathematical description, which is not necessary for the machine-vision treatment here.

Real industrial surfaces normally lie somewhere between these extremes. Their BRDF may contain a concentrated **specular component** together with a broader **diffuse component**, and defects such as scratches, roughness or contamination can change either component. This change in directional reflection is precisely what many machine-vision illumination techniques exploit to create contrast between features of interest and backgrounds.

Surface roughness provides an intuitive transition between strongly specular and more broadly distributed reflection. When a surface is very smooth relative to the wavelength of the illumination and its local slopes vary little, much of the reflected energy remains concentrated close to the specular direction. As microscopic height and slope variations increase, different parts of the surface redirect light in slightly different directions and the reflected energy becomes distributed over a wider angular range.

This relationship should be treated as a physical principle rather than as a simple conversion from one roughness number to one reflection pattern. The angular distribution also depends on the spatial statistics and structure of the surface, the wavelength, material properties and illumination geometry. Nevertheless, the central machine-vision idea is straightforward: changes in surface microgeometry change where the reflected light goes.

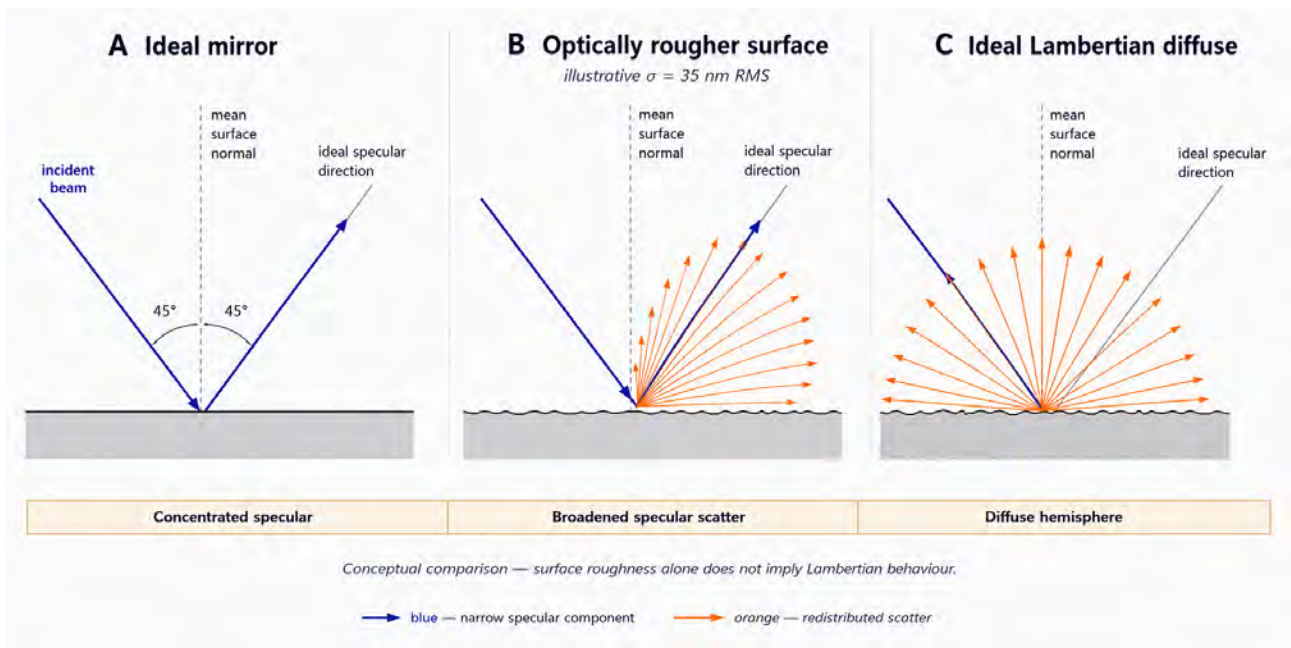


Figure 3.7.: Surface roughness and directional reflection. The online edition provides an interactive illustration of how increasing RMS roughness redistributes optical energy from the narrow specular component into angle-resolved scatter. Static formats compare an ideal mirror, an illustrative optically rougher surface and an ideal Lambertian diffuse surface. The comparison is conceptual; increasing surface roughness does not by itself imply that a real surface becomes Lambertian.

i From machine vision to surface metrology

The same principle can be used quantitatively. The **National Institute of Standards and Technology (NIST)** studies angle-resolved and polarised optical scattering to characterise surfaces and distinguish scattering associated with microroughness, particles, residues and subsurface defects [161].

This provides a useful perspective for machine vision: scattered light is not necessarily unwanted noise. Its angular and polarisation characteristics can themselves contain information about the condition of a surface.

The reflection model developed above can now be connected to some familiar machine-vision images. The purpose here is not to select an illumination technique—that is developed in Chapter 5—but to understand why changing illumination geometry can change the appearance of the same physical feature so dramatically.

Consider a relatively smooth, reflective metal surface containing an engraved mark, a scratch or another local change in surface geometry. To build intuition, we can approximate the nominal surface as strongly specular and the feature as a region that redirects or scatters light over a broader range of directions. This is deliberately simplified—a real scratch or engraving is not necessarily Lambertian—but it makes the consequence of directional reflection easy to see.

Assume that the camera views the surface approximately along its normal. If illumination arrives from a direction close to the camera axis, and therefore at a relatively small angle to the surface normal, the law of reflection places much of the strong return from the smooth background close to the camera direction. Because the lens accepts a finite cone of outgoing directions rather than a single ray, a substantial part of this reflection can enter the camera and the nominal surface appears bright.

An engraved line, scratch, tapered hole or other local change in surface slope behaves differently. Its local surface normals and microstructure redirect part of the illumination away from the direction

followed by the nominal surface. Against a bright specular background, the feature can therefore produce a weaker signal and appear darker. This type of image geometry is commonly described as **bright field**.

Now keep the camera and part fixed but move the illumination towards a more oblique angle. The law of reflection moves the strong specular return from the smooth nominal surface correspondingly farther away from the camera. Once most of that return falls outside the angular range accepted by the lens, the background becomes dark. The engraved or damaged region still redirects light over additional directions, some of which can reach the camera. The same physical feature can consequently become bright against a dark background. This is the physical basis of **dark field**.

The important point at this stage is not which arrangement should be selected. It is that **the object has not changed**: the large change in image contrast results from changing which part of the surface's directional reflection reaches the camera.

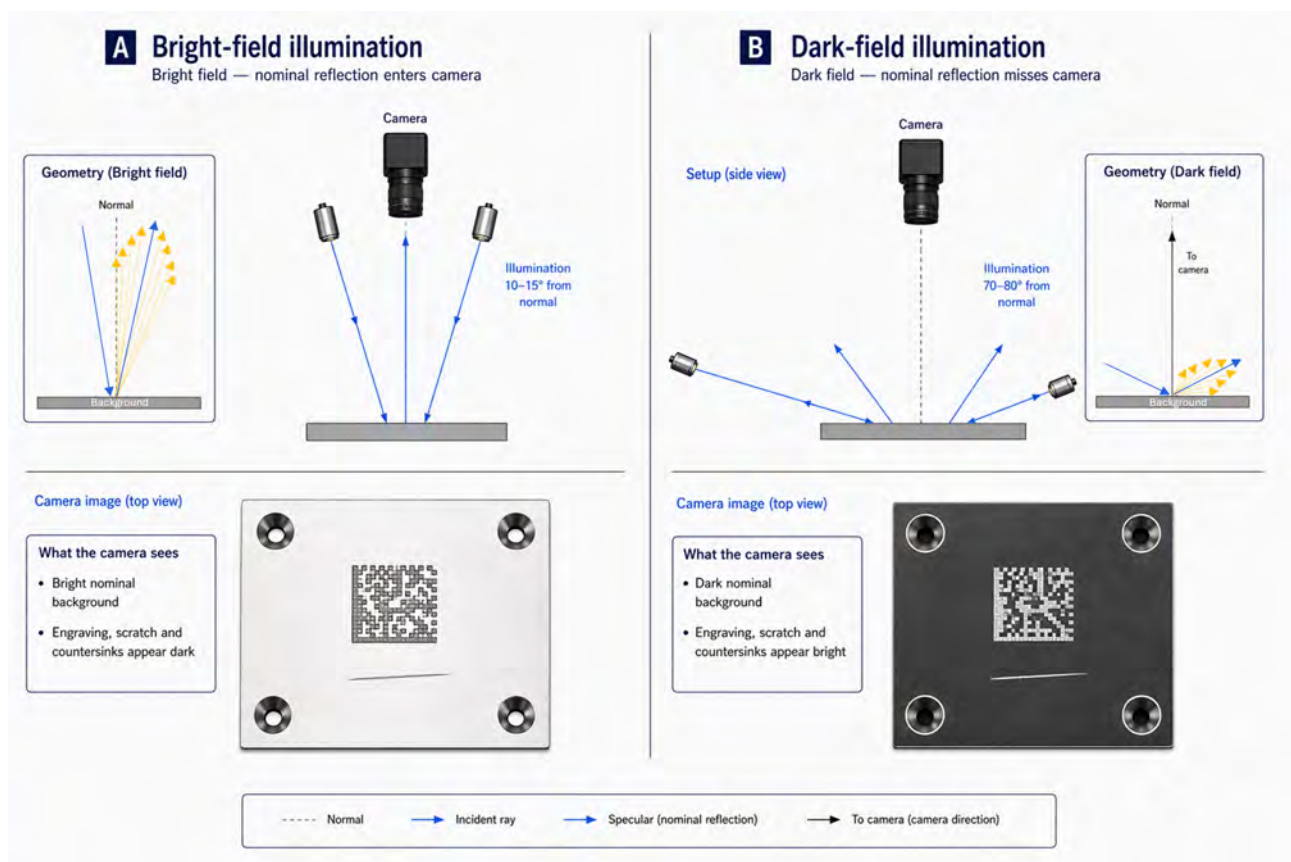


Figure 3.8.: Bright-field and dark-field illumination applied to the same reflective component. The online edition allows the illumination angle to be varied while keeping the camera and part fixed, showing how directional reflection alone can reverse feature contrast. Static formats show representative bright-field and dark-field configurations side by side. The transition is conceptual rather than a fixed angular threshold; the actual response depends on the surface BRDF, illumination geometry, source angular extent and camera acceptance angle.

Figure 3.8 provides a practical interpretation of the BRDF developed above. A scratch, engraving, edge or local slope change modifies the directional distribution of reflected light. Depending on illumination and observation geometry, that modification can either remove light from the camera or redirect light into it. This is why apparently small changes in lighting position can produce large changes in the image of a reflective component.

The same theory also explains a different machine-vision problem. Sometimes directional surface behaviour is not the feature of interest at all. Consider printed information on a glossy, wrinkled foil package. Every fold and wrinkle introduces a different local surface normal. Under illumination

arriving from a narrow range of directions, some regions satisfy the specular condition and produce strong highlights, while neighbouring regions direct their reflection away from the camera and become dark. The resulting brightness variation can be larger than the contrast of the print that the inspection is intended to observe.

This introduces an important distinction between **diffuse reflection** and **diffuse illumination**. Diffuse reflection, introduced earlier, describes what a surface does to incident light: surface structure or subsurface interactions distribute reflected radiation over a broad range of outgoing directions. Diffuse illumination instead describes the angular distribution of the *incoming* light. Rather than illuminating the part from one narrow direction, a diffuse source presents it with light arriving from many directions.

A dome provides a useful conceptual example. Placing a glossy object inside diffuse illumination does not make its surface Lambertian; locally, the surface can remain strongly specular. Instead, the broad range of incident directions means that different local surface normals can each receive illumination from a direction capable of reflecting towards the camera. Orientation-dependent bright and dark regions are therefore reduced and the appearance of a curved, wrinkled or glossy surface becomes more uniform.

The underlying print or material contrast has not been created by this process. In the packaging example, differences between ink and foil still originate from properties such as reflectance and absorption. Diffuse illumination has primarily reduced the image variation caused by geometry, allowing those other differences to become easier to observe.

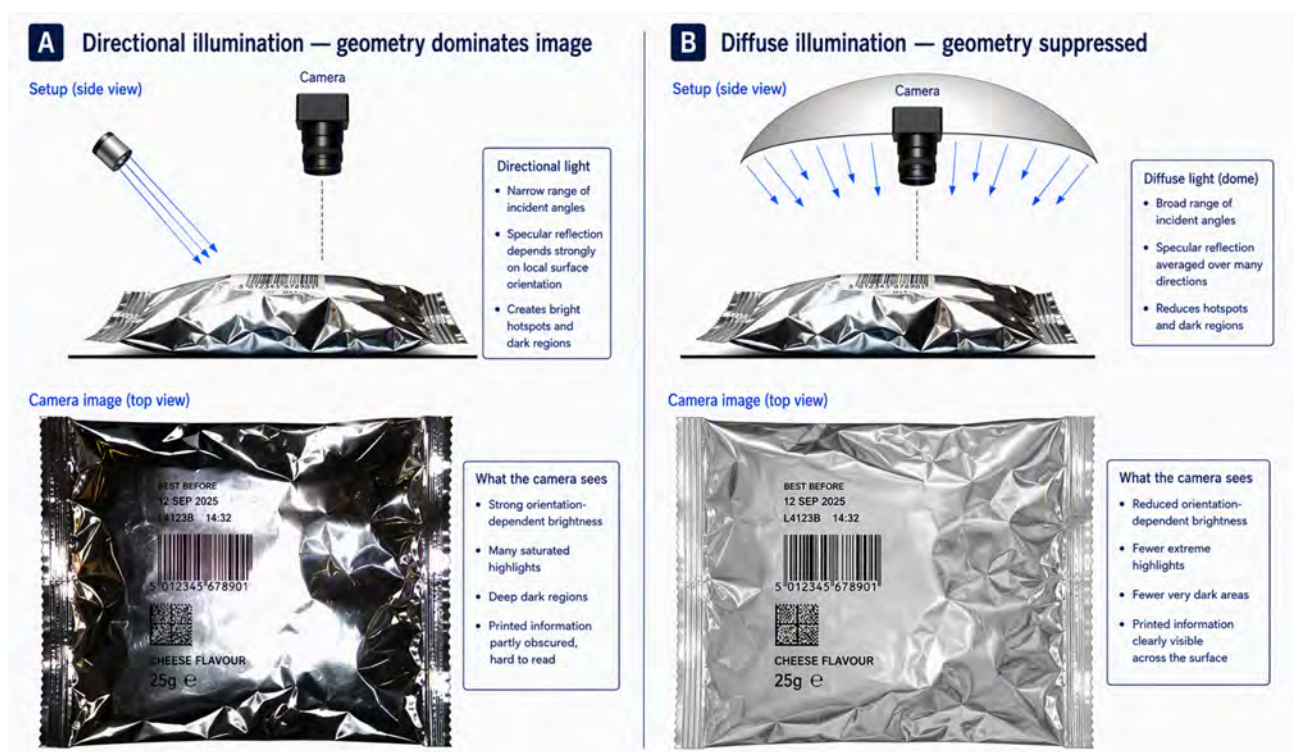


Figure 3.9.: Directional and diffuse illumination applied to printed information on a glossy wrinkled package. Directional illumination produces strong orientation-dependent highlights and dark regions that can obscure the print. Broad diffuse illumination reduces this geometrical variation, producing a more uniform background from which the printed feature can be distinguished.

Figure 3.9 therefore illustrates a different consequence of the same reflection theory. In Figure 3.8, directional reflection is responsible for the useful contrast. In Figure 3.9, sensitivity to local reflection geometry produces unwanted image variation and broad angular illumination reduces its influence.

The theoretical connection

From the viewpoint of the BRDF, these examples differ mainly in how the surface is illuminated and which outgoing directions are observed:

- **bright-field and dark-field examples** show how particular incident and observation directions can make directional surface differences highly visible;
- **diffuse illumination** shows what happens when the surface is exposed to a broad range of incident directions, reducing dependence on any single illumination direction.

These principles explain the images; the practical selection, design and validation of illumination techniques are developed in Chapter 5.

The practical lesson from this subsection is therefore deliberately limited. Surface appearance is not determined by reflectance alone: **illumination direction, observation direction and the directional scattering properties of the surface jointly determine the optical signal that reaches the camera**. Bright-field, dark-field and diffuse images are useful demonstrations of that principle, but they are not yet a lighting-selection framework.

For the formal definition and nomenclature of directional reflectance and BRDF quantities, see the NBS/NIST monograph by Nicodemus et al. [162]. Advanced Illumination [156], CCS [157] and the NI practical guide [160] provide application-oriented examples showing how these physical effects appear in real machine-vision images. Systematic illumination selection and hardware implementation are deferred to Chapter 5.

3.2.2. Transmission, refraction and Snell's law

So far, reflection has been treated mainly as light returning from a surface. Many machine-vision applications, however, involve components through which at least some of the illumination passes. Glass vials, transparent polymer packaging, protective windows, lenses and thin medical components such as catheter balloons are common examples. In these cases, the image depends not only on what is reflected at the first surface, but also on what happens to the light that enters the material, travels through it and eventually emerges again.

For the phenomena considered in this subsection, the most useful description is primarily the **ray model of light** introduced earlier. We will follow individual rays as they encounter material boundaries and ask how their direction changes. The underlying electromagnetic-wave description remains important: refractive index describes a change in wave propagation through the material, and it is ultimately this change that produces refraction. For most machine-vision geometry, however, representing that behaviour as a change in ray direction provides the clearest engineering picture.

When light reaches the boundary between two transparent materials, some of the incident optical power can be reflected while the remainder enters the second medium. If the second medium has a different refractive index, the transmitted light can also change direction. This change of direction is called **refraction**.

The refractive index, n , describes how light propagates through a material relative to vacuum. For a transparent medium,

$$n = \frac{c}{v},$$

where:

- n is the refractive index of the material and is dimensionless;
- c is the speed of light in vacuum, approximately $2.998 \times 10^8 \text{ m s}^{-1}$;

- v is the phase velocity of light in the material, measured in m s^{-1} .

A larger refractive index therefore corresponds to a lower phase velocity in the material. Air has a refractive index close to 1 under ordinary conditions, while common glasses and transparent polymers typically have larger values.

When light crosses an interface at an angle, the change in propagation speed results in a change in direction. The relationship is described by **Snell's law**:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2,$$

where:

- n_1 is the refractive index of the medium containing the incident light;
- n_2 is the refractive index of the medium into which the light is transmitted;
- θ_1 is the **angle of incidence**, measured between the incident ray and the surface normal;
- θ_2 is the **angle of refraction**, measured between the transmitted ray and the surface normal.

As with the law of reflection, both angles in Snell's law are measured from the **surface normal**, not from the surface itself. This makes the direction of refraction easier to interpret.

When light passes from a lower refractive index to a higher one, such as from air into glass, the transmitted ray bends **towards the normal**. When it passes in the opposite direction, from a higher refractive index to a lower one, such as from glass back into air, the transmitted ray bends **away from the normal**.

There is one simple limiting case worth making explicit. If the incoming ray travels exactly along the surface normal, it strikes the interface **perpendicularly**. Its incidence angle is then

$$\theta_1 = 0^\circ.$$

Snell's law gives

$$n_1 \sin(0^\circ) = n_2 \sin \theta_2.$$

Because $\sin(0^\circ) = 0$, the refracted angle must also be

$$\theta_2 = 0^\circ.$$

The ray therefore continues straight through the interface without changing direction. Refraction still occurs in the sense that the propagation speed and wavelength change inside the new material, but there is **no angular bending**. This is why a flat protective window viewed close to perpendicular to its surface can have relatively little geometrical effect even though its refractive index is substantially different from that of air.

For an oblique ray, the bending becomes visible. Consider light travelling from air, with $n_1 \approx 1.00$, into glass with $n_2 \approx 1.50$ at an incidence angle of 30° . Snell's law gives

$$1.00 \sin(30^\circ) = 1.50 \sin \theta_2,$$

and therefore

$$\theta_2 = \arcsin\left(\frac{1.00}{1.50} \sin 30^\circ\right) \approx 19.5^\circ.$$

3. Light, colour and radiometry

The ray has therefore bent towards the normal: its direction changes from 30° from the normal in air to approximately 19.5° inside the glass.

Now consider what happens when the ray later leaves the glass and returns to air through a second surface that is **parallel to the first**. The refractive-index change is reversed: the ray now travels from $n \approx 1.50$ back to $n \approx 1.00$, so it bends **away from the normal**. Applying Snell's law at the second interface restores the original external angle. In this example, the ray emerges back into air at 30° .

For a parallel-sided transparent plate surrounded by air, the complete sequence is therefore

$$30^\circ \xrightarrow{\text{air} \rightarrow \text{glass}} 19.5^\circ \xrightarrow{\text{glass} \rightarrow \text{air}} 30^\circ.$$

The two interfaces restore the original propagation angle, but they do not make the ray return to its original *path*. While travelling through the glass, the ray follows a different direction from the one it would have followed in air. When it emerges, it is therefore **parallel to the original ray but laterally displaced**.

A familiar example of this effect can be seen when looking through spectacles. Near the edge of a lens, features of the face can appear shifted relative to the same features viewed directly beside the lens. The effect is particularly easy to notice where the outline of the face appears discontinuous at the lens boundary, as illustrated in Figure 3.10. A spectacle lens is not a simple parallel-sided plate—its curved surfaces are deliberately designed to refract light and correct vision—but the visible displacement provides an intuitive example of how a transparent optical element can change the apparent position of what lies behind it.

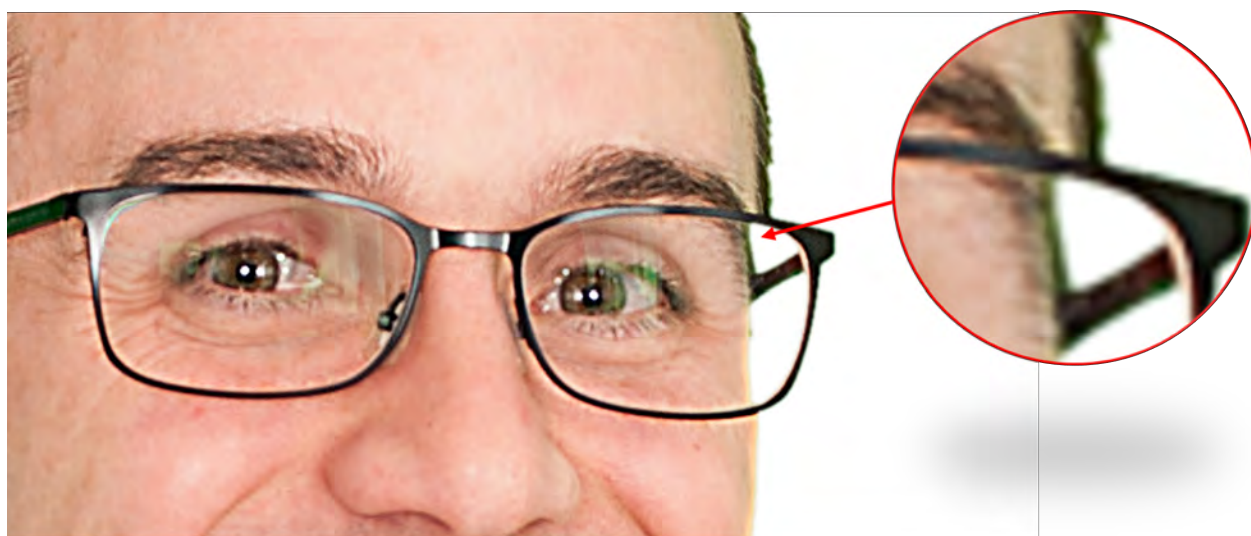


Figure 3.10.: A familiar visual consequence of refraction. Near the edge of the spectacle lens, the apparent position of the face is displaced relative to the portion viewed directly outside the lens. The enlarged region highlights the discontinuity at the lens boundary.

The spectacle example contains curved optical surfaces and is therefore more complicated than the simple geometry described by Snell's law above. To isolate the basic effect, Figure 3.11 considers instead a flat, parallel-sided transparent slab. This removes lens curvature and allows refractive index and material thickness to be varied independently.

The figure makes two effects easier to separate. Changing refractive index changes the angle followed by the ray inside the material: a higher refractive index bends the ray farther towards the normal. Changing slab thickness does not change that refraction angle; it changes how far the ray travels at that angle before reaching the second interface and therefore changes the resulting lateral displacement. In every case, because the entrance and exit surfaces are parallel and the surrounding medium is the

same on both sides, the emerging ray returns to the original external angle and remains parallel to the incident ray.

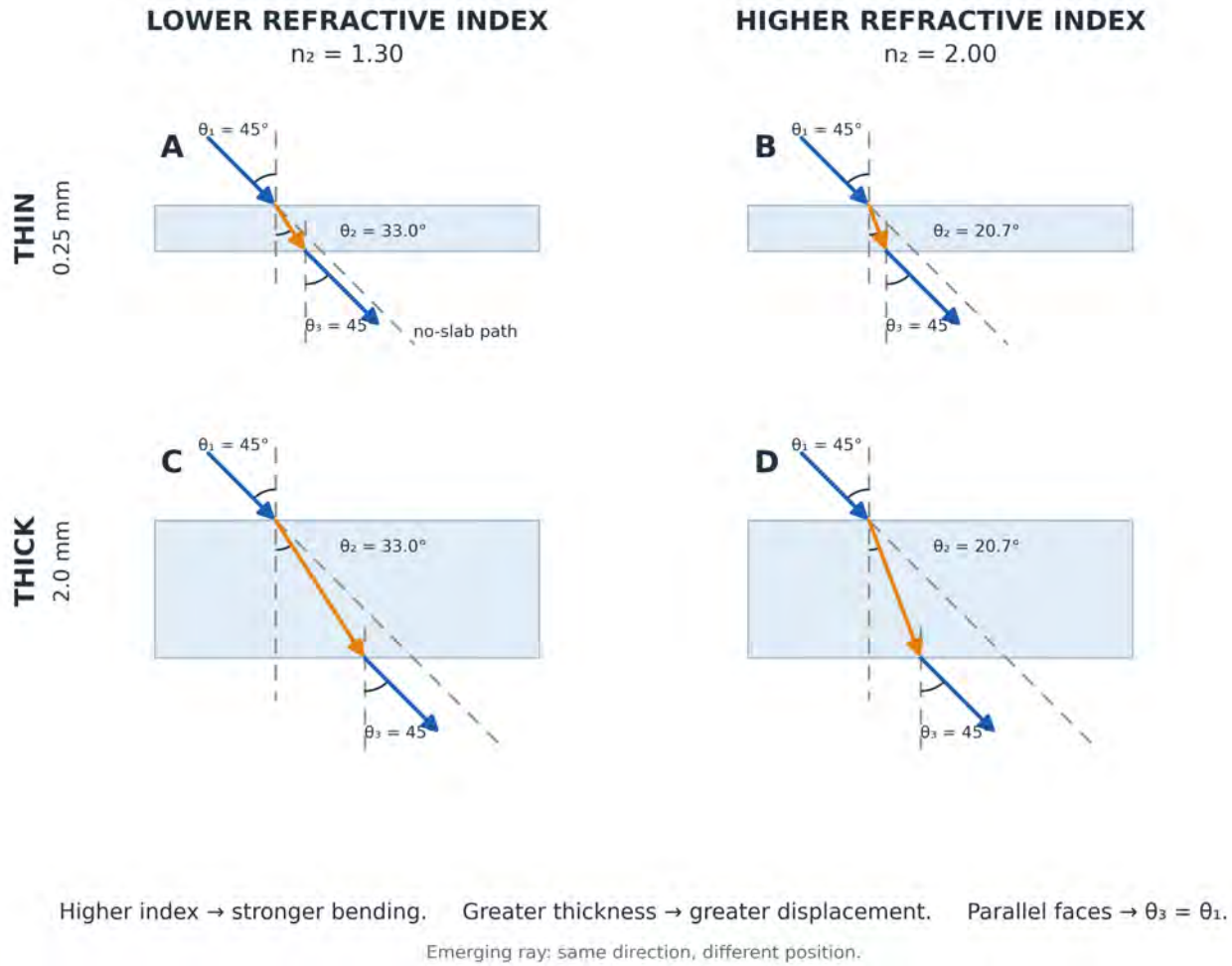


Figure 3.11.: Refraction through a parallel-sided transparent material. The online edition allows refractive index and material thickness to be varied independently. Increasing refractive index changes the angle of the ray inside the material; increasing thickness increases the lateral displacement of the emerging ray without changing that angle. Static formats compare representative combinations of refractive index and thickness.

This result follows directly from applying Snell's law at the two interfaces,

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

at the entrance and

$$n_2 \sin \theta_2 = n_1 \sin \theta_3$$

at the exit. Combining these relations gives

$$\theta_3 = \theta_1,$$

where:

- θ_1 is the original incidence angle in air;
- θ_2 is the refracted angle while the ray travels inside the transparent material;

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- θ_3 is the angle of the ray after it has emerged back into air;
- n_1 is the refractive index of the surrounding medium;
- n_2 is the refractive index of the transparent slab.

For flat, parallel interfaces surrounded by the same medium, the important result is that the emerging ray returns to the **same propagation angle** that it had before entering the slab. It is therefore parallel to the original ray, but it does not generally follow the same path: it has been **laterally displaced**. A larger refractive index produces stronger bending inside the material, while a thicker slab gives that altered direction a greater distance over which to accumulate displacement.

This distinction matters in machine vision because **lateral displacement is not the same as image distortion**. A thin, flat and uniform protective window can shift oblique rays slightly while allowing them to emerge in their original direction. Close to normal incidence, even that displacement becomes small. A transparent layer in front of the camera therefore does not automatically produce the obvious geometrical warping associated with looking through a curved bottle or a glass of water.

The situation becomes more interesting when the interfaces are not parallel. Curvature, wrinkles, wedges and local thickness variations change the surface normal encountered by different rays. The second refraction then no longer simply restores the original direction. A glass vial has curved inner and outer walls; a catheter balloon has thin but strongly curved polymer surfaces; transparent packaging can contain folds, stretched regions and locally non-parallel layers. Different parts of the image can consequently experience different ray directions and displacements, making features appear shifted, magnified, blurred or geometrically distorted.

The useful distinction is therefore:

- **parallel, uniform interfaces** primarily produce predictable refraction and lateral displacement;
- **curved, tilted, wedged or spatially varying interfaces** can produce spatially varying refraction and therefore apparent geometrical distortion.

Refractive index determines how strongly an interface bends a ray, but the effect observed by the camera also depends on material thickness, incidence angle and interface geometry.

There is another consequence that is particularly important for machine vision: the camera does not necessarily observe only light travelling directly through the transparent material. In a front-lit inspection, illumination can pass through the transparent layer, interact with a surface or material behind it, and then travel back through the transparent layer before reaching the camera. The transparent component is therefore part of both the **forward optical path** and the **return optical path**.

Consider a nominally clear polymer film in front of a reflective or diffusely reflecting backing. Light can refract as it enters the polymer, propagate through it, interact with the backing, and then cross the polymer interfaces again on its return towards the camera. A local defect can disturb this path in several ways. A fold changes local thickness and surface orientation; a bubble introduces additional interfaces; a burn or contamination can change absorption or scattering; and a pinhole removes part of the polymer optical path altogether. The feature can therefore create image contrast even when the transparent material itself has very little visible colour.

The same principle extends to curved transparent components. In a catheter balloon or transparent tube, local changes introduced during forming can alter wall thickness, curvature or surface orientation. Rays passing through these regions follow different paths from those passing through the nominal material. What the camera observes can consequently contain information about geometry and material uniformity encoded as changes in transmitted, refracted, reflected or scattered light.

A useful quantity for describing propagation through transparent material is the **optical path length**. For a uniform material and a geometrical path of length d ,

$$\text{OPL} = n d,$$

where:

- OPL is the optical path length;
- n is the refractive index of the material;
- d is the physical distance travelled through that material.

The optical path length expresses the equivalent distance that light would travel in vacuum while accumulating the same phase delay. Two neighbouring rays can therefore experience different optical paths because the material thickness differs, because the refractive index differs, or because their geometrical trajectories through the material differ. For ordinary intensity imaging this phase difference may not be measured directly, but the same physical variations often accompany **changes in refraction, focus, scattering or polarisation** that can become visible to a machine-vision camera. More specialised optical techniques can deliberately convert optical-path differences into measurable intensity contrast.

Refractive index itself can also depend on wavelength. More generally,

$$n = n(\lambda),$$

where:

- n is the refractive index of the material;
- λ is the wavelength of the optical radiation.

This wavelength dependence is called **dispersion**. Different wavelengths can therefore follow slightly different refracted paths through the same interface. The familiar separation of white light by a prism is a visible example: each wavelength experiences a slightly different refractive index and therefore a different refraction angle. In imaging optics, the same physics contributes to **chromatic aberration**, developed further in Chapter 4.

So far we have followed the part of the incident light that enters the transparent material. That is still not the complete interaction. At every interface, some optical power can also be **reflected** rather than transmitted. Even a perfectly clear material can therefore generate a visible reflected signal.

For light arriving perpendicular to an interface between two non-absorbing materials, the fraction of incident optical power reflected is given by the normal-incidence Fresnel relation:

$$R = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2,$$

where:

- R is the **power reflectance**, the fraction of incident optical power reflected by the interface;
- n_1 is the refractive index of the medium from which the light arrives;
- n_2 is the refractive index of the material on the other side of the interface.

Because R represents a fraction of optical power, it is dimensionless. For example, $R = 0.04$ means that approximately 4% of the incident optical power is reflected at that interface.

For an air-to-glass boundary with $n_1 \approx 1.00$ and $n_2 \approx 1.50$,

$$R = \left(\frac{1.00 - 1.50}{1.00 + 1.50} \right)^2 = 0.04.$$

Approximately **4% of the incident optical power is therefore reflected at a single uncoated air–glass interface at normal incidence**. Assuming negligible absorption, most of the remaining power is transmitted into the glass.

A transparent wall normally contains more than one interface. A simple glass plate has an air-to-glass boundary followed by a glass-to-air boundary; a vial introduces inner and outer curved walls; multilayer polymer packaging can contain several interfaces. Each boundary can divide the optical power again into reflected and transmitted components.

The camera can therefore receive several optical paths simultaneously. Some light may return from the first surface, some may travel through the transparent material and interact with whatever lies behind it, and some may undergo further reflections between interfaces before eventually reaching the lens. Depending on geometry, these paths can produce useful feature contrast or unwanted effects such as glare, secondary edges and faint displaced images.

i A transparent part is an optical system

For machine vision, it is useful to stop thinking of a transparent component as something that is simply *not opaque*. It is an optical element containing interfaces and propagation paths.

A feature can become visible because it changes **surface orientation, thickness, refractive index, transmission, absorption or scattering**, or because it adds or removes an interface.

The camera records the combined result of whichever of those optical paths reach its lens.

This perspective connects several apparently different inspection problems. A pinhole, fold or damaged seal region in transparent packaging can disturb the path through the film and the return from material behind it. A forming defect in a transparent catheter balloon can alter local curvature or wall thickness and therefore the paths followed by transmitted and returned rays. A scratch, coating defect or contamination on flat glass can change the balance between surface reflection, transmission and scattering. Glass inspection systems consequently make use of both reflected and transmitted optical information depending on which physical interaction contains the defect information.

The simple Fresnel equation above applies at **normal incidence**. At oblique incidence, reflectance changes with angle and also depends on the polarisation of the incident light. The Fresnel equations then treat the two orthogonal polarisation components separately. Transparent materials can also alter polarisation when internal stress or optical anisotropy is present; stressed glass, for example, can become birefringent and reveal otherwise difficult-to-observe internal stress patterns under polarised illumination. These effects provide the theoretical connection to Section 3.4.

The important conclusion is that **transparent does not mean optically absent**. A transparent component can change the position and direction of rays, divide optical power between reflected and transmitted paths, introduce optical-path differences and create additional interfaces from which light can return. Defects can become visible precisely because they perturb one or more of these behaviours.

The purpose here is to establish those physical mechanisms rather than prescribe an inspection arrangement. How illumination direction, background, transmission or reflection geometry, polarisation and filtering are combined to turn these effects into robust production contrast is developed later in Chapter 5.

For a fuller treatment of refraction, Fresnel coefficients and dispersion, see Hecht [154]. The consequences of refraction for lenses and geometrical image formation are developed in Chapter 4, while polarisation effects are developed in Section 3.4.

3.2.3. Absorption and fluorescence

The previous subsections were concerned mainly with **where light goes**. Reflection, refraction and transmission could therefore be understood efficiently using rays and, where necessary, the underlying wave description. Absorption requires us to ask a different question: **what happens when the material takes energy from the incident radiation instead of simply redirecting or transmitting it?**

For this discussion it is useful to return to the photon description introduced in Section 3.1.2. A photon carries an amount of energy determined by its frequency, or equivalently its wavelength,

$$E_{\gamma} = h\nu = \frac{hc}{\lambda},$$

where:

- E_{γ} is the energy carried by one photon, measured in joules (J);
- h is Planck's constant;
- ν is the optical frequency, measured in hertz (Hz);
- c is the speed of light in vacuum;
- λ is the vacuum wavelength.

When a photon is absorbed, its energy is transferred to the material rather than continuing along the original optical path. Depending on the material and photon energy, this can produce electronic or molecular excitation, with much of the absorbed energy eventually appearing as heat. In some materials, part of the absorbed energy can instead be released again as light; this is the basis of **fluorescence**, developed below.

Crucially for machine vision, absorption is normally **wavelength-dependent**. A material can absorb one part of the spectrum strongly while reflecting or transmitting another. Two materials that appear similar under visible illumination can therefore behave very differently in the ultraviolet (UV), near infrared (NIR) or short-wave infrared (SWIR). The camera can exploit this difference even when there is little useful contrast to the human eye.

This connects absorption to the signal ultimately recorded by the camera. Consider first a monochrome camera observing reflected light. The recorded grey level does not depend on the material alone: it depends on the complete spectral chain from the illumination, through the object and optics, to the sensor. A useful first-order model is

$$G \approx k t_{\text{exp}} \int S(\lambda) \rho(\lambda) \tau(\lambda) q(\lambda) d\lambda,$$

where:

- G is the resulting linearised pixel signal or grey level;
- k groups together factors that are not being described spectrally, such as illumination and imaging geometry and electronic scaling;
- t_{exp} is the camera exposure time;
- $S(\lambda)$ is the spectral distribution of the illumination reaching the object;
- $\rho(\lambda)$ is the spectral reflectance of the observed material or feature;
- $\tau(\lambda)$ is the wavelength-dependent transmission of the optical path, including the lens, windows and any filters;
- $q(\lambda)$ is the wavelength-dependent sensitivity of the image sensor;
- λ is wavelength;
- the integral adds the contributions from all wavelengths reaching the sensor.

The equation is less complicated than it first appears. At each wavelength, the illumination provides some optical energy, the material determines how much of it is reflected, the optical system determines how much reaches the sensor, and the sensor determines how efficiently that radiation contributes to the measured signal. The camera then accumulates those contributions across the spectral range to which the complete imaging chain responds.

For **transmitted-light imaging**, the same reasoning applies, but the material's spectral transmittance replaces its reflectance. In either case, image contrast appears when the feature and its background interact differently with the spectrum. For reflective imaging, for example, the relevant material difference is

$$\Delta\rho(\lambda) = \rho_{\text{feature}}(\lambda) - \rho_{\text{background}}(\lambda).$$

Here, $\rho_{\text{feature}}(\lambda)$ and $\rho_{\text{background}}(\lambda)$ describe how strongly the feature and its surrounding background reflect each wavelength. If their values are nearly equal over the wavelengths measured by the camera, little contrast is produced. If they differ strongly within part of the spectrum, that spectral region has the potential to produce much greater image separation.

This provides the machine-vision connection to absorption. A dark region in an image does not necessarily indicate a physically dark material; it may indicate that, at the wavelengths being measured, more of the incident optical energy has been **absorbed rather than returned towards the camera**. Conversely, a material that absorbs little at those wavelengths may produce a stronger reflected or transmitted signal. The resulting greyscale contrast is therefore a property of the complete interaction between source, material, optics and sensor rather than of the object's visible colour alone.

i Why wavelength can change a monochrome image

A monochrome camera does not remove the spectral behaviour of light; it removes the explicit separation into colour channels. Its grey level can still change substantially with wavelength.

A change in wavelength can alter:

- how much radiation the feature absorbs, reflects or transmits;
- how much of that radiation passes through the optics and filters;
- how efficiently the sensor detects it.

Two regions that produce almost the same grey level under one spectral band can therefore produce very different grey levels under another, even though neither the camera position nor the object geometry has changed.

The practical selection of illumination wavelength, source and filtering is developed later in Chapter 5.

The same spectral reasoning applies to unwanted light. The camera records task illumination together with ambient radiation from factory lighting, sunlight and neighbouring stations. Enclosures, spectral filtering and short strobed exposures can all change the relative contribution of that unwanted radiation, but the theoretical point here is simpler: **ambient light is part of the optical signal unless the imaging system excludes or rejects it** [160]. The practical design of that rejection is developed in Chapter 5.

3.2.3.1. From wavelength selection to multispectral and hyperspectral imaging

A conventional monochrome inspection often chooses one wavelength because one band provides the required contrast. Multispectral and hyperspectral imaging generalise that idea by retaining several or many spectral measurements at each image location. The result is not simply an RGB image with

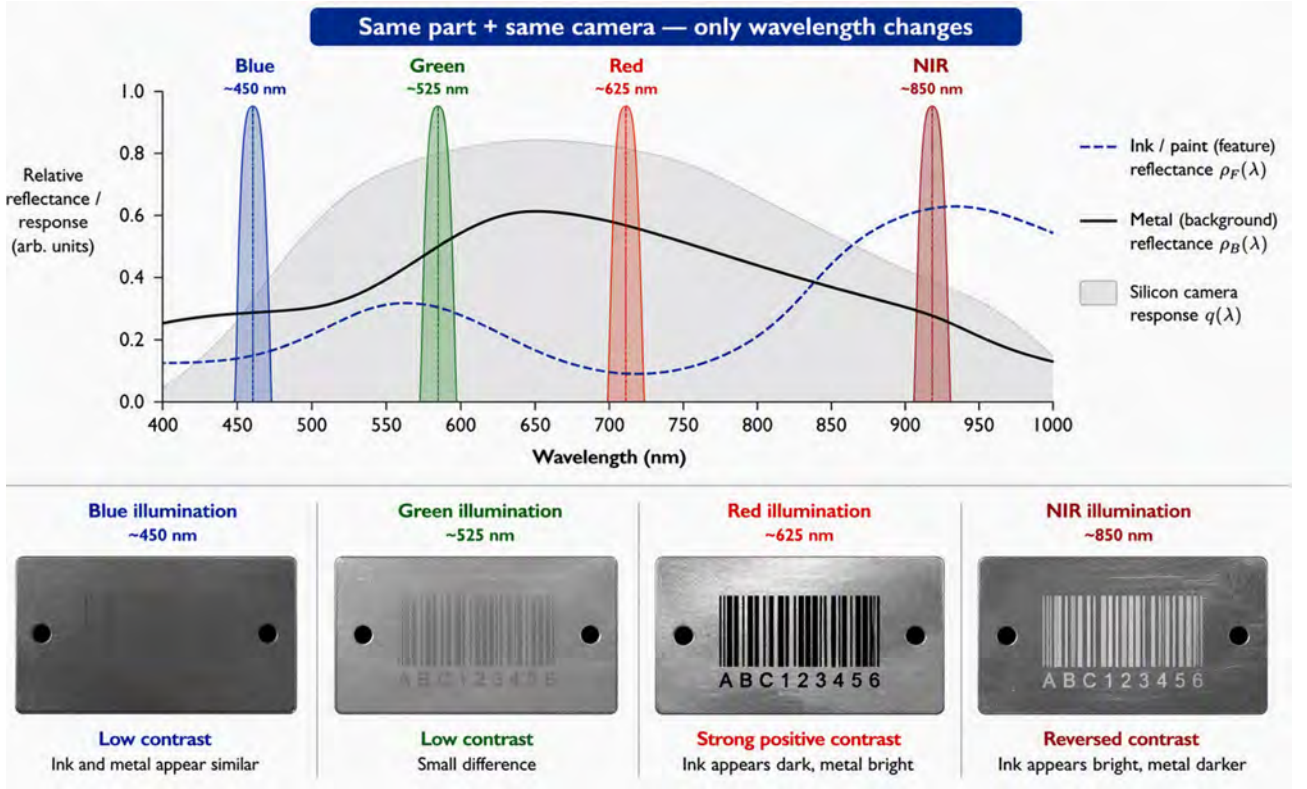


Figure 3.12.: Creating greyscale contrast by wavelength selection

more colours: it is an **imaging spectroscopy** problem in which spatial position and wavelength are both measured variables.

A hyperspectral image is commonly represented as a data cube

$$I = I(x, y, \lambda)$$

where each spatial pixel carries a sampled spectrum. A vertical slice through the cube is an ordinary image at one wavelength; a spectral vector through one pixel describes how that location responds across wavelength. There is no universal numerical boundary between multispectral and hyperspectral systems, but the engineering distinction is useful: multispectral systems normally use a relatively small set of deliberately selected bands, whereas hyperspectral systems acquire many comparatively narrow and often contiguous bands so that the shape of a spectrum becomes part of the measurement.

The phrase **spectral signature** is convenient, but it needs qualification in machine vision. The raw spectrum measured by the camera is not a property of the material alone. It is modulated by source spectral power, illumination/viewing geometry, surface BRDF, material reflectance or transmission, optical/filter transmission and detector sensitivity. If material reflectance is the desired quantity, a common first calibration uses dark and white references:

$$\hat{\rho}(x, y, \lambda) \approx \frac{I_s(x, y, \lambda) - I_d(\lambda)}{I_w(x, y, \lambda) - I_d(\lambda)} \rho_w(\lambda)$$

where I_s is the sample measurement, I_d is the dark reference, I_w is the measurement of a calibrated or characterised white reference under the same geometry, and ρ_w is the reference reflectance. This removes much of the source and detector spectral shape, but it does **not** remove geometry, specular reflection, scattering or calibration error. A useful spectral library is therefore meaningful only when acquisition conditions are documented and sufficiently controlled.

Hyperspectral imaging becomes powerful when materials that are difficult to distinguish geometrically or by visible colour exhibit different absorption or reflectance bands. Examples include polymers, textiles, foods, coatings, moisture content, pharmaceuticals and semiconductor materials. In the SWIR, molecular absorption features can provide contrast that simply does not exist in RGB. Conversely, adding hundreds of bands is not automatically an improvement: the available photons are divided spectrally, adjacent bands can be highly correlated, data rates increase sharply, and low-SNR bands may add more noise than information.

💡 Hyperspectral cameras still need the right light

Strictly speaking, **hyperspectral** usually describes the measurement, not the lamp. A system may illuminate with a broadband source and separate wavelengths in the camera, or it may illuminate sequentially with narrow/tunable bands.

- The illumination must provide usable power across every wavelength that the analysis depends on.
- A spectral gap in the source cannot be recovered later by a more sophisticated camera or algorithm.
- Filters, lenses, windows and sensor material must all transmit/detect the same spectral region.
- Stable geometry matters because angular reflectance can masquerade as a spectral difference, especially on glossy materials.
- More bands should be justified by a material-discrimination mechanism, not by the availability of a hyperspectral camera.

Human colour vision and conventional RGB cameras reduce a continuous spectrum to three broad responses. Many physically different spectra can therefore produce the same or very similar three-channel colour - the basis of metamerism discussed in the colour section. Hyperspectral imaging keeps far more of the spectral structure, which is why it can distinguish materials that look identical to the eye or to an RGB camera.

For further reading, imec [163] gives a concise overview of hyperspectral imaging and its use in industrial machine vision, while Specim [164] illustrates real-time material classification and spectral-signature workflows on production lines.

For a homogeneous absorbing medium, transmitted radiant power often follows an exponential attenuation law,

$$P(z) = P_0 e^{-\alpha z}$$

where α is the Napierian absorption coefficient. In spectrochemical form, the Beer-Lambert-Bouguer law relates decadic absorbance to concentration and path length:

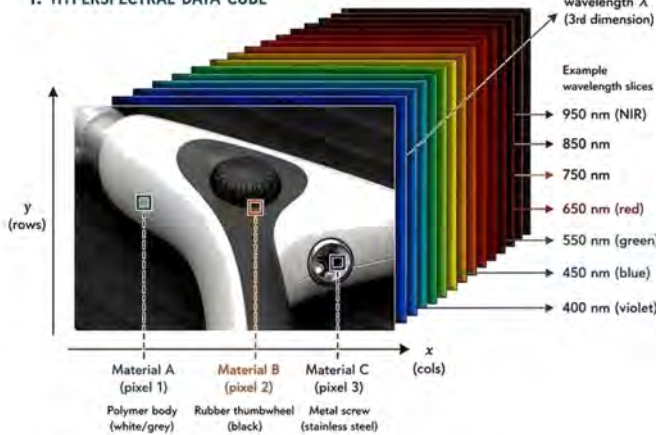
$$A = \log_{10} \left(\frac{P_0}{P} \right) = \varepsilon c \ell$$

The law assumes conditions such as monochromatic or sufficiently narrow-band radiation, a homogeneous medium and negligible complications from scattering or saturation. Industrial imaging through polymers, liquids, coatings and biological materials may depart from these assumptions, but the exponential model remains useful intuition: longer optical path length increases attenuation, and wavelength can change attenuation dramatically.

HYPERSPECTRAL IMAGING: FROM SCENE TO SPECTRA AND DATA CUBE

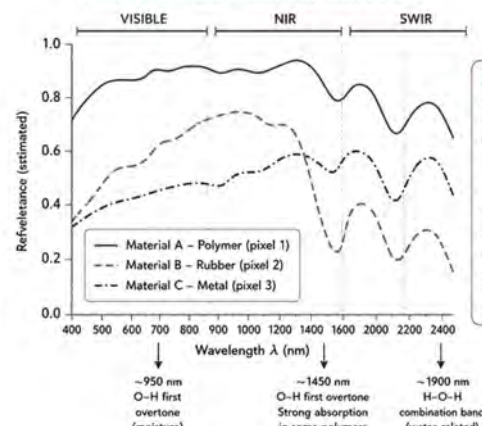
Hyperspectral imaging measures a full spectrum at every pixel, producing a 3-D data cube $I(x, y, \lambda)$.

1. HYPERSPECTRAL DATA CUBE



Stacking all spectral slices forms a 3-D data cube $I(x, y, \lambda)$ with two spatial dimensions (x, y) and one spectral dimension (λ).

2. SPECTRAL SIGNATURES AT THREE PIXELS

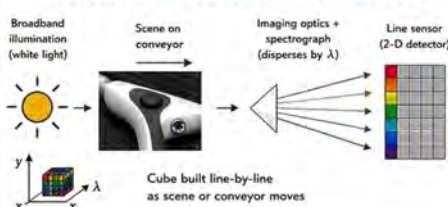


1 Biological materials often have distinctive spectral signatures caused by pigments, water, chemical composition and tissue structure. Hyperspectral imaging is therefore widely used in food and agricultural inspection.

$$G(x, y, \lambda) \propto S(\lambda) \rho(x, y, \lambda) \tau(\lambda) q(\lambda)$$

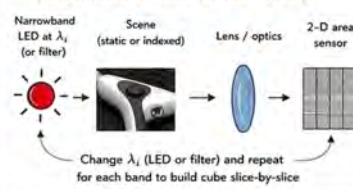
measured spectrum = source × material × geometry × optics × detector

3. PUSHBROOM, SPECTRALLY DISPERSED (HYPERSPECTRAL)



Captures many contiguous narrow bands simultaneously. Fast acquisition for hyperspectral data cubes (hundreds of bands).

B. SEQUENTIAL WAVELENGTH ILLUMINATION (FILTER-TUNED OR LED STEP-AND-REPEAT)



Captures one wavelength at a time (one exposure per band). Slower acquisition; typically used for multispectral systems (fewer, discrete bands with gaps).

COMPARISON

	Pushbroom (A)	Sequential (B)
Speed	All wavelengths captured in a single pass. High speed.	One wavelength at a time. Slower as bands increase.
Spectral sampling	Many contiguous narrow bands.	Fewer discrete bands (with gaps).
Best for	Hyperspectral analysis, classification, material ID, subtle spectral features.	Targeted measurements, inspection at selected wavelengths.
Data volume	High (large 3-D cube).	Lower (smaller 3-D cube).

i The measured spectrum depends on the entire system: source, material, geometry, optics and detector. It is not an intrinsic material property.

O-H = oxygen-hydrogen bond (common in moisture, hydroxyl groups)

H-O-H = water molecule (water combination band)

Figure 3.13.: Hyperspectral image cube and material spectral signatures

For the formal definition and limitations of the Beer-Lambert law, see the IUPAC Gold Book entry [165].

Fluorescence is different from ordinary reflection. A material absorbs photons, enters an excited state and subsequently emits photons, usually at longer wavelengths than the excitation. Because excitation and emission can be spectrally separated, fluorescence can produce exceptional contrast against a dark background. In machine vision this is useful for detecting optical brighteners, adhesives, coatings, inks, contamination and some biological materials.

A typical fluorescence imaging arrangement therefore uses an excitation source, an excitation filter, optical separation such as a dichroic element where needed, and an emission filter that strongly suppresses the excitation wavelength while passing the weaker fluorescence signal. The emitted signal can be orders of magnitude weaker than the excitation, so spectral blocking and stray-light control are often more important than raw source brightness.

For machine vision, the important physical consequence is **spectral separation**: excitation occurs in one wavelength region and useful emission is detected in another. The wavelength difference between excitation and emission is commonly called the Stokes shift. This separation explains why an emission filter can strongly reject excitation light while transmitting a comparatively weak fluorescent signal. The practical choice of excitation source, filters and optical layout is developed in Chapter 5.

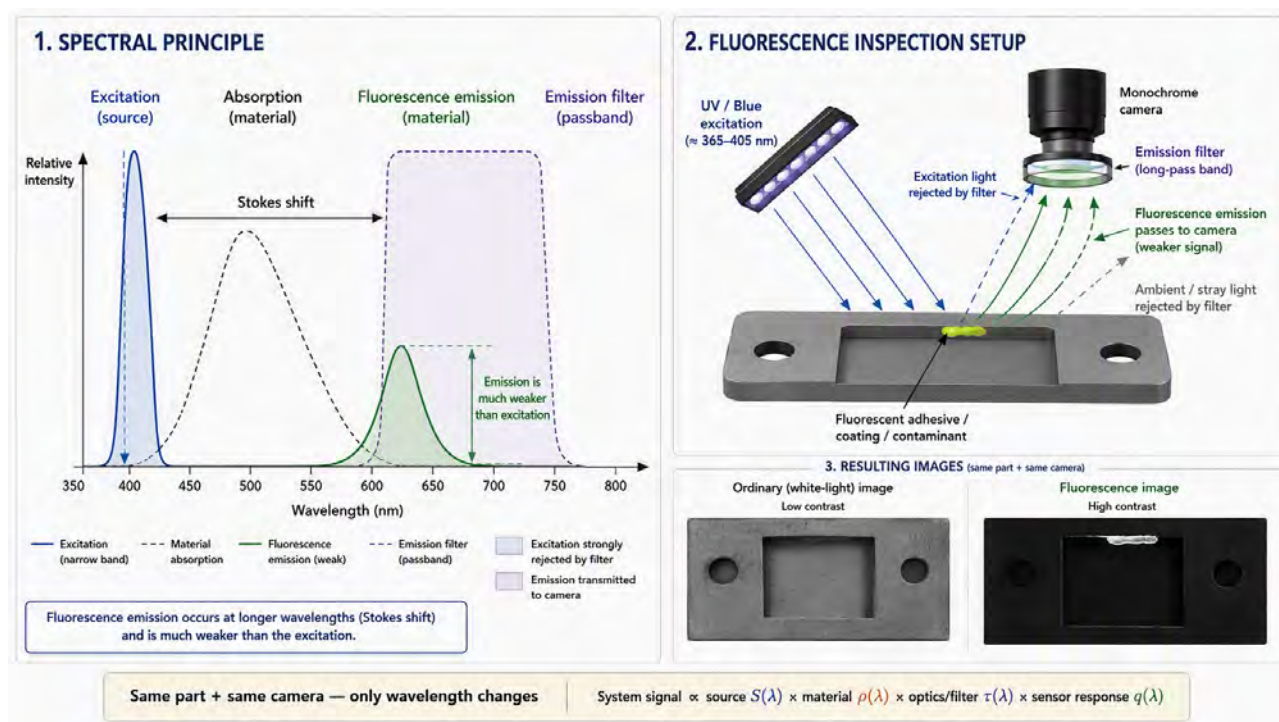


Figure 3.14.: Fluorescence as wavelength-shifted contrast

For additional practical examples of excitation/emission filtering and fluorescence imaging hardware, see Edmund Optics [166].

3.2.4. Scattering and translucency

Scattering redirects radiation because of spatial variations in refractive index. Unlike specular reflection at a smooth boundary, scattering can occur from particles, fibres, pores, surface roughness or internal microstructure. It can be elastic, preserving wavelength, or inelastic, changing photon energy; fluorescence is one important inelastic process but is normally treated separately because of its distinct mechanism.

Two limiting regimes are often used for intuition. Rayleigh scattering applies to particles much smaller than the wavelength and has a strong wavelength dependence, approximately proportional to $1/\lambda^4$ in the simplest case. Mie scattering applies to particles comparable to or larger than the wavelength and is more complex and often strongly forward-directed. Real industrial materials can contain broad size distributions, multiple scattering and absorption, so these models are qualitative guides rather than universal recipes.

Translucent materials combine transmission with substantial scattering. Light entering the surface can travel laterally and re-emerge elsewhere, so the apparent image point no longer corresponds only to a narrow surface location. This subsurface transport is why waxes, many polymers, skin, ceramics, foams and some foods can look softly glowing and why small features can lose contrast even when the camera is perfectly focused.

This is one reason longer wavelengths can sometimes reveal information through inks, polymers, haze or turbid materials that appear opaque in the visible: scattering often weakens as wavelength increases. It is not a universal “IR sees through things” rule, because absorption bands can rise at the same time and material chemistry can dominate. In practice, NIR or SWIR selection is therefore empirical but physics-guided: ask whether the desired contrast is expected from reduced scattering, increased transmission, or a material-specific absorption feature.

Scattering can be either a nuisance or the measurement mechanism. It reduces contrast when trying to see through haze, but changes in scatter can reveal surface roughness, particles, fibre orientation, coating uniformity or contamination. NIST surface-scattering metrology uses this principle explicitly.

 When more illumination makes the image worse

For translucent or strongly scattering materials, increasing source power may raise both the wanted signal and a broad background of multiply scattered light. Contrast can remain unchanged or even fall through flare and saturation. Changing wavelength, incidence angle, polarisation, collection geometry or using transmission/backlighting can be more effective than simply increasing intensity.

For a deeper treatment of scattering, diffraction and wave propagation, see Hecht [154].

3.2.5. Implications for surface inspection

The practical value of the preceding physics is that a physical feature becomes visible only when it changes the radiation reaching the camera. A defect may alter geometry, roughness, refractive index, absorption, fluorescence, scattering or polarisation. The role of this chapter is to establish those mechanisms; Chapter 5 later develops how illumination hardware and acquisition geometry are selected to exploit them reliably.

The mechanisms can be summarised without turning them into a lighting-selection recipe:

Table 3.6.: Physical light–matter interactions and their possible consequences in a machine-vision image.

Physical difference in the part	Effect on light	Possible image consequence
Surface slope, roughness or local geometry	Changes the directional distribution of reflection or scattering	A feature may become brighter or darker as illumination/viewing direction changes
Refractive index, thickness or interface geometry	Changes refraction, Fresnel reflection and optical path	Edges, transparent walls or internal boundaries can shift, distort or produce secondary reflections

Physical difference in the part	Effect on light	Possible image consequence
Spectral reflectance or absorption	Changes the fraction of radiation returned or transmitted as a function of wavelength	Materials or markings with similar visible appearance can separate strongly in another spectral band
Fluorescent response	Absorbs in one band and re-emits in another	A weakly visible material can generate wavelength-shifted emission against a low background
Particles, fibres, pores or internal structure	Redistributes radiation through scattering	Haze, translucency, contamination or texture can alter local contrast and spatial spread
Polarisation-dependent interaction	Changes field orientation or phase	Stress, glare or material anisotropy can alter the signal observed through polarisation analysis

The table is intentionally descriptive rather than prescriptive. The same physical feature can involve several mechanisms simultaneously, and the observed image also depends on source spectrum, geometry, optics and sensor response. The later illumination chapter turns these mechanisms into practical source, geometry, filtering and validation choices.

💡 A physical feature must become an optical feature

Before an algorithm can detect a feature, some physical difference in the part must first become a measurable difference in the radiation reaching the sensor. The mechanisms in this chapter explain **where that optical difference can come from**; Chapter 5 explains how an engineered illumination system can make it useful and repeatable.

3.2.5.1. Illumination direction as an information channel

The previous examples use illumination angle primarily to create contrast. The same geometry can be pushed further: **illumination direction can itself become a measured input**. This is the theoretical basis of shape-from-shading, photometric stereo and several multi-light surface-inspection methods that the illumination chapter will use in a more practical form.

For an ideal Lambertian surface viewed by a fixed camera and illuminated by a distant directional source, a simple image-formation model is

$$I_k(x, y) = c_k \rho(x, y) \max(0, \mathbf{n}(x, y)^T \mathbf{s}_k)$$

where ρ is diffuse albedo, $\mathbf{n}$ is the unit surface normal, $\mathbf{s}_k$ is the known illumination direction and c_k collects source/camera scale factors. The dot product is a cosine term: a surface patch is bright when its normal points towards the light and dark as it turns away. Shadows, specular reflection, interreflection and non-Lambertian behaviour violate this ideal model, but the equation captures why changing light direction changes the apparent shape of a surface.

A single shaded image does not generally provide a unique shape because brightness also depends on albedo, source direction and other unknowns. **Photometric stereo** reduces that ambiguity by keeping the camera fixed while acquiring several images under known illumination directions. If three

or more suitably independent light directions are used and the Lambertian assumptions are adequate, the image measurements at one pixel can be written

$$\mathbf{i} = \mathbf{S}\mathbf{g}, \quad \mathbf{g} = \rho\mathbf{n}$$

and estimated using a least-squares or pseudoinverse solution

$$\hat{\mathbf{g}} = \mathbf{S}^+\mathbf{i}, \quad \hat{\rho} = \|\hat{\mathbf{g}}\|, \quad \hat{\mathbf{n}} = \frac{\hat{\mathbf{g}}}{\|\hat{\mathbf{g}}\|}.$$

Using more than the minimum number of illumination directions provides additional measurements that can reduce sensitivity to shadows, saturation and imperfect reflectance. Even without reconstructing a metric surface, opposite-direction image pairs can be combined to form slope-sensitive or gradient-like images. A useful qualitative normalised directional contrast is

$$D_x = \frac{I_R - I_L}{I_R + I_L + \varepsilon}, \quad D_y = \frac{I_T - I_B}{I_T + I_B + \varepsilon},$$

where the normalisation reduces sensitivity to common albedo and overall brightness. This explains why shallow embossing, dents, scratches or raised characters can change appearance strongly as illumination direction changes. The practical implementation of multi-light inspection is treated in Chapter 5; here the important point is that illumination direction can carry information about local surface orientation.

⚠ Shape from shading is powerful because it uses assumptions - and fragile for the same reason

The simple equations assume known illumination, approximately Lambertian reflectance, no cast shadows, no saturation and a fixed relationship between images. Real polished, translucent or highly textured production parts violate these assumptions. The equations should therefore be interpreted as an ideal image-formation model. Practical multi-light implementation and validation are treated in Chapter 5.

Structured light is related but conceptually different. Instead of changing only the **direction** of otherwise uniform illumination, a structured-light system projects a known **spatial pattern** - a line, grid, dot field, Gray code or sinusoidal fringe pattern - onto the object. Surface geometry deforms the observed pattern. Once camera and projector geometry are calibrated and a projected feature is matched to its image observation, depth follows from triangulation. The projector therefore behaves like an active second viewpoint whose correspondences are deliberately encoded by light. Implementation choices are deferred to Chapter 5, while later 3D-vision material develops calibration and reconstruction in detail.

💡 Do not confuse multi-angle lighting with structured light

- **Multi-angle / photometric illumination:** the spatial field can be uniform; information comes from how brightness changes when the illumination direction changes.
- **Structured light:** information is encoded in a deliberately non-uniform spatial pattern and recovered from its displacement, deformation or phase.
- Both exploit controlled illumination as part of the measurement rather than treating light merely as a way to make the scene brighter.

For deeper study, Woodham's original photometric-stereo paper [167] develops the multiple-illumination surface-normal principle; Euresys [168] provides a concise industrial surface-inspection interpretation;

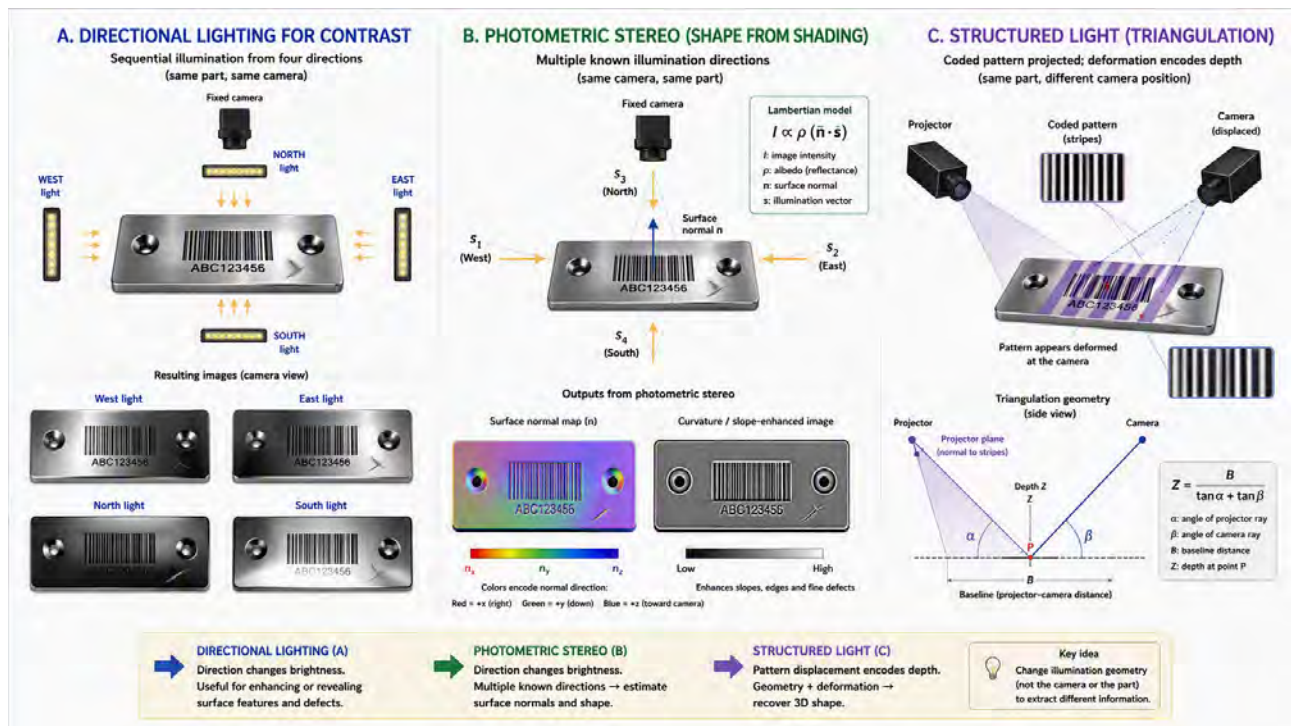


Figure 3.15.: Directional illumination, photometric stereo and structured light

and Salvi, Pagès and Batlle [169] review coded structured-light patterns and the correspondence/triangulation problem.

3.3. Colour

3.3.1. Trichromacy and the human visual system

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.3.2. Colour matching and the CIE XYZ system

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.3.3. Working colour spaces: RGB, sRGB, HSV, HSL, Lab

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.3.4. Colour temperature and illuminants

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.3.5. White balance and colour constancy

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.4. Polarisation

3.4.1. States of polarisation and Stokes parameters

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.4.2. Polarisers, retarders and the Fresnel equations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.4.3. Polarisation-resolving sensors

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.4.4. Industrial uses: glare suppression, stress imaging, transparent-material inspection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4. Optics and lens systems

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

4.1. Geometric optics

4.1.1. The pinhole model

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.1.2. Thin- and thick-lens models

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.1.3. Focal length, f-number, aperture

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.1.4. Field of view and working distance

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.1.5. Depth of field and hyperfocal distance

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.2. Real lens behaviour

4.2.1. Spherical and chromatic aberration

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.2.2. Geometric distortion: radial and tangential

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.2.3. Vignetting and cos fall-off

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.2.4. Modulation transfer function and resolving power

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.2.5. Diffraction limit

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.3. Specialty optics

4.3.1. Telecentric lenses for metrology

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.3.2. Macro and micro imaging

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.3.3. Line-scan and Scheimpflug configurations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.3.4. Liquid and variable-focus lenses

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.4. Optical filters

4.4.1. Bandpass, longpass and shortpass filters

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.4.2. Polarising filters in imaging

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.4.3. Neutral density, UV cut and IR cut filters

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.4.4. Interference filters and dichroic elements

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.4.5. Filter placement and stacking

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.5. Selecting an optical configuration

4.5.1. Matching focal length to sensor size, FOV and working distance

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.5.2. Resolution and DOF budgeting from the feature specification

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.5.3. When telecentricity, filtering or specialty optics are required

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.5.4. Mount standards and mechanical envelope: C, CS, F, M42, S-mount

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.5.5. A lens-selection workflow, cross-referenced to Chapters 2 and 5

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

Part II.

Image Formation

i **Part status:** Structural placeholder.

This part introduction will later summarise its scope, prerequisites and learning objectives.

5. Illumination for vision

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

5.1. Illumination principles

5.1.1. Light and geometry as the first solution

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.1.2. Contrast, feature enhancement and defect visibility

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.1.3. The lighting-first design method

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.2. Illumination geometries

5.2.1. Frontal and diffuse illumination

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.2.2. Directional and grazing illumination

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.2.3. Backlighting for silhouettes and dimensioning

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.2.4. Coaxial (on-axis) illumination

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.2.5. Dome and cloudy-day illumination

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.2.6. Dark-field illumination

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.2.7. Structured light

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.3. Light sources

5.3.1. LEDs: colour, drive, lifetime, thermal management

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.3.2. Fluorescent, halogen and xenon sources

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.3.3. Laser sources and safety classes

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.3.4. Continuous, strobed and pulsed operation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.4. Wavelength selection

5.4.1. Visible-band choices and colour separation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.4.2. Near-infrared imaging

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.4.3. Ultraviolet imaging

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.4.4. Short-wave infrared and thermal imaging

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.4.5. Multispectral and hyperspectral approaches

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.5. Environmental integration

5.5.1. Ambient light rejection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.5.2. Enclosures, shrouds and shielding

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.5.3. Vibration, contamination and thermal drift

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.6. Designing an illumination solution

5.6.1. Characterising the feature and the surface

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.6.2. Single-plane versus multi-plane and 3D features

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.6.3. Edge visibility for measurement versus contrast for detection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.6.4. Matching geometry to task: a decision guide

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.6.5. Combining geometry, wavelength and filtering

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.6.6. Worked examples: from part to lighting choice

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6. Image sensors and cameras

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

6.1. Sensor technologies

6.1.1. CCD architecture and readout

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.1.2. CMOS architecture: global vs rolling shutter

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.1.3. Line-scan and TDI sensors

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.1.4. SPAD, event-based and neuromorphic sensors

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.1.5. Multispectral and hyperspectral sensor architectures

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.2. Sensor performance and noise

6.2.1. Quantum efficiency and spectral response

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.2.2. Read noise, shot noise, dark current, fixed-pattern noise

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.2.3. Dynamic range and bit depth

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.2.4. Full-well capacity and saturation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.2.5. Signal-to-noise budgeting

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.3. Sampling, resolution and colour acquisition

6.3.1. Pixel pitch, sensor size and effective resolution

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.3.2. Bayer patterns and demosaicing

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.3.3. 3-CCD and multi-sensor colour

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.3.4. Monochrome versus colour trade-offs

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.4. Camera interfaces and image transport

6.4.1. Interface selection criteria: bandwidth, cable length, CPU load, cost

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.4.2. USB3 Vision: bandwidth, cable limits and use cases

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.4.3. Camera Link, CoaXPress and frame-grabber-based interfaces

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.4.4. Legacy and niche interfaces: FireWire, analogue, MIPI CSI

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.4.5. Triggering, exposure control and time stamping

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.5. GigE Vision in depth

6.5.1. Generational bandwidth: 1, 5, 10 and 25 GbE and practical reach

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.5.2. The GigE Vision protocol: discovery, stream and control channels

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.5.3. Network configuration: jumbo packets, dedicated NICs, receive buffers

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.5.4. Multi-camera topologies, switches and bandwidth budgeting

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.5.5. Power over Ethernet and cabling

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.5.6. GenICam and cross-vendor interoperability

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.6. Selecting a camera and interface

6.6.1. From resolution and speed requirement to sensor choice

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.6.2. Matching interface to bandwidth, distance and architecture

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.6.3. Area-scan versus line-scan decision

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.6.4. A camera-and-interface selection workflow

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

Part III.

Image Processing

i **Part status:** Structural placeholder.

This part introduction will later summarise its scope, prerequisites and learning objectives.

7. Digital images: representation and pixel-level processing

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

7.1. The digital image as a signal

7.1.1. Sampling, quantisation and the resulting data model

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.1.2. Coordinate systems, indexing and boundary conventions

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.1.3. Colour, greyscale and multi-channel representations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.1.4. File formats and lossless vs lossy compression

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.2. Point operations

7.2.1. Brightness, contrast and gain

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.2.2. Look-up tables, gamma and tone curves

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.2.3. Histogram equalisation and adaptive stretching

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.2.4. Thresholding as a point operation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.3. Arithmetic and logical operations

7.3.1. Pixel-wise arithmetic and blending

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.3.2. Absolute and signed differences

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.3.3. Masking and region-of-interest operations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.3.4. Bitwise operations on binary images

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.4. Noise models and denoising choices

7.4.1. Gaussian, Poisson and impulse noise models

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.4.2. Estimating noise from image statistics

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.4.3. Choosing a denoiser: signal preservation vs artefact suppression

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8. Neighbourhood operations and image filtering

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

8.1. Convolution and correlation

8.1.1. Discrete convolution in two dimensions

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.1.2. Separable and rank-one kernels

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.1.3. Boundary handling: zero, reflect, replicate, wrap

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.1.4. Computational cost and efficient implementations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.2. Smoothing filters

8.2.1. Box (mean) filter and integral images

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.2.2. Gaussian filter and selection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.2.3. Median and rank-order filters

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.2.4. Bilateral and edge-preserving filters

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.2.5. Non-local means

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.3. Derivative and edge filters

8.3.1. First-order operators: Sobel, Prewitt, Scharr

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.3.2. Second-order operators: Laplacian and Laplacian of Gaussian

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.3.3. Gradient magnitude and orientation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.3.4. Sub-pixel edge localisation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.4. The frequency domain

8.4.1. The 2D discrete Fourier transform

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.4.2. Magnitude, phase and their perceptual roles

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.4.3. The convolution theorem

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.4.4. Low-pass, high-pass and band-pass filtering

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.4.5. Frequency-domain inspection of periodic structure

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.4.6. The sampling theorem and aliasing

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.5. Scale space and image pyramids

8.5.1. Scale-space theory in outline

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.5.2. The Gaussian pyramid

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.5.3. The Laplacian pyramid

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.5.4. Multi-resolution processing patterns

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.6. Choosing a filtering strategy

8.6.1. Matching a filter to the noise model and the feature

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.6.2. Preserving edges while suppressing noise

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.6.3. Spatial versus frequency-domain: when each wins

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.6.4. Ordering operations in a preprocessing pipeline

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9. Segmentation, morphology and blob analysis

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

9.1. Thresholding methods

9.1.1. Global thresholds and manual selection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.1.2. Otsu's method

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.1.3. Adaptive and locally-adaptive thresholding

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.1.4. Multi-level and hysteresis thresholding

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.2. Region-based segmentation

9.2.1. Region growing and merging

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.2.2. Watershed segmentation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.2.3. Superpixels: SLIC and successors

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.2.4. Graph-cut segmentation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.3. Edge-based segmentation

9.3.1. The Canny detector

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.3.2. Edge linking and boundary tracing

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.3.3. Active contours (snakes) and level sets

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.4. Mathematical morphology

9.4.1. Structuring elements

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.4.2. Erosion and dilation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.4.3. Opening and closing

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.4.4. Hit-or-miss transform and skeletonisation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.4.5. Distance transforms and reconstruction

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.4.6. Greyscale morphology

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.5. Connected-components and blob analysis

9.5.1. Connectivity and labelling algorithms

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.5.2. Region properties: area, perimeter, centroid

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.5.3. Geometric moments and Hu invariants

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.5.4. Shape descriptors: circularity, eccentricity, compactness

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.5.5. Filtering blobs by property

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.6. Choosing a segmentation approach

9.6.1. Thresholding versus region versus edge versus learned

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.6.2. Reliability under illumination and contrast variation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.6.3. Combining segmentation with morphological clean-up

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10. Feature detection, description and matching

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

10.1. Interest-point detection

10.1.1. The Moravec detector and the corner concept

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.1.2. Harris and Shi–Tomasi corners

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.1.3. Scale-invariant detection: difference of Gaussians

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.1.4. Hessian-based detectors

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.1.5. FAST and AGAST corners for real-time use

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.2. Feature description

10.2.1. SIFT descriptor

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.2.2. SURF descriptor

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.2.3. Binary descriptors: BRIEF, ORB, BRISK, FREAK

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.2.4. Learned descriptors: SuperPoint and successors

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.3. Feature matching

10.3.1. Distance metrics: L2, Hamming

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.3.2. Brute-force and FLANN matching

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.3.3. The Lowe ratio test and cross-check

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.3.4. Robust matching with RANSAC and MSAC

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.4. Template and pattern matching

10.4.1. Sum of squared and absolute differences

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.4.2. Normalised cross-correlation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.4.3. Rotation- and scale-invariant template matching

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.4.4. Geometric pattern matching (edge-model methods)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.5. Line, circle and shape detection

10.5.1. The Hough transform for lines

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.5.2. The Hough transform for circles

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.5.3. The generalised Hough transform

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.5.4. Direct fitting: least-squares lines, circles, ellipses

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.5.5. Robust fitting with RANSAC

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.6. Choosing a feature method

10.6.1. Repeatability, invariance and speed trade-offs

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.6.2. Hand-crafted versus learned features

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.6.3. Matching the detector–descriptor pair to the task

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11. Geometric transformations, calibration and registration

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

11.1. Coordinate systems and homogeneous coordinates

11.1.1. Image, camera, world and object frames

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.1.2. Homogeneous representation of points and lines

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.1.3. Matrix representation of transformations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.2. The 2D transformation hierarchy

11.2.1. Rigid, similarity and affine transformations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.2.2. Projective transformations (homographies)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.2.3. Estimation from point correspondences

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.2.4. Robust estimation of homographies

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.3. Image warping and interpolation

11.3.1. Forward vs inverse mapping

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.3.2. Nearest-neighbour, bilinear and bicubic interpolation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.3.3. Anti-aliased warping

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.4. The pinhole camera model

11.4.1. Intrinsic parameters and the camera matrix

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.4.2. Extrinsic parameters

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.4.3. Radial and tangential distortion models

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.5. Camera calibration

11.5.1. Zhang's method and the planar-target approach

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.5.2. Calibration targets and image-acquisition strategy

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.5.3. Reprojection error and calibration diagnostics

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.5.4. Uncertainty and re-calibration policy

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.5.5. Hand-eye calibration for robot vision

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.6. Image registration

11.6.1. Feature-based registration pipelines

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.6.2. Intensity-based (direct) registration

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.6.3. Registration for inspection: golden templates

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.6.4. Mosaicing and stitching

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.7. Image-comparison metrics

11.7.1. Absolute and squared differences; PSNR

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.7.2. SSIM and multi-scale SSIM

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.7.3. Histogram-distance metrics

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.7.4. Learned perceptual metrics (outline)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

Part IV.

3D Vision and Motion

i **Part status:** Structural placeholder.

This part introduction will later summarise its scope, prerequisites and learning objectives.

12. Stereo and 3D vision

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

12.1. Two-view geometry

12.1.1. The epipolar constraint

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.1.2. The essential and fundamental matrices

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.1.3. Estimation from correspondences

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.1.4. Stereo rectification

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.2. Stereo matching

12.2.1. Block-matching costs: SAD, SSD, NCC, census

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.2.2. Aggregation and disparity selection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.2.3. Semi-global matching

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.2.4. Learned stereo matchers (outline)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.2.5. Disparity to depth and triangulation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.3. Uncertainty in 3D measurement

12.3.1. Depth-error propagation from disparity

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.3.2. Task uncertainty versus datasheet accuracy

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.3.3. Uncertainty budgets for 3D measurement

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.4. 3D imaging modalities

12.4.1. Passive stereo

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.4.2. Structured-light triangulation and fringe projection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.4.3. Time-of-flight cameras

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.4.4. LiDAR and long-range sensing

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.4.5. Photogrammetry, structure from motion and MVS

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.4.6. Comparison and selection guide

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.5. Point clouds and 3D representations

12.5.1. Point-cloud data structures and file formats

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.5.2. Filtering, downsampling and normal estimation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.5.3. Meshing and surface reconstruction

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.5.4. Voxelisation and occupancy representations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.6. Point-cloud registration and alignment

12.6.1. Iterative closest point (ICP) and variants

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.6.2. Feature-based 3D registration

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.6.3. Global alignment strategies

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.7. 3D measurement in practice

12.7.1. Sensor acceptance testing with calibrated artefacts

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.7.2. Surface interaction: specular and translucent materials

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.7.3. Warm-up drift and environmental control

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13. Motion, tracking and event-based vision

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

13.1. Motion representation

13.1.1. The brightness-constancy assumption

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.1.2. Sparse vs dense motion fields

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.2. Optical flow

13.2.1. The Lucas–Kanade method

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.2.2. The Horn–Schunck method

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.2.3. Dense flow: Farnebäck and pyramidal schemes

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.2.4. Learned optical flow (outline)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.3. Background modelling and change detection

13.3.1. Frame differencing

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.3.2. Gaussian mixture models

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.3.3. Robust background subtraction

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.4. Object tracking

13.4.1. Template and feature tracking

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.4.2. Mean-shift and correlation-filter trackers

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.4.3. Kalman filters and particle filters

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.4.4. Multi-object tracking and data association

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.5. Event-based vision

13.5.1. The event-camera imaging principle

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.5.2. Representations: event frames, time surfaces, voxel grids

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.5.3. Algorithms on event streams

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.5.4. Industrial fit and integration considerations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

Part V.

Deep Learning

i Part status: Structural placeholder.
This part introduction will later summarise its scope, prerequisites and learning objectives.

14. Machine learning foundations for vision

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

14.1. The learning paradigm

14.1.1. Supervised, unsupervised and self-supervised learning

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.1.2. Training, validation and test sets

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.1.3. Bias, variance and generalisation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.1.4. Cross-validation and model selection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.2. Classical classifiers

14.2.1. Linear and logistic regression

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.2.2. K-nearest neighbours

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.2.3. Support vector machines and kernels

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.2.4. Decision trees, random forests and boosting

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.2.5. Bayesian classifiers

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.3. Feature engineering and dimensionality reduction

14.3.1. Hand-crafted image features

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.3.2. Principal component analysis

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.3.3. Linear discriminant analysis

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.3.4. Manifold learning (outline)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.4. Loss functions and optimisation

14.4.1. Squared, absolute and Huber loss

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.4.2. Cross-entropy and log-loss

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.4.3. Gradient descent and its variants

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.4.4. Regularisation: L1, L2, elastic net

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.5. Evaluation of classifiers

14.5.1. The confusion matrix

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.5.2. Accuracy, precision, recall, F-score

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.5.3. ROC and precision-recall curves; AUC

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.5.4. Class imbalance and industrial reality

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.6. Choosing classical versus deep approaches

14.6.1. Data availability and labelling cost

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.6.2. Interpretability and validation burden

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.6.3. When hand-crafted features still win

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15. Neural networks and deep learning

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

15.1. Feed-forward networks

15.1.1. Neurons, layers and activation functions

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.1.2. ReLU, sigmoid, tanh, GELU

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.1.3. Softmax outputs and cross-entropy

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.1.4. Universal approximation and the role of depth

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.2. Backpropagation and automatic differentiation

15.2.1. The chain rule on the computational graph

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.2.2. Forward and reverse-mode differentiation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.2.3. Practical implementation in modern frameworks

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.3. Optimisation of deep models

15.3.1. Stochastic gradient descent and mini-batching

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.3.2. Momentum, Nesterov and Adam

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.3.3. Learning-rate schedules and warm-up

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.3.4. Weight decay, dropout, batch and layer normalisation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.3.5. Data augmentation for images

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.3.6. Early stopping and validation monitoring

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.4. Convolutional neural networks

15.4.1. The convolutional layer and parameter sharing

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.4.2. Pooling, stride and receptive field

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.4.3. Padding conventions and output-size arithmetic

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.4.4. Feature hierarchies and interpretability

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.5. Canonical architectures

15.5.1. LeNet-5, AlexNet, VGG

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.5.2. GoogLeNet / Inception

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.5.3. ResNet and residual connections

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.5.4. DenseNet, MobileNet, EfficientNet

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.5.5. Vision transformers and hybrid models

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.6. Training strategy for industrial data

15.6.1. Transfer learning and fine-tuning

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.6.2. Small-data regimes and few-shot learning

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.6.3. Class-imbalance countermeasures

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.6.4. Reproducibility: seeds, versioning, provenance

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16. Deep learning for classification, detection, segmentation and anomaly

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

16.1. Image classification

16.1.1. Whole-image classification pipelines

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.1.2. Multi-label classification

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.1.3. Fine-grained classification

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.1.4. Evaluation with imbalanced industrial datasets

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.2. Object detection

16.2.1. Two-stage detectors: R-CNN, Fast R-CNN, Faster R-CNN

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.2.2. One-stage detectors: YOLO family, SSD, RetinaNet

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.2.3. Anchor-free detectors: FCOS, CenterNet

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.2.4. Transformer-based detection: DETR

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.2.5. Evaluation: IoU, mAP, per-class analysis

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.3. Semantic and instance segmentation

16.3.1. Fully convolutional networks

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.3.2. U-Net and encoder–decoder architectures

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.3.3. DeepLab and dilated convolutions

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.3.4. Mask R-CNN for instance segmentation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.3.5. Panoptic segmentation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.4. Anomaly and defect detection

16.4.1. Reconstruction-based methods: autoencoders and GANs

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.4.2. Embedding-based methods: PatchCore, PaDiM, SPADE

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.4.3. One-class classifiers and Deep SVDD

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.4.4. Benchmarks: MVTec AD and industrial extensions

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.4.5. Choosing between anomaly detection and supervised approaches

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.5. Related tasks in industrial vision

16.5.1. Keypoint detection and pose estimation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.5.2. Deep OCR and OCV

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.5.3. Depth estimation from a single image

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.5.4. Self-supervised representation learning

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

Part VI.

Deployment and Compliance

i Part status: Structural placeholder.

This part introduction will later summarise its scope, prerequisites and learning objectives.

17. From theory to tools: selection, cross-platform mapping and inspection-flow design

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

17.1. The industrial vision software landscape

17.1.1. Configurable tool sets versus code libraries

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.1.2. Open-source libraries: OpenCV, scikit-image, PyTorch/TF ecosystems

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.1.3. Commercial libraries and SDKs: Halcon, VisionPro, Sapera

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.1.4. Smart-sensor and no-code environments: Keyence, Zebra Aurora, In-Sight

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.1.5. Configuring versus programming: trade-offs and consequences

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.2. Cross-platform tool mapping

17.2.1. How to read the mapping tables: theory concept to platform tool

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.2.2. Acquisition and preprocessing across platforms

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.2.3. Edge, line and circle detection: OpenCV / VisionPro / Halcon / In-Sight / Keyence / Zebra Aurora

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.2.4. Pattern location and alignment across platforms

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.2.5. Measurement and gauging tools across platforms

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.2.6. Defect and surface-inspection tools across platforms

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.2.7. Identification: barcode, Data Matrix, OCR/OCV across platforms

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.2.8. Deep-learning tools across platforms

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.2.9. Limits of equivalence: where tools do not map cleanly

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.3. Selecting and prioritising tools

17.3.1. From feature and task to candidate tools

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.3.2. Prioritisation: robustness, speed, determinism, maintainability

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.3.3. Rule-based versus learned tools for a given problem

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.3.4. A tool-selection decision guide, cross-referenced to theory chapters

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.4. Designing an inspection as a sequence of steps

17.4.1. Decomposing an objective into ordered steps

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.4.2. The step pattern: objective, tools, region, output

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.4.3. Locate–mask–measure–decide as a canonical flow

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.4.4. Worked example: scratch inspection decomposed into steps

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.5. Thresholds, checks and pass/fail logic

17.5.1. Per-step checks: presence, contrast, dimension, position limits

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.5.2. Setting limits and thresholds on tool outputs

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.5.3. Building the overall pass/fail decision from step results

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.5.4. Failure attribution: knowing which step failed and why

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.6. Exposing KPIs and data-driven threshold setting

17.6.1. Choosing KPIs and measured outputs per tool and step

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.6.2. Logging tool outputs for offline analysis

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.6.3. Batch statistical analysis to set thresholds and margins

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.6.4. Separating good/bad populations and setting operating points

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.6.5. Linking threshold setting to the validation of Chapter 18

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.7. Software design practice for vision applications

17.7.1. Modularity, reuse and configuration management

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.7.2. Robustness: handling absent, occluded and out-of-spec parts

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.7.3. Defensive checks, error states and safe defaults

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.7.4. Version control, documentation and change control of vision jobs

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.7.5. Performance, cycle time and resource budgeting

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18. Systems integration and industrial interfacing

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

18.1. Industrial communication

18.1.1. EtherNet/IP and PROFINET

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.1.2. Modbus TCP and serial protocols

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.1.3. OPC-UA and information modelling

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.1.4. MQTT and vision-to-cloud patterns

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.2. Integration with PLCs

18.2.1. Handshaking and trigger management

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.2.2. Result data structures and tags

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.2.3. Timing budgets and cycle time

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.3. Integration with robots

18.3.1. Coordinate-frame transformations and hand-eye calibration

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.3.2. Fixed and moving cameras

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.3.3. Pick-and-place and vision-guided assembly

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.3.4. Bin picking and 3D-guided robotics

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.4. Line integration

18.4.1. Reject mechanisms and part tracking

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.4.2. Statistical process-control feedback

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.4.3. Traceability and data-logging architecture

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.4.4. Human–machine interfaces for vision stations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19. Validation, measurement uncertainty and sampling for inspection

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

19.1. Defining acceptance

19.1.1. False accept and false reject rates

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.1.2. The confusion matrix in industrial terms

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.1.3. Producer and consumer risk

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.1.4. Setting acceptance criteria from requirements

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.2. Repeatability and reproducibility

19.2.1. Sources of variation in a vision measurement

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.2.2. Attribute Gauge R&R: method and calculation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.2.3. Variable Gauge R&R by the ANOVA method

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.2.4. Kappa statistics and inter-rater/inter-system agreement

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.2.5. Interpreting %R&R, %tolerance and number of distinct categories

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.3. Confidence, tolerance and statistical intervals

19.3.1. Confidence intervals for proportions and means

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.3.2. Tolerance intervals and coverage

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.3.3. Confidence–reliability statements (e.g. 95/95)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.3.4. Propagation of uncertainty through a measurement chain

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.4. Sampling plans for inspection validation

19.4.1. Attributes versus variables sampling

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.4.2. Binomial and hypergeometric bases for attribute plans

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.4.3. Operating-characteristic curves, AQL and RQL/LTPD

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.4.4. Zero-acceptance ($c=0$) plans and their use

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.4.5. Confidence–reliability sample-size formulae

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.4.6. ANSI/ASQ Z1.4, Z1.9 and ISO 2859 in practice

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.5. Measurement-system analysis for vision

19.5.1. Bias, linearity and stability studies

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.5.2. Uncertainty budgets and traceability

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.5.3. Total measurement uncertainty for vision gauges

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.6. Verification, troubleshooting and re-validation

19.6.1. A systematic troubleshooting method

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.6.2. Isolating optical, lighting, sensor and software failures

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.6.3. Recording and root-cause analysis

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.6.4. Change control and re-validation triggers

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20. Deployment, monitoring and human-in-the-loop for learning-based vision

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

20.1. From trained model to production system

20.1.1. Model export and interchange (ONNX)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.1.2. Quantisation, pruning and operator fusion

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.1.3. Runtime targets: CPU, GPU, edge accelerators

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.1.4. Revalidation of the deployed artefact

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.2. Dataset governance

20.2.1. Collection planning against the operating envelope

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.2.2. Labelling protocols and inter-annotator agreement

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.2.3. Curation, exclusion records and provenance

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.2.4. Versioning and model-to-data traceability

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.3. Post-deployment monitoring

20.3.1. Domain shift and concept drift

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.3.2. Statistical drift detectors

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.3.3. Retraining loops and change control

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.4. Human-in-the-loop systems

20.4.1. Why humans remain in critical and deep-learning loops

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.4.2. Review triggers: low confidence, drift, novelty, borderline cases

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.4.3. Interface and workflow design for human review

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.4.4. Human-in-the-loop as a labelling and improvement engine

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.4.5. Levels of autonomy and human oversight

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.4.6. Automation bias and the limits of human checking

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.5. Explainability and interpretability

20.5.1. Saliency and Grad-CAM

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.5.2. LIME and local surrogates

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.5.3. SHAP and Shapley attributions

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.5.4. Limits of post-hoc explanation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.5.5. Explanation as debugging versus explanation for audit

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.6. Bias and fairness in industrial vision

20.6.1. Subgroups in industrial data: cavities, suppliers, shifts

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.6.2. Subgroup analysis and reporting

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.6.3. Mitigations: data, loss, imaging

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.6.4. Fairness when the system observes people

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21. Process validation and regulation of vision inspections in regulated industries

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

21.1. Process validation of inspection systems

21.1.1. IQ, OQ and PQ for vision systems

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.1.2. Validating an inspection against its acceptance criteria

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.1.3. Linking validation to the sampling and R&R methods of Chapter 18

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.1.4. Software validation and computer-system validation (CSV/CSA)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.1.5. Validating learning-based inspections: special considerations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.2. The EU Artificial Intelligence Act

21.2.1. Scope, definitions and territorial reach

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.2.2. Prohibited practices

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.2.3. The high-risk category and Annex III

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.2.4. Obligations for providers and deployers of high-risk AI

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.2.5. Limited- and minimal-risk tiers

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.2.6. Timelines, enforcement and penalties

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.3. US regulation for AI and inspection

21.3.1. The FDA framework for AI/ML-enabled devices

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.3.2. Predetermined change-control plans and the total-product-lifecycle approach

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.3.3. 21 CFR Part 11: electronic records and signatures

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.3.4. NIST AI Risk Management Framework

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.3.5. Comparison of EU and US approaches

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.4. Medical-device and pharmaceutical inspection

21.4.1. MDR and IVDR requirements relevant to vision inspection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.4.2. FDA 21 CFR 820 / QSR and process validation expectations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.4.3. GMP and pharma visual-inspection requirements

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.4.4. Validating automated visual inspection against manual inspection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.5. Other high-risk and safety-critical sectors

21.5.1. Aerospace inspection and qualification expectations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.5.2. Automotive quality and functional-safety context

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.5.3. Common threads across regulated industries

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

Bibliography

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